MATHEMATICS

DETERMINANT

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DETERMINANT

JEE EXERCISES

Exercise J-1

BASED ON EXPANSION , PROPERTIES OF DETERMINANT

1. The value of the determinant $\begin{vmatrix} a^2 & a & 1 \\ \cos(nx) & \cos(n+1)x & \cos(n+2)x \\ \sin(nx) & \sin(n+1)x & \sin(n+2)x \end{vmatrix}$ is independent of :

(A) n (B) a (C) x (D) a , n and x

सारणिक $\begin{vmatrix} a^2 & a & 1 \\ \cos(nx) & \cos(n+1)x & \cos(n+2)x \\ \sin(nx) & \sin(n+1)x & \sin(n+2)x \end{vmatrix}$ का मान किससे स्वतन्त्र होगा :

(A) n (B) a (C) x (D) a , n तथा x

Ans. A

Sol. Directly open by R_1 to get a form of sin (A – B) etc.

2. If a, b, c are all different from zero and $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$, then the value of $a^{-1}+b^{-1}+c^{-1}$ is :

यदि a, b, c शून्य से भिन्न वास्तविक संख्यायें तथा $\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = 0$, तो $a^{-1}+b^{-1}+c^{-1}$ का मान होगा :

(A) abc (B) $a^{-1}b^{-1}c^{-1}$ (C) $-a-b-c$ (D) -1

Ans. D

Sol. $C_1 \to C_1 - C_2$ and $C_2 \to C_2 - C_3$ and then open by R_1 to get ab + abc + ac + bc = 0 ; divided by abc

3. If α, β and γ are the roots of the equation $x^3+px+q=0$, then the value of the determinant $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ =

(A) p (B) q (C) p^2-2q (D) None

यदि समीकरण $x^3 + px + q = 0$ के मूल α, β व γ है, तो सारणिक $\begin{vmatrix} \alpha & \beta & \gamma \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix}$ का मान होगा :

(A) p (B) q (C) $p^2 - 2q$ (D) कोई नहीं

Ans. D

Sol. $D = \alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma$

Suppose given $\alpha + \beta + \gamma = 0$

as coefficient of $x^2 = 0$

Now $\det(\alpha + \beta + \gamma)\begin{vmatrix} 1 & 1 & 1 \\ \beta & \gamma & \alpha \\ \gamma & \alpha & \beta \end{vmatrix} = 0 \Rightarrow$ (D)

4. If p^{th}, q^{th}, r^{th} term of a GP are ℓ, m, n then the value of $\begin{vmatrix} \log \ell & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix}$ is equal to :

(A) 0 (B) 1 (C) $\ell + m + n$ (D) None of these

यदि एक गुणोत्तर श्रेणी के pवें, qवें, rवें पद ℓ, m, n हों तो $\begin{vmatrix} \log \ell & p & 1 \\ \log m & q & 1 \\ \log n & r & 1 \end{vmatrix}$ का मान है :

(A) 0 (B) 1 (C) $\ell + m + n$ (D) इनमें से कोई नहीं

Ans. A

Sol. $\begin{vmatrix} \log a & p & 1 \\ \log b & q & 1 \\ \log c & r & 1 \end{vmatrix} = 0$

Here $a = AR^{p-1}$, $b = AR^{q-1}$, $C = AR^{r-1}$

$$\begin{vmatrix} \log AR^{p-1} & p & 1 \\ \log AR^{q-1} & q & 1 \\ \log AR^{r-1} & r & 1 \end{vmatrix} = \begin{vmatrix} \log A + (p-1)\log R & p & 1 \\ \log A + (q-1)\log R & q & 1 \\ \log A + (r-1)\log R & r & 1 \end{vmatrix}$$

Applying $R_2 \to R_2 \to R_1$ & $R_3 \to R_3 - R_1$

$$\begin{vmatrix} \log A + (p-1)\log R & p & 1 \\ \log R(q-p) & q-p & 0 \\ \log R(r-p) & r-p & 0 \end{vmatrix} = (r-p)(q-p)\begin{vmatrix} \log A + (p-1)\log R & p & 1 \\ \log R & 1 & 0 \\ \log R & 1 & 0 \end{vmatrix} = 0$$

5. If $a_1, a_2, a_3, \ldots\ldots, a_n, \ldots\ldots$ are in G.P., then the value of the determinant $\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$, is :

यदि $a_1, a_2, a_3, \ldots\ldots, a_n, \ldots\ldots$ गु.श्रे. में हों, तो सारणिक $\begin{vmatrix} \log a_n & \log a_{n+1} & \log a_{n+2} \\ \log a_{n+3} & \log a_{n+4} & \log a_{n+5} \\ \log a_{n+6} & \log a_{n+7} & \log a_{n+8} \end{vmatrix}$ का मान है :

(A) 0 (B) 1 (C) 2 (D) – 2

Ans. A

Sol. If $a_1, a_2, \ldots\ldots a_n, \ldots\ldots$ are in G.P.

then $\log a_1, \log a_2, \ldots\ldots \log a_n, \ldots\ldots$ are in A.P.

$A, \quad A + D, \ldots\ldots\ldots\ldots$

Let common difference of A.P. is D

so $\begin{vmatrix} \log a_n & \log a_{n+2} & \log a_{n+4} \\ \log a_{n+6} & \log a_{n+8} & \log a_{n+10} \\ \log a_{n+12} & \log a_{n+14} & \log a_{n+16} \end{vmatrix}$

$$\begin{vmatrix} A & A+2D & A+4D \\ A+6D & A+8D & A+10D \\ A+12D & A+14D & A+16D \end{vmatrix}$$

$C_2 \to 2C_2 - (C_1 + C_3)$

$$\begin{vmatrix} A & 0 & A+4D \\ A+6D & 0 & A+10D \\ A+12D & 0 & A+16D \end{vmatrix} = 0$$

6. If $a^2+b^2+c^2=-2$ and $f(x)=\begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$ then f(x) is a polynomial of degree :

यदि $a^2+b^2+c^2=-2$ तथा $f(x)=\begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$ तो f(x) एक ऐसा बहुपद है, जिसकी घात है :

(A) 1 (B) 0 (C) 3 (D) 2

Ans. D

Sol. On applying $C_1 \to C_1 + C_2 + C_3$

we get $\begin{vmatrix} 1 & x+b^2x & c^2x+x \\ 1 & 1+b^2x & c^2x+x \\ 1 & x+b^2x & c^2x+1 \end{vmatrix} x = f(x)$

Now solve it.

7. If $D=\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ for $x \neq 0, y \neq 0$ then D is :

(A) divisible by neither x nor y
(B) divisible by both x and y
(C) divisible by x but not y
(D) divisible by y but not x

यदि $D=\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$ $x \neq 0, y \neq 0$ तब D होगा :

(A) x तथा y दोनों से विभाज्य नहीं है
(B) x तथा y दोनों से विभाज्य है
(C) x से विभाज्य पर y से विभाज्य नहीं है
(D) y से विभाज्य पर x से विभाज्य नहीं

Ans. B

Sol. $D=\begin{vmatrix} 1 & 1 & 1 \\ 1 & 1+x & 1 \\ 1 & 1 & 1+y \end{vmatrix}$

$C_1 \to C_1 - C_2$, $C_2 \to C_2 - C_3$

$$D = \begin{vmatrix} 0 & 0 & 1 \\ -x & x & 1 \\ 0 & -y & 1+y \end{vmatrix} = xy$$

$\therefore$ D is divisible by both x and y $\therefore$ अतः D, x एवं y दोनों से विभाजित है।

8. Let a, b, c be such that $b(a + c) \neq 0$. If :

$$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix} = 0$$, then the value of n is : [AIEEE-2009]

(A) any even integer (B) any odd integer (C) any integer (D) zero

माना a, b, c के लिए $b(a + c) \neq 0$ यदि $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^n c \end{vmatrix} = 0$, तब n का मान है :

[AIEEE-2009]

(A) कोई भी समपूर्णांक (B) कोई भी विषम पूर्णांक (C) कोई भी पूर्णांक (D) शून्य

Ans. B

Sol. $$\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ a & -b & c \end{vmatrix}^{1} = \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^n \begin{vmatrix} a+1 & a-1 & a \\ b+1 & b-1 & -b \\ c-1 & c+1 & c \end{vmatrix}$$

$$= \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + (-1)^{n+1} \begin{vmatrix} a+1 & a & a-1 \\ b+1 & -b & b-1 \\ c-1 & c & c+1 \end{vmatrix}$$

$$C_2 \leftrightarrow C_3$$

$$= (1 + (-1)^{n+2}) \begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix}$$

This is equal to zero only if n + 2 is odd i.e. n is an odd integer.

यह तभी शून्य है जब n + 2 विषम है अर्थात् n विषम पूर्णांक है।

MULTIPLICATION OF DETERMINANT

9. If $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$, then Δ^2 is equal to :

(A) $\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$

(B) $\begin{vmatrix} bc+a^2 & ca+b^2 & ab+c^2 \\ ca+b^2 & ab+c^2 & bc+a^2 \\ ab+c^2 & bc+a^2 & ca+b^2 \end{vmatrix}$

(C) $\begin{vmatrix} a^2-bc & b^2-ca & c^2-ab \\ b^2-ca & c^2-ab & a^2-bc \\ c^2-ab & a^2-bc & b^2-ca \end{vmatrix}$

(D) None of these

यदि $\Delta = \begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$, तो Δ^2 बराबर होगा :

(A) $\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$

(B) $\begin{vmatrix} bc+a^2 & ca+b^2 & ab+c^2 \\ ca+b^2 & ab+c^2 & bc+a^2 \\ ab+c^2 & bc+a^2 & ca+b^2 \end{vmatrix}$

(C) $\begin{vmatrix} a^2-bc & b^2-ca & c^2-ab \\ b^2-ca & c^2-ab & a^2-bc \\ c^2-ab & a^2-bc & b^2-ca \end{vmatrix}$

(D) इनमें से कोई नहीं

Ans. A

Sol. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

det (cofactors) $= \Delta^{n-1} = \Delta^{3-1} = \Delta^2$

$$\det(\text{cofactors}) = \begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$$

10. If $ax + by + cz = 1, bx + cy + az = 0 = cx + ay + bz$, then $\begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix} \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$ is equal to :

यदि $ax + by + cz = 1, bx + cy + az = 0 = cx + ay + bz$, तो $\begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix} \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$ बराबर है :

(A) 0 (B) 1 (C) – 1 (D) 2

Ans. B

Sol. $\begin{vmatrix} x & y & z \\ z & x & y \\ y & z & x \end{vmatrix} \begin{vmatrix} a & c & b \\ b & a & c \\ c & b & a \end{vmatrix}$

$$\begin{vmatrix} ax+by+cz & cx+ay+bz & bx+cy+az \\ az+bx+cy & cz+ax+by & bz+cx+ay \\ ay+bz+cx & cy+za+bx & by+zc+ax \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{vmatrix} = 1$$

11. If $\alpha, \beta \neq 0$ and $f(n) = \alpha^n + \beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$ then K is equal to:

[JEE-(Main)-2014]

यदि $\alpha, \beta \neq 0$ तथा $f(n) = \alpha^n + \beta^n$ तथा $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K(1-\alpha)^2(1-\beta)^2(\alpha-\beta)^2$ है तो K बरारबर होगा :

[JEE-(Main)-2014]

(A) 1 (B) –1 (C) $\alpha\beta$ (D) $1/\alpha\beta$

Ans. A

Sol. $$\begin{vmatrix} 1+1+1 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix}^2$$

$= (1-\alpha)^2 (1-\beta)^2 (\alpha-\beta)^2$

SYSTEM OF EQUATIONS

12. Number of value of 'a' for which the system of equations,
$a^2x + (2-a)y = 4 + a^2$
$ax + (2a-1)y = a^5 - 2$ possess no solution is :
(A) 0 (B) 1 (C) 2 (D) infinite

'a' के कुल कितने मानों के लिये समीकरणों के निकाय
$a^2x + (2-a)y = 4 + a^2$
$ax + (2a-1)y = a^5 - 2$ का कोई हल नहीं होगा :
(A) 0 (B) 1 (C) 2 (D) अनन्त

Ans. C

Sol. $\dfrac{a^2}{a} = \dfrac{2-a}{2a-1} \neq \dfrac{4+a^2}{a^5-2}$ [Ans. $\{-1, 1\}$]

13. The following system of equations $3x - 7y + 5z = 3$; $3x + y + 5z = 7$ and $2x + 3y + 5z = 5$ are :
(A) consistent with trivial solution (B) consistent with unique non trivial solution
(C) consistent with infinite solution (D) inconsistent with no solution

समीकरण निकाय $3x - 7y + 5z = 3$; $3x + y + 5z = 7$ तथा $2x + 3y + 5z = 5$:
(A) शून्य हल के साथ संगत (B) अद्वितीय अशून्य हल के साथ संगत
(C) अन्नत हलों के साथ संगत (D) असंगत तथा कोई हल नहीं

Ans. B

Sol. $D \neq 0 \Rightarrow$ consistency

14. If the system of equations
$x - 2y + z = a$
$2x + y - 2z = b$
and $x + 3y - 3z = c$
have atleast one solution, then the relationship between a, b and c is :

यदि समीकरणों के निकाय

$x - 2y + z = a$

$2x + y - 2z = b$

तथा $x + 3y - 3z = c$

का कम से कम एक हल है तो a, b तथा c से सम्बन्ध बताओ :

(A) $a + b + c = 0$ (B) $a - b + c = 0$ (C) $-a + b + c = 0$ (D) $a + b - c = 0$

Ans. B

Sol. $D = \begin{vmatrix} 1 & -2 & 1 \\ 2 & 1 & -2 \\ 1 & 3 & -3 \end{vmatrix}$ which vanishes; hence for atleast one solution $D_1 = D_2 = D_3 = 0$

$\therefore \quad D_1 = \begin{vmatrix} a & -2 & 1 \\ b & 1 & -2 \\ c & 3 & -3 \end{vmatrix} = 0 \quad \Rightarrow \quad a - b + c = 0$ Ans.

15. Let α, β, γ are the real roots of the equation $x^3 + ax^2 + bx + c = 0$ ($a, b, c \in R$ and $a \neq 0$). If the system of equations (in u, v and w) given by

$\alpha u + \beta v + \gamma w = 0$

$\beta u + \gamma v + \alpha w = 0$

$\gamma u + \alpha v + \beta w = 0$

has non-trivial solutions, then a^2 equals :

माना α, β, γ समीकरण $x^3 + ax^2 + bx + c = 0$ के वास्तविक हल है। ($a, b, c \in R$ तथा $a \neq 0$) यदि समीकरण निकाय (u, v तथा w में)

$\alpha u + \beta v + \gamma w = 0$

$\beta u + \gamma v + \alpha w = 0$

$\gamma u + \alpha v + \beta w = 0$

सार्थक हल रखता हो, तो a^2 का मान होगा :

(A) b (B) 2b (C) 3b (D) 4b

Ans. C

Sol. We have $x^3 + ax^2 + bx + c = 0 \begin{cases} \alpha \\ \beta \\ \gamma \end{cases}$

$\therefore \quad \alpha + \beta + \gamma = -a$

$\alpha\beta + \beta\gamma + \gamma\alpha = b$

$$\begin{vmatrix}\alpha & \beta & \gamma\\ \beta & \gamma & \alpha\\ \gamma & \alpha & \beta\end{vmatrix}$$

$$\Rightarrow -\left(\alpha^3+\beta^3+\gamma^3-3\alpha\beta\gamma\right)=0$$

$$\text{or } -(\alpha+\beta+\gamma)\left(\left(\alpha^2+\beta^2+\gamma^2\right)-(\alpha\beta+\beta\gamma+\gamma\alpha)\right)=0$$

$$-(\alpha+\beta+\gamma)\left((\alpha+\beta+\gamma)^2-2(\alpha\beta+\beta\gamma+\gamma\alpha)-(\alpha\beta+\beta\gamma+\gamma\alpha)\right)=0$$

$a\,(a^2-3b)=0$

$a^2=3b$

16. If the system of linear equations

$x+2ay+az=0$

$x+3by+bz=0$

$x+4cy+cz=0$

has a non-zero solution, then a, b, c :

(A) are in G..P. (B) are in H.P.

(C) satisfy $a+2b+3c=0$ (D) are in A.P.

यदि रेखीय समीकरण निकाय

$x+2ay+az=0$

$x+3by+bz=0$

$x+4cy+cz=0$

का कोई अशून्य हल हो तो a, b, c :

(A) गु.श्रे. में है (B) ह.श्रे. में है

(C) $a+2b+3c=0$ को संतुष्ट करते है (D) स.श्रे. में है

Ans. B

Sol. For non zero solution

$$\begin{vmatrix}1 & 2a & a\\ 1 & 3b & b\\ 1 & 4c & c\end{vmatrix}=0$$

$$R_3\to R_3-R_2 \;\&\; R_2\to R_2-R_1$$

$$\begin{vmatrix}1 & 2a & a\\ 0 & 3b-2a & b-a\\ 0 & 4c-3b & c-b\end{vmatrix}=0$$

$3bc - 3b^2 - 2ac + 2ab - (abc - 3b^2 - 4ac + 3ab) = 0$

$-bc + 2ac - ab = 0$

divided by abc, then

$$\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$$

17. Equations $x + 2y + 3z = 1$, $2x + y + 3z = 2$ and $5x + 5y + 9z = 4$ have :

(A) unique solution (B) infinite solutions (C) inconsistent (D) None of these

समीकरण निकाय $x + 2y + 3z = 1$, $2x + y + 3z = 2$ तथा $5x + 5y + 9z = 4$ के हैं :

(A) अद्वितीय हल (B) अनन्त हल (C) असंगत हल (D) इनमें से कोई नही

Ans. A

Sol. $$D = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 5 & 5 & 9 \end{vmatrix}$$

$D = 1(9 - 15) - 2(18 - 13) + 3(10 - 5)$

$= -6 - 6 + 15$

$= 3 \neq 0$

$$D_1 = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 5 & 5 & 9 \end{vmatrix}$$

$D_1 = 1(9 - 15) - 2(18 - 12) + 3(10 - 4) = -6 - 12 + 18 = 0$

$$D_2 = \begin{vmatrix} 1 & 1 & 3 \\ 2 & 2 & 3 \\ 5 & 4 & 9 \end{vmatrix}$$

$D_2 = 1(18 - 12) - 1(18 - 15) + (8 - 10) = 0$

$= 6 - 3 - 6 \neq 0$

18. The existence of unique solution of the system $x + y + z = b$, $2x + 3y - z = 6$, $5x - y + az = 10$ depends on :

(A) b only (B) a only (C) a and b (D) neither a nor b

समीकरण निकाय $x + y + z = b$, $2x + 3y - z = 6$, $5x - y + az = 10$ का अद्वितीय हल निर्भर करेगा :

(A) केवल b पर (B) केवल a पर (C) a तथा b पर (D) न a पर और न ही b पर

Ans. B

Sol. $D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & -1 \\ 5 & -1 & a \end{vmatrix}$

$D = 1\,(3a - 1) - 1(2a + 5) + 1(-2 + 15)$

$D = 3a - 1 - 2a - 5 + 13$

$D = a + 7$

a only

19. The system equations $\alpha x + y + z = \alpha - 1$, $x + \alpha y + z = \alpha - 1$, $x + y + \alpha z = \alpha - 1$ has no solution, if α is :

(A) -2 (B) either -2 or 1 (C) not -2 (D) 1

समीकरण निकाय $\alpha x + y + z = \alpha - 1$; $x + \alpha y + z = \alpha - 1$; $x + y + \alpha z = \alpha - 1$ का कोई हल नहीं हैं, यदि α का मान :

(A) -2 है (B) -2 अथवा 1 है (C) -2 नहीं है (D) 1 है

Ans. A

Sol. For no solution

$\Delta = 0$ and Δ_x or Δ_y or Δ_z at least one is not zero.

$$\Delta = \begin{vmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{vmatrix} = 0$$

at $\alpha = 1$ their are 3 row are identical so factor of determinant $(\alpha - 1)^2$

and other factor will be find out by $R_1 \to R_1 + R_3 + R_3$

$\lambda(\alpha + 2)\,(\alpha - 1)^2 = 0$

$\alpha = -2, 1$

but at $\alpha = 1$

all equation are same so at $\alpha = 1$ system of equation infinite solution and

at $\alpha = -2$

$$\Delta x = \begin{vmatrix} -3 & 1 & 1 \\ -3 & -2 & 1 \\ -3 & 1 & -2 \end{vmatrix}$$

$-3(4 - 1) - 1(6 + 3) + 1\,(-3 + 6)$

$-9 - 9 + 3 = -15 \neq 0$

so at $\alpha = -2$ system have no solution.

20. The number of values of k, for which the system of equations

$(k + 1)x + 8y = 4k$,

$kx + (k + 3)y = 3k - 1$

has no solution, is :

(A) infinite (B) 1 (C) 2 (D) 3

k के उन मानों की संख्या, जिनके लिए निम्न समीकरण निकाय

$(k + 1)x + 8y = 4k$,

$kx + (k + 3)y = 3k - 1$

का कोई हल नहीं है : [JEE-(Main)-2013]

(A) infinite (B) 1 (C) 2 (D) 3

Ans. B

Sol. $\frac{k+1}{k} = \frac{8}{k+3} \neq \frac{4k}{3k-1} \Rightarrow k^2 + 4k + 3 = 8k \Rightarrow k^2 - 4k + 3 = 0$

$k = 1, 3$

If $k = 1$ then $\frac{8}{1+3} \neq \frac{4.1}{2}$ False

And If $k = 3$ then $\frac{8}{6} \neq \frac{4.3}{9-1}$ True

therefore $k = 3$/ Hence only one value of k.

Hindi. (2)

$\frac{k+1}{k} = \frac{8}{k+3} \neq \frac{4k}{3k-1} \Rightarrow k^2 + 4k + 3 = 8k \Rightarrow k^2 - 4k + 3 = 0$

$k = 1, 3$

यदि $k = 1$ हो तब $\frac{8}{1+3} \neq \frac{4.1}{2}$ असत्य

और यदि $k = 3$ तब $\frac{8}{6} \neq \frac{4.3}{9-1}$ सत्य

इसलिये $k = 3$ अतः k का केवल एक मान ही संभव है।

21. The set of all values of λ for which the system of linear equation

$2x_1 - 2x_2 + x_3 = \lambda x_1$

$2x_1 - 3x_2 + 2x_3 = \lambda x_2$

$-x_1 + 2x_2 = \lambda x_3$ has a non-trivial solution :

(A) is an empty set (B) is a singleton

(C) contains two elemetns (D) contains more two elements

λ के सभी मानों का समुच्चय, जिनके लिए रैखिक समीकरण निकाय

$2x_1 - 2x_2 + x_3 = \lambda x_1$

$2x_1 - 3x_2 + 2x_3 = \lambda x_2$

$-x_1 + 2x_2 = \lambda x_3$ का एक अतुच्छ हल है : [JEE-(Main)-2015]

(A) एक रिक्त समुच्चय है (B) एक एकल समुच्चय है

(C) में दो अवयव है (D) में दो से अधिक अवयव है

Ans. C

Sol. $\begin{vmatrix} 2-\lambda & -2 & 1 \\ 2 & -3-\lambda & 2 \\ -1 & 2 & -\lambda \end{vmatrix} = 0$

22. If S is the set of distinct values of 'b' for which the following system of linear equations $x + y + z = 1$, $x + ay + z = 1$, $ax + by + z = 0$ has no solution, then S is [JEE-(Main)-2017]

(A) a finite set containing two or more elements

(B) a singleton

(C) an empty set

(D) an infinite set

यदि S, 'b' की उन विभिन्न मानों का समुच्चय है जिनके लिए निम्न रैखिक समीकरण निकाय

$x + y + z = 1$, $x + ay + z = 1$, $ax + by + z = 0$ का कोई हल नहीं है, तो S : [JEE-(Main)-2017]

(A) एक परिमित समुच्चय है जिसमें दो या अधिक अवयव है

(B) एक ही अवयव वाला समुच्चय है

(C) एक रिक्त समुच्चय है

(D) एक अपरिमित समुच्चय है

Ans. B

Sol. $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & a & 1 \\ a & b & 1 \end{vmatrix} = -(a-1)^2 = 0$

$a = 1$

If $a = 1$ two planes are parallel

So for no solution

Third plane must be parallel

$\Rightarrow b = 1$

Exercise J-2

SUBJECTIVE / INTEGER TYPE QUESTIONS

BASED ON EXPANSION , PROPERTIES OF DETERMINANT

1. Without expanding as far as possible, prove that :

जहाँ तक संभव हो बिना विस्तार किये सिद्ध कीजिए कि :

(a) $\begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (a-1)^3$ (b) $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = [(x-y)(y-z)(z-x)(x+y+z)]$

Ans.

Sol. (a) $\begin{vmatrix} a^2+2a & 2a+1 & 1 \\ 2a+1 & a+2 & 1 \\ 3 & 3 & 1 \end{vmatrix} = (a-1)^3$

$C_2 \to C_2 - C_1$

$$\begin{vmatrix} a^2+2a & 1+a & 1 \\ 2a+1 & 1-a & 1 \\ 3 & 0 & 1 \end{vmatrix}$$

$$(1-a)\begin{vmatrix} a^2+2a & 1+a & 1 \\ 2a+1 & 1 & 1 \\ 3 & 0 & 1 \end{vmatrix}$$

$C_1 \to C_1 - 2C_2$

$$(1-a)\begin{vmatrix} a^2-2 & 1+a & 1 \\ 2a-1 & 1 & 1 \\ 3 & 0 & 1 \end{vmatrix}$$

$(1 - a)(a^2 - 2(1 - 0) - (1 + 0)(2a - 1 - 3) + 1(0 - 3))$
$(1 - a)(a^2 - 2 - (1 + a)(2a - 4) - 3)$
$(1 - a)(a^2 - 2 - 2a + 4 - 2a^2 + 4a - 3)$
$(1 - a)(-a^2 + 2a - 1)$
$-(1 - a)(a^2 - 2a + 1)$

$(1-a)(a-1)^2$

$\Rightarrow (a-1)^3$

(b) $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = (x-y)(y-z)(z-x)(x+y+z)$

$C_2 \to C_2 - C_1$ $\qquad$ $C_3 \to C_3 - C_1$

2. If $\begin{vmatrix} x^3+1 & x^2 & x \\ y^3+1 & y^2 & y \\ z^3+1 & z^2 & z \end{vmatrix} = 0$ and x, y, z are all different then, prove that $xyz = -1$.

यदि $\begin{vmatrix} x^3+1 & x^2 & x \\ y^3+1 & y^2 & y \\ z^3+1 & z^2 & z \end{vmatrix} = 0$ तथा x, y, z सभी भिन्न है तो सिद्ध कीजिए $xyz = -1$

Ans.

Sol. $\begin{vmatrix} x^3+1 & x^2 & x \\ y^3+1 & y^2 & y \\ z^3+1 & z^2 & z \end{vmatrix} = 0$

$$\begin{vmatrix} x^3 & x^2 & x \\ y^3 & y^2 & y \\ z^3 & z^2 & z \end{vmatrix} + \begin{vmatrix} 1 & x^2 & x \\ 1 & y^2 & y \\ 1 & z^2 & z \end{vmatrix}$$

$$xyz\begin{vmatrix} x^2 & x & 1 \\ y^2 & y & 1 \\ z^2 & z & 1 \end{vmatrix} + \begin{vmatrix} 1 & x^2 & x \\ 1 & y^2 & y \\ 1 & z^2 & z \end{vmatrix} = 0$$

$$xyz\begin{vmatrix} 1 & x^2 & x \\ 1 & y^2 & y \\ 1 & z^2 & z \end{vmatrix} + \begin{vmatrix} 1 & x^2 & x \\ 1 & y^2 & y \\ 1 & z^2 & z \end{vmatrix} = 0$$

$$\begin{vmatrix} 1 & x^2 & x \\ 1 & y^2 & y \end{vmatrix}(xyz+1)=0$$

$$\begin{vmatrix} 1 & x & x^2 \\ 1 & y & y^2 \end{vmatrix}(xyz+1)=0$$

x, y, z are all diff. then $xyz + 1 = 0 \Rightarrow xyz = -1$

3. Using properties of determinants or otherwise evaluate $\begin{vmatrix} 18 & 40 & 89 \\ 40 & 89 & 198 \\ 89 & 198 & 440 \end{vmatrix}$.

$\begin{vmatrix} 18 & 40 & 89 \\ 40 & 89 & 198 \\ 89 & 198 & 440 \end{vmatrix}$ का मान ज्ञात कीजिए।

Ans. –1

Sol. $\begin{vmatrix} 18 & 40 & 89 \\ 40 & 89 & 198 \\ 89 & 198 & 440 \end{vmatrix}$

$R_2 \to R_2 - 2R_1$ $\quad$ $R_3 \to R_3 - 2R_1$

$$\begin{vmatrix} 13 & 40 & 89 \\ 4 & 9 & 20 \\ 9 & 20 & 44 \end{vmatrix}$$

$R_1 \to R_1 - (2R_2 + R_3)$

$$= \begin{vmatrix} 1 & 2 & 5 \\ 4 & 9 & 20 \\ 9 & 20 & 44 \end{vmatrix}$$

$R_3 \to R_3 - 2R_1$

$$\begin{vmatrix} 1 & 2 & 5 \\ 4 & 9 & 20 \\ 1 & 2 & 4 \end{vmatrix}$$

$1(36 - 40) - 2(16 - 20) + 5(8 - 9)$

$-4 + 8 - 5 = -1$

4. Prove that (सिद्ध कीजिए) $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$.

Ans.

Sol. $R_1 \to R_1 + R_2 + R_3$

$$\begin{vmatrix} a+b+c & a+b+c & a+b+c \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$$(a+b+c)\begin{vmatrix} 1 & 1 & 1 \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix}$$

$C_2 \to C_2 - C_1 \qquad C_3 \to C_3 - C_1$

$$(a+b+c)\begin{vmatrix} 1 & 0 & 0 \\ 2b & -(a+b+c) & 0 \\ 2c & 0 & -(a+b+c) \end{vmatrix}$$

$$(a+b+c)^3\begin{vmatrix} 1 & 0 & 0 \\ 2b & -1 & 0 \\ 2c & 0 & -1 \end{vmatrix} \Rightarrow (a+b+c)^3$$

5. Prove that for all values of θ, $\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ \sin\left(\theta+\frac{2\pi}{3}\right) & \cos\left(\theta+\frac{2\pi}{3}\right) & \sin\left(2\theta+\frac{4\pi}{3}\right) \\ \sin\left(\theta-\frac{2\pi}{3}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta-\frac{4\pi}{3}\right) \end{vmatrix} = 0$

सिद्ध कीजिए कि θ के सभी मानों के लिए, $\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ \sin\left(\theta+\frac{2\pi}{3}\right) & \cos\left(\theta+\frac{2\pi}{3}\right) & \sin\left(2\theta+\frac{4\pi}{3}\right) \\ \sin\left(\theta-\frac{2\pi}{3}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta-\frac{4\pi}{3}\right) \end{vmatrix} = 0$ होगा।

Ans.

Sol. $R_2 \to R_2 + R_3$

$$\begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ 2\sin\theta\cos\frac{2\pi}{3} & 2\cos\theta\cos\frac{2\pi}{3} & 2\sin 2\theta\cos\frac{4\pi}{3} \\ \sin\left(\theta-\frac{2\pi}{3}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta-\frac{4\pi}{3}\right) \end{vmatrix}$$

$$\Rightarrow \begin{vmatrix} \sin\theta & \cos\theta & \sin 2\theta \\ -\sin\theta & -\cos\theta & -\sin 2\theta \\ \sin\left(\theta-\frac{2\pi}{3}\right) & \cos\left(\theta-\frac{2\pi}{3}\right) & \sin\left(2\theta-\frac{4\pi}{3}\right) \end{vmatrix}$$

0

6. Prove that (सिद्ध कीजिए) $\begin{vmatrix} (\beta+\gamma-\alpha-\delta)^4 & (\beta+\gamma-\alpha-\delta)^2 & 1 \\ (\gamma+\alpha-\beta-\delta)^4 & (\gamma+\alpha-\beta-\delta)^2 & 1 \\ (\alpha+\beta-\gamma-\delta)^4 & (\alpha+\beta-\gamma-\delta)^2 & 1 \end{vmatrix} = -64(\alpha-\beta)(\alpha-\gamma)(\alpha-\delta)(\beta-\gamma)(\beta-\delta)(\gamma-\delta)$

Ans.

Sol. Let $a = \beta + \gamma - \delta - \alpha$, $b = \gamma + \alpha - \beta - \delta$, $c = \alpha + \beta - \gamma - \delta$

we get $\begin{vmatrix} a^4 & a^2 & 1 \\ b^4 & b^2 & 1 \\ c^4 & c^2 & 1 \end{vmatrix}$

$R_1 \to R_1 - R_2$, $R_2 \to R_2 - R_3$

$= \begin{vmatrix} a^4-b^4 & a^2-b^2 & 0 \\ b^4-c^4 & b^2-c^2 & 0 \\ c^4 & c^2 & 1 \end{vmatrix}$

$= (a^2 - b^2)(b^2 - c^2) \begin{vmatrix} a^2+b^2 & 1 & 0 \\ b^2+c^2 & 1 & 0 \\ c^4 & c^2 & 1 \end{vmatrix}$

$= (a^2 - b^2)(b^2 - c^2)(a^2 - c^2)$

$= (a - b)(a + b)(b - c)(b + c)(a + c)(a - c)$

$= -2^6 (\alpha - \beta)(\alpha - \gamma)(\beta - \gamma)(\alpha - \delta)(\beta - \delta)(\gamma - \delta)$

7. Solve for x the equation $\begin{vmatrix} a^2 & a & 1 \\ \sin(n+1)x & \sin nx & \sin(n-1)x \\ \cos(n+1)x & \cos nx & \cos(n-1)x \end{vmatrix} = 0.$

x के लिये निम्न समीकरण को सरल कीजिए $\begin{vmatrix} a^2 & a & 1 \\ \sin(n+1)x & \sin nx & \sin(n-1)x \\ \cos(n+1)x & \cos nx & \cos(n-1)x \end{vmatrix} = 0.$

Ans. $x = n\pi$, $n \in I$

Sol. already done

8. Solve for x

x के लिये हल कीजिए

(a) $\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+5 & 5x+8 & 10x+17 \end{vmatrix} = 0$ (b) $\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0$

Ans. (a) $x = -1$ or $x = -2$; (b) $x = 4$

Sol. (a) $\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 2x+3 & 3x+4 & 4x+5 \\ 3x+5 & 5x+8 & 10x+17 \end{vmatrix} = 0$

$R_2 \to R_2 - 2R_1$ $R_3 \to R_3 - 3R_1$

$$\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ -1 & -x-2 & -2x-3 \\ -1 & -x-1 & x+5 \end{vmatrix}$$

$R_2 \to R_2 - R_3$

$$\begin{vmatrix} x+2 & 2x+3 & 3x+4 \\ 0 & -1 & -3x-8 \\ -1 & -x-1 & x+5 \end{vmatrix}$$

$(x + 1)(x^2 + 3x + 2) = 0$

$x = -1, -2$

(b) $\begin{vmatrix} x-2 & 2x-3 & 3x-4 \\ x-4 & 2x-9 & 3x-16 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0$

$R_1 \to R_1 - R_2$

$R_2 \to R_2 - R_3$

$$\begin{vmatrix} 2 & 6 & 12 \\ 4 & 18 & 48 \\ x-8 & 2x-27 & 3x-64 \end{vmatrix} = 0$$

$R_3 \to R_3 - (R_1 + R_2)$

$$\begin{vmatrix} 2 & 6 & 12 \\ 4 & 18 & 48 \\ 2x-4 & 2x-3 & 3x-4 \end{vmatrix} = 0$$

$\Rightarrow x = 4$

9. If $a+b+c=0$, solve for x : $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0.$

यदि $a+b+c=0$ है, तो x के लिये हल कीजिये : $\begin{vmatrix} a-x & c & b \\ c & b-x & a \\ b & a & c-x \end{vmatrix} = 0.$

Ans. $x = 0$ or $x = \pm\sqrt{\frac{3}{2}\left(a^2 + b^2 + c^2\right)}$

Sol. Hint : On applying $R_1 \to R_1 + R_2 + R_3$
and then solving the determinant we get

$x = 0$ or $x^2 = a^2 + b^2 + c^2 - ab - bc - ca$

$$x = \pm\sqrt{\frac{3}{2}(a^2 + b^2 + c^2)}$$

10. If $p+q+r=0$, prove that $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix} = pqr \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$.

यदि $p+q+r=0$, सिद्ध कीजिये $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix} = pqr \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$

Ans.

Sol. $\begin{vmatrix} pa & qb & rc \\ qc & ra & pb \\ rb & pc & qa \end{vmatrix}$

$\Rightarrow pa(a^2qr - p^2bc) - qb(q^2ac - prb^2) + rc(pqc^2 - r^2ab)$

$\Rightarrow a^3pqr - p^3abc - q^3abc + b^3pqr + c^3pqr - r^3abc$

$\Rightarrow pqr(a^3 + b^3 + c^3) - abc(p^3 + q^3 + r^3)$

$pqr(a^3 + b^3 + c^3) - abc(3pqr)$

$(pqr)(a^3 + b^3 + c^3 - 3abc)$

$$-pqr\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} + pqr\begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$$

11. If a, b, c are all different and $\begin{vmatrix} a & a^3 & a^4-1 \\ b & b^3 & b^4-1 \\ c & c^3 & c^4-1 \end{vmatrix} = 0$, then prove that, $abc(ab+bc+ca) = a+b+c$.

यदि a, b, c सभी भिन्न है तथा $\begin{vmatrix} a & a^3 & a^4-1 \\ b & b^3 & b^4-1 \\ c & c^3 & c^4-1 \end{vmatrix} = 0$, तो सिद्ध कीजिए $abc(ab+bc+ca) = a+b+c$

Ans.

Sol. $\begin{vmatrix} a & a^3 & a^4-1 \\ b & b^3 & b^4-1 \\ c & c^3 & c^4-1 \end{vmatrix} = 0$

$$\begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} - \begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix}$$

$$(abc)\begin{vmatrix}1 & a^2 & a^3\\ 1 & b^2 & b^3\\ 1 & c^2 & c^3\end{vmatrix}-\begin{vmatrix}1 & a & a^3\\ 1 & b & b^3\\ 1 & c & c^3\end{vmatrix}=0$$

$abc(a-b)(b-c)(c-a)(ab+bc+ca)-(a-b)(b-c)(c-a)(a+b+c)=0$

$(a-b)(b-c)(c-a)\{abc(ab+bc+ca)-(a+b+c)\}=0$

a, b, c are all diff. so,

$abc(ab+bc+ca)=a+b+c$

12. Show that, $\begin{vmatrix}a^2+\lambda & ab & ac\\ ab & b^2+\lambda & bc\\ ac & bc & c^2+\lambda\end{vmatrix}$ is divisible by λ^2 and find the other factor.

दर्शाये कि सारणिक, $\begin{vmatrix}a^2+\lambda & ab & ac\\ ab & b^2+\lambda & bc\\ ac & bc & c^2+\lambda\end{vmatrix}$, λ^2 से भाज्य है तथा इसका अन्य गुणनखण्ड भी ज्ञात कीजिए।

Ans. $\lambda^2(a^2+b^2+c^2+\lambda)$

Sol. $$(abc)\begin{vmatrix}a+\frac{\alpha}{a} & a & a\\ b & b+\frac{\lambda}{b} & b\\ c & c & c+\frac{\lambda}{c}\end{vmatrix}$$

$$\begin{vmatrix}a^2+\lambda & a^2 & a^2\\ b^2 & b^2+\lambda & b^2\\ c^2 & c^2 & c^2+\lambda\end{vmatrix}$$

$R_1 \to R_1+R_2+R_3$

$$\begin{vmatrix}\lambda+a^2+b^2+c^2 & \lambda^2+a^2+b^2+c^2 & \lambda+a^2+b^2+c^2\\ b^2 & b^2+\lambda & b^2\\ c^2 & c^2 & c^2+\lambda\end{vmatrix}$$

$$\Rightarrow \left(\lambda + a^2 + b^2 + c^2\right)\begin{vmatrix} 1 & 1 & 1 \\ b^2 & b^2+\lambda & b^2 \\ c^2 & c^2 & c^2+\lambda \end{vmatrix}$$

$C_2 \to C_2 - C_1 \qquad C_3 \to C_3 - C_1$

$$\Rightarrow \left(\lambda + a^2 + b^2 + c^2\right)\begin{vmatrix} 1 & 0 & 0 \\ b^2 & \lambda & 0 \\ c^2 & 0 & \lambda \end{vmatrix}$$

$$\Rightarrow \left(\lambda + a^2 + b^2 + c^2\right)\left(\lambda^2\right)$$

13. Evaluate $\begin{vmatrix} \sqrt{13}+\sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15}+\sqrt{26} & 5 & \sqrt{10} \\ 3+\sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$

$\begin{vmatrix} \sqrt{13}+\sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15}+\sqrt{26} & 5 & \sqrt{10} \\ 3+\sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$ मान ज्ञात कीजिए।

Ans. $5\left(3\sqrt{2}-5\sqrt{3}\right)$

Sol. $\begin{vmatrix} \sqrt{13}+\sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15}+\sqrt{26} & 5 & \sqrt{10} \\ 3+\sqrt{65} & \sqrt{15} & 5 \end{vmatrix} = \begin{vmatrix} \sqrt{3} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{15} & 5 & \sqrt{10} \\ 3 & \sqrt{15} & 5 \end{vmatrix} + \begin{vmatrix} \sqrt{13} & 2\sqrt{5} & \sqrt{5} \\ \sqrt{26} & 5 & \sqrt{10} \\ \sqrt{65} & \sqrt{15} & 5 \end{vmatrix}$

$$= \sqrt{3}\sqrt{5}\sqrt{5}\begin{vmatrix} 1 & 2 & 1 \\ \sqrt{5} & \sqrt{5} & \sqrt{2} \\ \sqrt{3} & \sqrt{3} & \sqrt{5} \end{vmatrix} + \sqrt{13}\sqrt{5}\sqrt{5}\begin{vmatrix} 1 & 2 & 1 \\ \sqrt{2} & \sqrt{5} & \sqrt{2} \\ \sqrt{5} & \sqrt{3} & \sqrt{5} \end{vmatrix}$$

$C_1 \to C_1 - C_2$

$$= 5\sqrt{3}\begin{vmatrix} -1 & 2 & 1 \\ 0 & \sqrt{5} & \sqrt{2} \\ 0 & \sqrt{3} & \sqrt{5} \end{vmatrix} = 5(3\sqrt{2} - 5\sqrt{3})$$

14. Let $\Delta_r = \begin{vmatrix} r-1 & n & 6 \\ (r-1)^2 & 2n^2 & 4n-2 \\ (r-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$. Find the sum $\sum_{r=1}^{n} \Delta_r$

माना $\Delta_r = \begin{vmatrix} r-1 & n & 6 \\ (r-1)^2 & 2n^2 & 4n-2 \\ (r-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$ योग $\sum_{r=1}^{n} \Delta_r$ का मान ज्ञात कीजिए।

Ans. 0

Sol.
$$\begin{vmatrix} \sum_{r=1}^{n} r-1 & n & 6 \\ \sum_{r=1}^{n} (r-1)^2 & 2n^2 & 4n-2 \\ \sum_{r=1}^{n} (r-1)^3 & 3n^3 & 3n^2-3n \end{vmatrix}$$

$$\begin{vmatrix} \frac{n(n-1)}{2} & n & 6 \\ \frac{n(n-1)(2n-1)}{6} & 2n^2 & 4n-2 \\ \left(\frac{n(n-1)}{2}\right)^2 & 3n^3 & 3n^2-3n \end{vmatrix}$$

$$\frac{n(n-1)}{2}\begin{vmatrix} 1 & n & 6 \\ \frac{2n-1}{3} & 2n^2 & 4n-2 \\ \frac{n(n-1)}{2} & 3n^3 & 3n^2-3n \end{vmatrix}$$

$$\frac{n(n-1)}{2\times 6}\begin{vmatrix} 6 & n & 6 \\ (2n-1)2 & 2n^2 & 4n-2 \\ 3n(n-1) & 3n^3 & 3n^2-3n \end{vmatrix}$$

$C_1 = C_3$

MULTIPLICATION OF DETERMINANT

15. Prove that (सिद्ध कीजिए) $\begin{vmatrix} (a_1-b_1)^2 & (a_1-b_2)^2 & (a_1-b_3)^2 \\ (a_2-b_1)^2 & (a_2-b_2)^2 & (a_2-b_3)^2 \\ (a_3-b_1)^2 & (a_3-b_2)^2 & (a_3-b_3)^2 \end{vmatrix} = 2(a_1-a_2)(a_2-a_3)(a_3-a_1)(b_1-b_2)(b_2-b_3)(b_3-b_1)$

Ans.

Sol. $\begin{vmatrix} a_1^2-2a_1b_1+b_1^2 & a_1^2-2a_1b_2+b_2^2 & a_1^2+b_2^3-2a_1b_3 \\ a_2^2+b_1^2-2a_2b_1 & a_2^2+b_2^2-2a_2b_2 & a_2^2+b_3^2-2a_2b_3 \\ a_3^2+b_1^2-2a_3b_1 & a_3^2+b_2^2-3a_3b_2 & a_3^2+b_3^2-3a_3b_3 \end{vmatrix}$

$$\begin{vmatrix} a_1^2 & -2a_1 & 1 \\ a_2^2 & -2a_2 & 1 \\ a_3^2 & -2a_3 & 1 \end{vmatrix}\begin{vmatrix} 1 & 1 & 1 \\ b_1 & b_2 & b_3 \\ b_1^2 & b_2^2 & b_3^2 \end{vmatrix}$$

$2(a_1 - a_2)\,(a_2 - a_3)\,(a_3 - a_1)\,(b_1 - b_2)\,(b_2 - b_3)\,(b_3 - b_1)$

16. Prove that (सिद्ध कीजिए) $\begin{vmatrix} 2 & \alpha+\beta+\gamma+\delta & \alpha\beta+\gamma\delta \\ \alpha+\beta+\gamma+\delta & 2(\alpha+\beta)(\gamma+\delta) & \alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta) \\ \alpha\beta+\gamma\delta & \alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta) & 2\alpha\beta\gamma\delta \end{vmatrix} = 0.$

Ans.

Sol. $\begin{vmatrix} 1+1 & (\alpha+\beta)+(\gamma+\delta) & \alpha\beta+\gamma\delta \\ (\alpha+\beta)+(\gamma+\delta) & 2(\alpha+\beta)(\gamma+\delta) & \alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta) \\ \alpha\beta+\gamma\delta & \alpha\beta(\gamma+\delta)+\gamma\delta(\alpha+\beta) & 2\alpha\beta\gamma\delta \end{vmatrix}$

$$= \begin{vmatrix} 1 & 1 & 0 \\ (\gamma+\delta) & \alpha+\beta & 0 \\ \gamma\delta & \alpha\beta & 0 \end{vmatrix} \begin{vmatrix} 1 & \alpha+\beta & \alpha\beta \\ 1 & \gamma+\delta & \gamma\delta \\ 0 & 1 & 1 \end{vmatrix}$$

17. Prove that (सिद्ध कीजिए) : $\begin{vmatrix} (a-p)^2 & (a-q)^2 & (a-r)^2 \\ (b-p)^2 & (b-q)^2 & (b-r)^2 \\ (c-p)^2 & (c-q)^2 & (c-r)^2 \end{vmatrix} = \begin{vmatrix} (1+ap)^2 & (1+aq)^2 & (1+ar)^2 \\ (1+bp)^2 & (1+bq)^2 & (1+br)^2 \\ (1+cp)^2 & (1+cq)^2 & (1+cr)^2 \end{vmatrix}$

Ans.

Sol. $\begin{vmatrix} a^2+p^2-2ap & a^2+q^2-2aq & a^2+r^2-2ar \\ b^2+p^2-2bp & b^2+q^2-2bq & b^2+r^2-2br \\ c^2+p^2-2pc & c^2+q^2-2cq & c^2+r^2-2cr \end{vmatrix}$

$$\begin{vmatrix} a^2 & -2a & 1 \\ b^2 & -2b & 1 \\ c^2 & -2c & 1 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ p & q & r \\ p^2 & q^2 & r^2 \end{vmatrix}$$

$$-\begin{vmatrix} 1 & -2a & a^2 \\ 1 & -2b & b^2 \\ 1 & -2c & c^2 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ p & q & r \\ p^2 & q^2 & r^2 \end{vmatrix} \Rightarrow \begin{vmatrix} 1 & 2a & a^2 \\ 1 & 2b & b^2 \\ 1 & 2c & c^2 \end{vmatrix} \begin{vmatrix} 1 & 1 & 1 \\ p & q & r \\ p^2 & q^2 & r^2 \end{vmatrix}$$

$$\begin{vmatrix} 1+2ap+a^2p^2 & 1+2aq+a^2q^2 & 1+2ar+a^2r^2 \\ - & - & - \\ - & - & - \end{vmatrix}$$

$$= \begin{vmatrix} (1+ap)^2 & (1+aq)^2 & (1+ar)^2 \\ (1+bp)^2 & (1+bq)^2 & (1+br)^2 \\ (1+cp)^2 & (1+cq)^2 & (1+cr)^2 \end{vmatrix}$$

18. If $S_r = \alpha^r + \beta^r + \gamma^r$ then show that $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix} = (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2$.

यदि $S_r = \alpha^r + \beta^r + \gamma^r$ हो, तो दर्शाइये कि $\begin{vmatrix} S_0 & S_1 & S_2 \\ S_1 & S_2 & S_3 \\ S_2 & S_3 & S_4 \end{vmatrix} = (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2$

Ans.

Sol. $\begin{vmatrix} 3 & \alpha+\beta+\gamma & \alpha^2+\beta^2+\gamma^2 \\ \alpha+\beta+\gamma & \alpha^2+\beta^2+\gamma^2 & \alpha^3+\beta^3+\gamma^3 \\ \alpha^2+\beta^2+\gamma^2 & \alpha^3+\beta^3+\gamma^3 & \alpha^4+\beta^4+\gamma^4 \end{vmatrix}$

$= \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix} \times \begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix}$

$= (\alpha - \beta)^2 (\beta - \gamma)^2 (\gamma - \alpha)^2$

SYSTEM OF EQUATIONS

19. Find the real values of r for which the following system of linear equations has a non-trivial solution. Also find the non-trivial solutions :

$2rx - 2y + 3z = 0$
$x + ry + 2z = 0$
$2x + rz = 0$

r के किस वास्तविक मान के लिये रेखीय समीकरण निकाय :

$2rx - 2y + 3z = 0$
$x + ry + 2z = 0$
$2x + rz = 0$ के अशून्य हल होंगे।

Ans. $r = 2; x = k; y = k/2; z = -k$ where $k \in R - \{0\}$

Sol. $D = \begin{vmatrix} 2r & -2 & 3 \\ 1 & r & 2 \\ 2 & 0 & r \end{vmatrix} = 0$

$2r(r^2 - 0) + 2(r - 4) + 3(0 - 2r) = 0$

$2r^3 + 2r - 8 - 6r = 0$

$2r^3 - 4r - 8 = 0$

$r^3 - 2r - 4 = 0$

$r^2 (r - 2) + 2r (r - 2) + 2 (r - 2)$

$(r - 2) (r^2 + 2r + 2) = 0$

$(r - 2)$

$r = 2$

Let $x = k$

$2k + 2z = 0$

$z = - k$

$k + 2y - 2k = 0$

$2y = k$

$$y = \frac{k}{2}$$

non trivial solution

$$\left(k, \frac{k}{2}, -k \right)$$

20. Solve the following using Cramer's rule and state whether consistent or not :

क्रेमर के नियम का प्रयोग करते हुए निम्न समीकरण निकायों को हल कीजिये तथा बताइयें कि यह संगत हैं या नहीं :

(a) $\begin{matrix} x + y + z - 6 = 0 \\ 2x + y - z - 1 = 0 \\ x + y - 2z + 3 = 0 \end{matrix}$ (b) $\begin{matrix} x + 2y + z = 1 \\ 3x + y + z = 6 \\ x + 2y = 0 \end{matrix}$ (c) $\begin{matrix} 7x - 7y + 5z = 3 \\ 3x + y + 5z = 7 \\ 2x + 3y + 5z = 5 \end{matrix}$

Ans. (a) $x = 1, y = 2, z = 3$; consistent (b) $x = 2, y = -1, z = 1$; consistent, (c) inconsistent

Sol. (a) $$D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 1 & 1 & -2 \end{vmatrix} = 3$$

$$\Rightarrow D_1 = \begin{vmatrix} 6 & 1 & 1 \\ 1 & 1 & -1 \\ -3 & 1 & -2 \end{vmatrix} = 3$$

$$D_2 = \begin{vmatrix} 1 & 6 & 1 \\ 2 & 1 & -1 \\ 1 & -3 & -2 \end{vmatrix} = 6$$

Similarly $D_3 = 9$

so consistent having $x = 1, y = 2, z = 3$

(c) $$D = \begin{vmatrix} 7 & -7 & 5 \\ 3 & 1 & 5 \\ 2 & 3 & 5 \end{vmatrix} = 0$$

$$D_1 = \begin{vmatrix} 3 & -7 & 1 \\ 7 & 1 & 1 \\ 5 & 3 & 1 \end{vmatrix} = 24$$

Inconsistent system

21. For what value of K do the following system of equations possess a non trivial (i.e. not all zero) solution over the set of rationals Q?

$x + Ky + 3z = 0$, $3x + Ky - 2z = 0$, $2x + 3y - 4z = 0$.

For that value of K , find all the solutions of the system.

K के किन मानों के लिये निम्नलिखित समीकरणों के निकाय का परिमेय संख्याओं के समुच्चय Q में अनिरर्थक हल (अर्थात सभी शून्य नहीं) होंगे?

$x + Ky + 3z = 0$, $3x + Ky - 2z = 0$, $2x + 3y - 4z = 0$.

K के उस मान कें लिये निकाय के सभी हल भी ज्ञात कीजिये।

Ans. $K = \frac{33}{2}$, $x : y : z = -\frac{15}{2} : 1 : -3$

Sol. $$D = \begin{vmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{vmatrix} = 0$$

$1(-4k + 6) - k(-12 + 4) + 3(9 - 2k) = 0$

$-4k + 6 + 12k - 4k + 27 - 6k = 0$

$33 - 2k = 0$

$k = \frac{33}{2}$

Let $x = k$

$ky + 3z = -k$

$ky - 2z = -3k$

$- \quad + \quad +$

$5z = 2k$

$$z = \frac{2}{5}k$$

$$2k + 3y - 4\left(\frac{2k}{5}\right) = 0$$

$$3y = \frac{8k}{5} - 2k$$

$$3y = -\frac{2k}{5}$$

$$y = -\frac{2k}{15}$$

$x : y : z$

$$k : -\frac{2}{15}k : \frac{2}{5}k$$

22. The system of equations

$\alpha x + y + z = \alpha - 1$

$x + \alpha y + z = \alpha - 1$

$x + y + \alpha z = \alpha - 1$ has no solution. Find α.

यदि समीकरणों के निकाय

$\alpha x + y + z = \alpha - 1$

$x + \alpha y + z = \alpha - 1$

$x + y + \alpha z = \alpha - 1$ का कोई हल नहीं हो, तो α का मान ज्ञात कीजिये।

Ans. –2

Sol.
$$D = \begin{vmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{vmatrix} = 0$$

$$\alpha\left(\alpha^2 - 1\right) - 1\left(\alpha - 1\right) + 1\left(1 - \alpha\right) = 0$$

$\alpha^3 - \alpha - \alpha + 1 + 1 - \alpha = 0$

$\alpha^3 - 3\alpha + 2 = 0$

$\alpha^2(\alpha - 1) + \alpha(\alpha - 1) - 2(\alpha - 1) = 0$

$(\alpha - 1)(\alpha^2 + \alpha - 2)$

$(\alpha - 1)(\alpha^2 + 2\alpha - \alpha - 2)$

$(\alpha - 1)(\alpha(\alpha + 2) - 1(\alpha + 2))$

$\alpha = 1$ $\alpha = -2$

23. Given $x = cy + bz$; $y = az + cx$; $z = bx + ay$ where x , y, z are not all zero, prove that $a^2 + b^2 + c^2 + 2abc = 1$.

दिया गया है $x = cy + bz$; $y = az + cx$; $z = bx + ay$ जहाँ x , y, z सभी शून्य नहीं है, सिद्ध कीजिये कि $a^2 + b^2 + c^2 + 2abc = 1$.

Ans.

Sol. $x - cy - bz = 0$

$cx - y + az = 0$ $x \neq 0$; $y \neq 0, z \neq 0$

$bx + ay - z = 0$

these system is homogeneous

so $\Delta_x = \Delta_y = \Delta_z = 0$

and at $\Delta = 0 \rightarrow$ system have non zero solution.

$$\Delta = \begin{vmatrix} 1 & -c & -b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0$$

$1 - a^2 + c(-c - ab) - b(ac + b) = 0$

$1 - a^2 - b^2 - c^2 - abc - abc = 0$

$a^2 + b^2 + c^2 + 2abc = 1$

24. For what values of p , the equations : $x + y + z = 1$; $x + 2y + 4z = p$ and $x + 4y + 10z = p^2$ have a solution? Solve them completely in each case.

p के किन मानों के लिये समीकरणों $x + y + z = 1$; $x + 2y + 4z = p$ तथा $x + 4y + 10z = p^2$ का हल होगा ? प्रत्येक मान के लिये इन्हे हल भी कीजिये।

Ans. $x = 1 + 2K$, $y = -3K$, $z = K$, when $p = 1$; $x = 2K$, $y = 1 - 3K$, $z = K$ when $p = 2$; where $K \in R$

Sol. $D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 4 & 10 \end{vmatrix}$

$D = 1\,(20 - 16) - 1(10 - 4) + 1\,(4 - 2)$

$D = 4 - 6 + 2 = 0$

$D_1 = D_2 = D_3 = 0$

$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ p & 2 & 4 \\ p^2 & 4 & 10 \end{vmatrix}$

$D_1 = 1\,(20 - 16) - 1(10p - 4p^2) + 1(4p - 2p^2) = 0$

$D_1 = 4 - 10p + 4p^2 + 4p - 2p^2 = 0$

$= 2p^2 - 6p + 4 = 0$

$= p^2 - 3p + 2 = 0$

$p^2 - 2p - p + 2 = 0$

$p(p - 2) - 1\,(p - 2) = 0$

$(p - 1)\,(p - 2) = 0$

$p = 1, 2$

for $p = 1$ $\quad x + y + z = 1$

$x + 2y + 4z = 1$

$x + 4y + 10z = 1$

for $p = 2$ $\quad x + y + z = 1 \quad z = k$

$x + 2y + 4z = 2 \quad y = 1 - 3k$

$x + 4y + 10z = 4 \quad x = 2k$

let $z = k$

$x + y = 1 - k$

$x + 2y = 1 - 4k$

$- \quad - \quad - \quad +$

$-y = 3k$

$y = -3k$

$x - 3k + k = 0$

$x - 2k = 1$

$x = 2k + 1$

25. If the following system of equations $(a-t)x+by+cz=0$, $bx+(c-t)y+az=0$ and $cx+ay+(b-t)z=0$ has non–trivial solutions for different values of t, then show that we can express product of these values of t in the form of determinant.

यदि समीकरणो के निकाय $(a-t)x+by+cz=0$, $bx+(c-t)y+az=0$ तथा $cx+ay+(b-t)z=0$ के t के विभिन्न मानो के लिये अनिरर्थक हल है, तो दशाईये की t के सभी मानों के गुणन को एक सारणिक के रूप में प्रदर्शित किया जा सकता है।

Ans. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$

Sol. $(A-t)\,x + by + cz = 0$

$bx + (c-t)y + az = 0$

$cx + ay + (b-t)z = 0$

$$D = \begin{vmatrix} a-t & b & c \\ b & c-t & a \\ c & a & b-t \end{vmatrix}$$

$R_1 \to R_1 + R_2 + R_3$

$$\begin{vmatrix} a+b+c-t & a+b+c-t & a+b+c-t \\ b & c-t & a \\ c & a & b-t \end{vmatrix} = 0$$

$$(a+b+c-t)\begin{vmatrix} 1 & 1 & 1 \\ b & c-t & a \\ c & a & b-t \end{vmatrix} = 0$$

$C_2 \to C_2 - C_1$ $\qquad$ $C_3 \to C_3 - C_1$

$$(a+b+c-t)\begin{vmatrix} 1 & 0 & 0 \\ b & c-t-b & a-b \\ c & a-c & b-t-c \end{vmatrix} = 0$$

$(a + b + c - t)((c - t - b)(b - t - c) - (a - c)(a - b)) = 0$

$(a + b + c - t)(bc - ct - c^2 - tb + t^2 + tc - b^2 + bt + bc - a^2 + ab + ac - bc) = 0$

$(a + b + c - t)(t^2 - a^2 - b^2 - c^2 + ab + bc + ca) = 0$

$t = a + b + c$

$t^2 = a^2 + b^2 + c^2 - ab - bc - ca$

$$t^3 = \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix} = a^3 + b^3 + c^3 - 3abc$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 26 to 28

Consider the determinant $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix}$

M_{ij} = Minor of the element of ith row and j^{th} column
C_{ij} = Cofactor of the element of ith row and j^{th} column

माना सारणिक $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ d_1 & d_2 & d_3 \end{vmatrix}$

M_{ij} = i^{th} पक्ति तथा j^{th} स्तम्भ के अवयव का उपसारणिक
Cij = i^{th} पक्ति तथा j^{th} स्तम्भ के अवयव का सहखण्ड

26. Value of $b_1 . C_{31} + b_2 . C_{32} + b_3 . C_{33}$ is :

$b_1 . C_{31} + b_2 . C_{32} + b_3 . C_{33}$ का मान होगा :

(A) 0 (B) Δ (C) 2Δ (D) Δ^2

Ans. A

Sol. On multiplying elements of any row with co-factor of corresponding elements of other row then added then the resultant sum will be zero.

27. If all the elements of the determinant are multiplied by 2, then the value of new determinant is :

यदि सारणिक के सभी अवयवों को 2 से गुणा किया जाये तो नये सारणिक का मान होगा :

(A) 0 (B) 8Δ (C) 2Δ (D) $2^9\Delta$

Ans. B

Sol. If all the elements of a det. are multiplied by k then value of that det. gets multiplied by k^n where n is the order of det.

28. $a_3 M_{13} - b_3 . M_{23} + d_3 . M_{33}$ is equal to :

$a_3 M_{13} - b_3 . M_{23} + d_3 . M_{33}$ का मान होगा :

(A) 0 (B) 4Δ (C) 2Δ (D) Δ

Ans. D

Sol. By multiplying elements of a row with corresponding cofactors and then added, resultant sum will be equal to the value of det.

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

29. The set of equations $x - y + 3z = 2$, $2x - y + z = 4$, $x - 2y + \alpha z = 3$ has :

(A) unique soluton only for $\alpha = 0$ (B) unique solution for $\alpha \neq 8$

(C) infinite number of solutions for $\alpha = 8$ (D) no solution for $\alpha = 8$

समीकरणों का समुच्चय $x - y + 3z = 2$, $2x - y + z = 4$ तथा $x - 2y + \alpha z = 3$ के पास :

(A) अद्वितीय हल केवल $\alpha = 0$ के लिये होगा (B) अद्वितीयक हल केवल $\alpha \neq 8$ के लिये होगा

(C) $\alpha = 8$ के लिये अनन्त हल होगा (D) $\alpha = 8$ के लिये कोई हल नही है

Ans. BD

Sol. $D = \alpha - 8 \Rightarrow$ (B) ; If $\alpha = 8$, $D_x = 2\alpha - 14 \quad \Rightarrow$ no solution

30. If A_{11}, A_{12}, A_{13} are the cofactors of the elements of the first row of the determinant $A = (a_{ij})$ and Δ is the value of the determinant then :

यदि A_{11} , A_{12} , A_{13} सारणिक $A = (a_{ij})$ के प्रथम पंक्ति के अवयवों के सहखण्ड है, एवं Δ सारणिक का मान हैं, तो :

(A) $a_{21} A_{11} + a_{22} A_{12} + a_{23} A_{13} = 0$ (B) $a_{11} A_{11} + a_{12} A_{12} + a_{13} A_{13} = \Delta$

(C) $a_{11} A_{11} - a_{12} A_{12} + a_{13} A_{13} = \Delta$ (D) $a_{31}A_{11} + a_{32}A_{12} + a_{33}A_{13} = 0$

Ans. ABD

Sol. By property

31. The determinant $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$ if :

(A) a, b, c are in A.P.
(B) a, b, c are in G.P.
(C) α is a roots of $ax^2 + bx + c = 0$
(D) $(x - \alpha)$ is a factor of $ax^2 + 2bx + c = 0$

सारणिक $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$ है, यदि :

(A) a, b, c स.श्रे. में है
(B) a, b, c गु.श्रे. में है
(C) α समीकरण $ax^2 + bx + c = 0$ का मूल है
(D) $ax^2 + 2bx + c = 0$ का गुणनखण्ड $(x - \alpha)$ है

Ans. BD

Sol. $\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ a\alpha + b & b\alpha + c & 0 \end{vmatrix} = 0$

$R_3 \to R_3 - (\alpha R_1 + R_2)$

$\begin{vmatrix} a & b & a\alpha + b \\ b & c & b\alpha + c \\ 0 & 0 & -a\alpha^2 - 2b\alpha - c \end{vmatrix} = 0$

$\left(-a\alpha^{2=0} - 2b\alpha - c\right)\left(ac - b^2\right)$

$\left(a\alpha^2 + 2b\alpha + c\right)\left(ac - b^2\right) = 0$

$a\alpha^2 + 2b\alpha + c = 0$

α root of equation.

$ac = b^2$ G.P.

32. The parameter, on which the value of $\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$ depend upon, is :

वह प्राचाल जिस पर $\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$ का मान निर्भर करता है :

(A) a (B) p (C) d (D) x

Ans. ACD

Sol. $\sin((p+d)x - px) - a(\sin((p+d)x - (p-d)x) + a^2(\sin(px - (p-d)x))$

$\sin dx - a\sin(2dx) + a^2 \sin dx$

MATCH THE COLUMN TYPE QUESTION(S)

33. Match the following Columns :

Column-I	**Column-II**
(A) Let $\|A\| = \|a_{ij}\|_{3\times3} \neq 0$. Each element a_{ij} is multiplied by k^{i-j}. Let $\|B\|$ the resulting Determinant, where $k_1\|A\| + k_2\|B\| = 0$. Then $k_1 + k_2 =$	(P) 0
(B) The maximum value of a third order determinant each of its entries are ± 1 equals	(Q) 4
(C) If $\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$ then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma =$	(R) 1
(D) $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = Ax + B$ where A and B are determinants of order 3. Then $A + 2B =$	(S) 2

निम्न Columns को सुमेलित कीजिए :

Column-I	**Column-II**
(A) माना $\|A\| = \|a_{ij}\|_{3\times3} \neq 0$ प्रत्येक अवयवों a_{ij}, k^{i-j} से गुणा किया जाये तो यदि $\|B\|$ परिणामी सारणिक है, तो यदि $k_1\|A\| + k_2\|B\| = 0$ तो $k_1 + k_2 =$	(P) 0
(B) एक क्रम तीन के सारणिक का प्रत्येक अवयव ±1 हो सकता है, तो सारणिक का अधिकतम मान होगा	(Q) 4
(C) यदि $\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$ तो $\cos^2\alpha + \cos^2\beta + \cos^2\gamma =$	(R) 1
(D) $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = Ax + B$ जहाँ A तथा B क्रम 3 के	(S) 2

सारणिक है, तो A + 2B =

Ans. (A) P (B) Q (C) R (D) P

Sol. (i) $B=\begin{vmatrix} 1- & k^{-1}- & \frac{1}{k^2}- \\ k- & 1- & \frac{1}{k}- \\ k^2- & k- & 1- \end{vmatrix}$

$$B=\frac{1}{k^2\cdot k}\begin{vmatrix} k^2- & k- & 1- \\ k^2- & k- & 1- \\ k^2- & k- & 1- \end{vmatrix}$$

$$B=\begin{vmatrix} - & - & - \\ - & - & - \\ - & - & - \end{vmatrix}$$

$|B| = |A|$

$|A| - |B| = 0$

$k_1 = 1 \qquad k_2 = -1$

$k_1 + k_2 = 0$

(iii) $\begin{vmatrix} 1 & \cos\alpha & \cos\beta \\ \cos\alpha & 1 & \cos\gamma \\ \cos\beta & \cos\gamma & 1 \end{vmatrix} = \begin{vmatrix} 0 & \cos\alpha & \cos\beta \\ \cos\alpha & 0 & \cos\gamma \\ \cos\beta & \cos\gamma & 0 \end{vmatrix}$

$$\left(1-\cos^2\gamma\right)-\cos\alpha\left(\cos\alpha-\cos\beta\cos\gamma\right)+\cos\beta\left|\cos\alpha\cos\gamma-\cos\beta\right|$$

$$=0-\cos\alpha\left(0-\cos\beta\cos\gamma\right)+\cos\beta\left(\cos\alpha\cos\gamma\right)$$

$$1-\cos^2\gamma-\cos^2\alpha+\cos\alpha\cos\beta\cos\gamma+\cos\alpha\cos\beta\cos\gamma-\cos^2\beta=2\cos\alpha\cos\beta\cos\gamma$$

$$\cos^2\alpha+\cos^2\beta+\cos^2\gamma=1$$

(d) $\begin{vmatrix} x^2+x & x+1 & x-2 \\ 2x^2+3x-1 & 3x & 3x-3 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = Ax + B$

$R_2 \to R_2 - (R_1 + R_3)$

$\begin{vmatrix} x^2+x & x+1 & x-2 \\ -4 & 0 & 0 \\ x^2+2x+3 & 2x-1 & 2x-1 \end{vmatrix} = Ax + B$

$4((x + 1)(2x - 1) - (2x - 1)(x - 2)) = Ax + B$

$4(2x^2 - x + 2x - 1 - 2x^2 + 4x + x - 2) = Ax + B$

$4(6x - 3) = Ax + B$

$24x - 12 = Ax + B$

$A = 24, B = -12$

$A + 2B$

$24 - 2 \times 12 \Rightarrow 0$

Exercise J-M

1. If the system of equations
$x + 2y + 3z = 3$
$4x + 3y - 4z = 4$
$8x + 4y - \lambda z = 9 + \mu$
has infinitely many solutions, then the ordered pair (λ, μ) is equal to : **[JEE-Main,Jan.2023]**

यदि समीकरणों
$x + 2y + 3z = 3$
$4x + 3y - 4z = 4$
$8x + 4y - \lambda z = 9 + \mu$
के अनंत हल है, तब क्रमित युग्म (λ, μ) बराबर है : **[JEE-Main,Jan.2023]**

(A) $\left(\frac{72}{5}, \frac{21}{5}\right)$ (B) $\left(-\frac{72}{5}, -\frac{21}{5}\right)$ (C) $\left(\frac{72}{5}, -\frac{21}{5}\right)$ (D) $\left(-\frac{72}{5}, \frac{21}{5}\right)$

Ans. C

Sol. $x + 2y + 3z = 3$ _____(i)
$4x + 3y - 4z = 4$ _____(ii)
$8x + 4y - \lambda z = 9 + \mu$ _____(iii)
(i) $\times 4$ – (ii) $\Rightarrow 5y + 16z = 8$ _____(iv)
(ii) $\times 2$ – (iii) $\Rightarrow 2y + (\lambda - 8)z = -1 - \mu$ _____(v)
(iv) $\times 2$ – (iii) $\times 5 \Rightarrow (32 - 5(\lambda - 8))z = 16 - 5(-1 - \mu)$

For infinite solutions $\Rightarrow 72 - 5\lambda = 0 \Rightarrow \lambda = \frac{72}{5}$

$21 + 5\mu = 0 \Rightarrow \mu = \frac{-21}{5}$

$\Rightarrow (\lambda, \mu) \equiv \left(\frac{72}{5}, \frac{-21}{5}\right)$

2. Let S_1 and S_2 be respectively the sets of all $a \in R - \{0\}$ for which the system of linear equations :
$ax + 2ay - 3az = 1$
$(2a + 1)x + (2a + 3)y + (a + 1)z = 2$
$(3a + 5)x + (a + 5)y + (a + 2)z = 3$
has unique solution and infinitely many solutions. Then : **[JEE-Main,Jan.2023]**

(A) S_1 is an infinite set and $n(S_2) = 2$ (B) $S_1 = R - \{0\}$ and $S_2 = \phi$
(C) $n(S_1) = 2$ and S_2 is an infinite set (D) $S_1 = \phi$ and $S_2 = R - \{0\}$

माना सभी $a \in R - \{0\}$, जिनके लिए रैखिक समीकरण निकाय
$ax + 2ay - 3az = 1$
$(2a + 1)x + (2a + 3)y + (a + 1)z = 2$
$(3a + 5)x + (a + 5)y + (a + 2)z = 3$

का केवल एक हल है तथा अनंत हल है, के समुच्चय क्रमशः S_1 तथा S_2 है। तो **[JEE-Main,Jan.2023]**

(A) S_1 एक अपरिमित समुच्चय है तथा $n(S_2) = 2$ है

(B) $S_1 = R - \{0\}$ तथा $S_2 = \phi$

(C) $n(S_1) = 2$ तथा S_2 एक अपरिमित समुच्चय है

(D) $S_1 = \phi$ तथा $S_2 = R - \{0\}$

Ans. B

Sol. $$D = \begin{bmatrix} a & 2a & -3a \\ 2a+1 & 2a+3 & 2a+1 \\ 3a+5 & a+5 & a+2 \end{bmatrix} = a\left(15a^2 + 31a + 37\right)$$

$\Rightarrow D \neq a$ for all $a \in R - \{0\}$

3. Consider the following system of questions

$$\alpha x + 2y + z = 1$$
$$2\alpha x + 3y + z = 1$$
$$3x + \alpha y + 2z = \beta$$

for some $\alpha, \beta \in R$. Then which of the following is NOT correct. **[JEE-Main,Jan.2023]**

(A) It has no solution for $\alpha = -1$ and for all $\beta \in R$

(B) It has no solution for $\alpha = 3$ and for all $\beta \neq 2$

(C) It has no solution if $\alpha = -1$ and $\beta \neq 2$

(D) It has no solution for all $\alpha \neq -1$ and $\beta = 2$

किसी $\alpha, \beta \in R$ के लिए निम्न समीकरण निकाय का विचार कीजिए :

$$\alpha x + 2y + z = 1$$
$$2\alpha x + 3y + z = 1$$
$$3x + \alpha y + 2z = \beta$$

तब निम्न में से कौन सा सही नहीं है। **[JEE-Main,Jan.2023]**

(A) $\alpha = -1$ एवं सभी $\beta \in R$ के लिए इसका कोई हल नहीं है

(B) $\alpha = 3$ एवं सभी $\beta \neq 2$ के लिए इसका कोई हल नहीं है

(C) इसका कोई हल नहीं है यदि $\alpha = -1$ एवं $\beta \neq 2$ हैं

(D) सभी $\alpha \neq -1$ एवं $\beta = 2$ के लिए इसका एक हल है

Ans. A

Sol. $D=\begin{vmatrix}\alpha & 2 & 1\\ 2\alpha & 3 & 1\\ 3 & \alpha & 2\end{vmatrix}=0\Rightarrow\alpha=-1,3$

$$D_x=\begin{vmatrix}2 & 1 & 1\\ 3 & 1 & 1\\ \alpha & 2 & \beta\end{vmatrix}=0\Rightarrow\beta=2$$

$$D_y=\begin{vmatrix}\alpha & 1 & 1\\ 2\alpha & 1 & 1\\ 3 & 2 & \beta\end{vmatrix}=0$$

$$D_z=\begin{vmatrix}\alpha & 2 & 1\\ 2\alpha & 3 & 1\\ 3 & \alpha & \beta\end{vmatrix}=0$$

$\beta = 2,\ \alpha = -1$
$\alpha = -1,\ \beta = 2$Infinite solution

4. Let the system of linear equations

$x + y + kz = 2$

$2x + 3y - z = 1$

$3x + 4y + 2z = k$

have infinitely many solutions. Then the system

$(k + 1)x + (2k - 1)y = 7$

$(2k + 1)x + (k + 5)y = 10$

has : **[JEE-Main,Jan.2023]**

(A) no solution

(B) unique slution satisfying $x - y = 1$

(C) infinitely many solutions

(D) unique solution satisfying $x + y = 1$

माना रैखिक समीकरण निकाय

$x + y + kz = 2$

$2x + 3y - z = 1$

$3x + 4y + 2z = k$

के अनंत हल है। तो निकाय

$(k + 1)x + (2k - 1)y = 7$

$(2k + 1)\,x + (k + 5)\,y = 10$ **[JEE-Main,Jan.2023]**

(A) का कोई हल नहीं हैं

(B) केवल एक हल है जो $x - y = 1$ को संतुष्ट करता है

(C) के अनंत हल है

(D) केवल एक हल है जो $x + y = 1$ को संतुष्ट करता है

Ans. D

Sol. $\begin{vmatrix} 1 & 1 & k \\ 2 & 3 & -1 \\ 3 & 4 & 2 \end{vmatrix} = 0$

$\Rightarrow 1(10) - 1(7) + k(-1) - 0$

$\Rightarrow k = 3$

For $k = 3$, 2nd system is

$$4x + 5y = 7 \quad ...(1)$$

and $$7x + 8y = 10 \quad ...(2)$$

Clearly, they have a unique solution

(B) – (1) $\Rightarrow 3x + 3y = 3$

$\Rightarrow x + y = 1$

5. For $\alpha, \beta \in R$, suppose system of linear equations

$x - y + z = 5$

$2a + 2y + \alpha z = 8$

$3x - y + 4z = \beta$

has infinitely many solutions. Then α and β are the roots of: **[JEE-Main,Jan.2023]**

$\alpha, \beta \in R$ के लिए, माना समीकरण निकाय

$x - y + z = 5$

$2a + 2y + \alpha z = 8$

$3x - y + 4z = \beta$

के अनंत हल है। तब α व β निम्न में से किसके मूल है : **[JEE-Main,Jan.2023]**

(A) $x^2 - 10x + 16 = 0$ (B) $x^2 + 18x + 56 = 0$

(C) $x^2 + 14x + 24 = 0$ (D) $x^2 - 18x + 56 = 0$

Ans. D

Sol. $\begin{vmatrix} 1 & -1 & 1 \\ 2 & 2 & \alpha \\ 3 & -1 & 4 \end{vmatrix} = 0$; $8 + \alpha - 2(-4 + 1) + 3(-\alpha - 2) = 0$

$8 + \alpha + 6 - 3\alpha - 6 = 0$

$\alpha = 4$

6. For the system of linear equations

$x + y + z = 6$

$\alpha x + \beta y + 7z = 3$

$x + 2y + 3z = 14,$

which of the following is **NOT** true? **[JEE-Main,Jan.2023]**

(A) If $\alpha = \beta = 7$, then the system has no solution.

(B) If $\alpha = \beta$ and $\alpha \neq 7$, then the system has a unique solution.

(C) There is a unique point (α, β) on the line $x + 2y + 18 = 0$ for which the system has infinitely many solutions.

(D) For every point $(\alpha, \beta) \neq (7, 7)$ on the line $x - 2y + 7 = 0$, the system has infinitely many solutions.

रैखिक समीकरण निकाय

$x + y + z = 6$

$\alpha x + \beta y + 7z = 3$

$x + 2y + 3z = 14,$

के लिए निम्न में से कौन सा सत्य नहीं है ? **[JEE-Main,Jan.2023]**

(A) यदि $\alpha = \beta = 7$ है, तो निकाय का कोई हल नहीं है

(B) यदि $\alpha = \beta$ तथा $\alpha \neq 7$ है, तो निकाय का केवल एक हल है

(C) रेखा $x + 2y + 18 = 0$ पर केवल एक बिंदु (α, β) है जिसके लिए निकाय के अनंत हल है

(D) रेखा $x - 2y + 7 = 0$ पर प्रत्येक बिंदु $(\alpha, \beta) \neq (7, 7)$ के लिए निकाय के अनंत हल है

Ans. D

Sol. By equation 1 and 3 $y + 2z = 8$

$y = 8 - 2z$

And $x = -2 + z$

Now putting in equation 2

$\alpha(z - 2) + \beta(-2z + 8) + 7z = 3$

$\Rightarrow (\alpha - 2\beta + 7)z = 2\alpha - 8\beta + 3$

So equations have unique solution if

$\alpha - 2\beta + 7 \neq 0$

And equations have no solution if

$\alpha - 2\beta + 7 = 0$ and $2\alpha - 8\beta + 3 \neq 0$

And equations have infinite solution if

$\alpha - 2\beta + 7 = 0$ and $2\alpha - 8\beta + 3 = 0$

7. Let S denote the set of all real values of λ such that the system of equations

$\lambda x + y + z = 1$
$x + \lambda y + z = 1$
$x + y + \lambda z = 1$

is inconsistent, then $\sum_{\lambda \in S}\left(|\lambda|^2 + |\lambda|\right)$ is equal to : **[JEE-Main,Jan.2023]**

माना λ के सभी वास्तविक मानों, जिसके लिए समीकरण निकाय

$\lambda x + y + z = 1$
$x + \lambda y + z = 1$
$x + y + \lambda z = 1$

असंगत है, का समुच्चय S है, तब $\sum_{\lambda \in S}\left(|\lambda|^2 + |\lambda|\right)$ का मान है : **[JEE-Main,Jan.2023]**

(A) 4 (B) 6 (C) 2 (D) 12

Ans. B

Sol. $D = \begin{vmatrix} \lambda & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = (\lambda - 1)^2(\lambda + 2)$ If $D = 0 \Rightarrow \lambda = 1, -2$

$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = (\lambda - 1)^2$ When $\lambda = -2$, $D \neq 0$

when $\lambda = 1$, all equations are identical so number of solutions are infinite.
So for inconsistent system, λ can be equal to -2 only.

$\therefore \sum_{\lambda \to 5}\left(|\lambda|^2 + |\lambda|\right) = |-2|^2 + |-2| = 4 + 2 = 6$

8. Let $f(x) = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & \sin 2x \\ \sin^2 x & 1+\cos^2 x & \sin 2x \\ \sin^2 x & \cos^2 x & 1+\sin 2x \end{vmatrix}$, $x \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right]$. If α and β respectively are the maximum and the minimum values of f, then : **[JEE-Main,Jan.2023]**

माना $f(x)=\begin{vmatrix}1+\sin^2 x & \cos^2 x & \sin 2x\\ \sin^2 x & 1+\cos^2 x & \sin 2x\\ \sin^2 x & \cos^2 x & 1+\sin 2x\end{vmatrix}$, $x\in\left[\frac{\pi}{6},\frac{\pi}{3}\right]$ हैं यदि f के अधिकतम व न्यूनतम मान क्रमशः α व β हैं, तब :

(A) $\beta^2+2\sqrt{\alpha}=\frac{19}{4}$ (B) $\alpha^2-\beta^2=4\sqrt{3}$ (C) $\beta^2-2\sqrt{\alpha}=\frac{19}{4}$ (D) $\alpha^2+\beta^2=\frac{19}{2}$ **[JEE-Main,Jan.2023]**

Ans. C

Sol. $f(x)=\begin{vmatrix}1+\sin^2 x & \cos^2 x & \sin 2x\\ \sin^2 x & 1+\cos^2 x & \sin(2x)\\ \sin^2 x & \cos^2 x & 1+\sin(2x)\end{vmatrix}, x\in\left[\frac{\pi}{6},\frac{\pi}{3}\right]$

$R_1\to R_1-R_2$
$R_2\to R_2-R_3$

$$f(x)=\begin{vmatrix}1 & -1 & 0\\ 0 & 1 & -1\\ \sin^2 x & \cos^2 x & 1+\sin(2x)\end{vmatrix}$$

$= 1\ (1+\sin(2x))+\cos^2x+\sin^2x$

$f(x)=2+\sin 2x$ $\quad\left[x\in\left(\frac{\pi}{6},\frac{\pi}{3}\right)\right.$
$\left.2x\in\left(\frac{\pi}{3},\frac{2\pi}{3}\right)\right]$

$\max(f(x))=3=\alpha$

$\text{Min}\,(f(x))=2+\frac{\sqrt{3}}{2}=\beta$

$\therefore\beta^2=\left(2+\frac{\sqrt{3}}{2}\right)^2=4+\frac{3}{4}+2\sqrt{3}=\frac{19}{4}+2\sqrt{3}$ $\therefore\beta^2-2\sqrt{\alpha}=\frac{19}{4}+2\sqrt{3}-2\sqrt{3}=\frac{19}{4}$

9. For the system of linear equations $\alpha x+y+z=1$, $x+\alpha y+z=1$, $x+y+\alpha z=\beta$, which one of the following statements is NOT true? **[JEE-Main,Jan.2023]**

(A) It has infinitely many solutions if $\alpha=1$ and $\beta=1$

(2) $x+y+z=\frac{3}{4}$ if $\alpha=2$ and $\beta=1$

(3) It has no solution if $\alpha=-2$ and $\beta=1$

(4) It has infinitely many solutions if $\alpha=2$ and $\beta=-1$

रैखिक समीकरण निकाय $\alpha x + y + z = 1,\ x + \alpha y + z = 1,\ x + y + \alpha z = \beta$ के लिए निम्न में से कौन सा कथन सही नहीं है ?

[JEE-Main,Jan.2023]

(A) इसके अनंत हल है यदि $\alpha = 1$ तथा $\beta = 1$ है

(2) $x + y + z = \frac{3}{4}$ यदि $\alpha = 2$ तथा $\beta = 1$ हैं

(3) इसका कोई हल नहीं है यदि $\alpha = -2$ तथा $\beta = 1$ हैं

(4) इसके अनंत हल है यदि $\alpha = 2$ तथा $\beta = -1$ हैं

Ans. D

Sol. $\begin{vmatrix} \alpha & 1 & 1 \\ 1 & \alpha & 1 \\ 1 & 1 & \alpha \end{vmatrix} = \alpha\left(\alpha^2 - 1\right) - (\alpha - 1) + (1 - \alpha) = \alpha^3 - 3\alpha + 2 = (\alpha - 1)\left(\alpha^2 + \alpha - 2\right)$

$(\alpha - 1)(\alpha - 1)(\alpha + 2)$ θ.f $\alpha = 2, \beta = -1$ then

for $\alpha = 1$ system of equations is θ$x + y + z = 1$

$x + y + z = 1$ $x + 2y + z = 1$

$x + y + z = 1$ $x + y + 2z = -1$ No solution

$x + y + z = \beta \Rightarrow \beta = 1$ then infinitely many solutions

10. If the system of equations

$x + y + az = b$

$2x + 5y + 2z = 6$

$x + 2y + 3z = 3$

has infinitely many solutions, then $2a + 3b$ is equal to : **[JEE-Main,April,2023]**

यदि समीकरण निकाय

$x + y + az = b$

$2x + 5y + 2z = 6$

$x + 2y + 3z = 3$

के अनंत हल है। तब $2a + 3b$ बराबर है। **[JEE-Main,April,2023]**

(A) 20 (B) 25 (C) 28 (D) 23

Ans. D

Sol. $x + y + az = b$

$2x + 5y + 2z = 6$

$x + 2y + 3z = 3$

For ∞ solution

$\Delta = 0, \Delta_x = 0, \Delta_y = 0, \Delta_z = 0$

$\Delta = \begin{vmatrix} 1 & 1 & a \\ 2 & 5 & 2 \\ 1 & 2 & 3 \end{vmatrix} = 0 \Rightarrow 11 - 4 - a = 0 \Rightarrow a = 7$

$$\Delta_z = \begin{vmatrix} 1 & 1 & b \\ 2 & 5 & 6 \\ 1 & 2 & 3 \end{vmatrix} = 0 \Rightarrow 3 - 0 - b = 0 \Rightarrow b = 3$$

Hence $2a + 3b = 23$

11. For the system of equations

$x + y + z = 6$

$x + 2y + \alpha z = 10$

$x + 3y + 5z = \beta$, which one of the following is NOT true? **[JEE-Main,April,2023]**

(A) System has a unique solution for $\alpha = -3, \beta = 14$

(B) System has a unique solution for $\alpha = -3, \beta \neq 14$

(C) System has no solution for $\alpha = 3, \beta = 24$

(D) System has infinitely many solutions for $\alpha = 3, \beta = 14$

समीकरण निकाय

$x + y + z = 6$

$x + 2y + \alpha z = 10$

$x + 3y + 5z = \beta$, के लिए निम्न में से कौन सा सत्य नहीं है ? **[JEE-Main,April,2023]**

(A) $\alpha = -3, \beta = 14$ के लिए समीकरण निकाय का अद्वितीय हल नहीं

(B) $\alpha = -3, \beta \neq 14$ के लिए समीकरण निकाय का अद्वितीय हल है

(C) $\alpha = 3, \beta = 24$ के लिए समीकरण निकाय का कोई हल नहीं है

(D) $\alpha = 3, \beta = 14$ के लिए समीकरण निकाय के अनंत हल है

Ans. D

Sol. $$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & \alpha \\ 1 & 3 & 5 \end{vmatrix}$$

$= 1(10 - 3\alpha) - (5 - \alpha) + (3 - 2)$

$= 6 - 2\alpha$

$D \neq 0 \Rightarrow \alpha \neq 3$

Unique solution $\Rightarrow \alpha \neq 3$

12. Let S be the set of all values of $\theta \in [-\pi, \pi]$ for which the system of linear equations

$x + y + \sqrt{3}z = 0$

$-x + (\tan\theta)y + \sqrt{7}z = 0$

$x + y + (\tan\theta)z = 0$

has non-trivial solution. Then $\frac{120}{\pi}\sum_{\theta \in S}\theta$ is equal to : **[JEE-Main,April,2023]**

माना $\theta \in [-\pi, \pi]$ के सभी मानों, जिनके लिए रैखिक समीकरण निकाय

$x + y + \sqrt{3}z = 0$

$-x + (\tan\theta)y + \sqrt{7}z = 0$

$x + y + (\tan\theta)z = 0$

का अतुच्छ हल है, का समुच्च S है। तो $\frac{120}{\pi}\sum_{\theta\in S}\theta$ बराबर है : **[JEE-Main,April,2023]**

(A) 30 (B) 20 (C) 10 (D) 40

Ans. B

Sol. For non trivial solutions

$D = 0$

$$\begin{vmatrix} 1 & 1 & \sqrt{3} \\ -1 & \tan\theta & \sqrt{7} \\ 1 & 1 & \tan\theta \end{vmatrix} = 0$$

$\tan^2\theta - \left(\sqrt{3}-1\right)\tan\theta - \sqrt{3} = 0$

$\tan\theta = \sqrt{3}, -1$

$\theta = \left\{\frac{\pi}{3}, \frac{-2\pi}{3}, \frac{-\pi}{4}, \frac{3\pi}{4}\right\}$

$\frac{120}{\pi}\left(\sum\theta\right) = \frac{120}{\pi}\times\frac{\pi}{6} = 20$

13. For the system of linear equations

$2x - y + 3z = 5$
$3x + 2y - z = 7$
$4x + 5y + \alpha z = \beta,$

which of the following is NOT correct? **[JEE-Main,April,2023]**

(A) The system has infinitely many solutions for $\alpha = -5$ and $\beta = 9$

(B) The system has infinitely many solutions for $\alpha = -6$ and $\beta = 9$

(C) The system has a unique solution for $\alpha \neq -5$ and $\beta = 8$

(D) The system is inconsistent for $\alpha = -5$ and $\beta = 8$

रैखिक समीकरण निकाय

$2x - y + 3z = 5$
$3x + 2y - z = 7$
$4x + 5y + \alpha z = \beta,$

के लिए निम्न में से कौन सा सही नहीं है ? **[JEE-Main,April,2023]**

(A) $\alpha = -5$ तथा $\beta = 9$ के लिए समीकरण निकाय के अनंत हल हैं

(B) $\alpha = -6$ तथा $\beta = 9$ के लिए समीकरण निकाय के अनंत हल हैं

(C) $\alpha \neq -5$ तथा $\beta = 8$ के लिए समीकरण निकाय का एक अद्वितीय हल है

(D) $\alpha = -5$ तथा $\beta = 8$ के लिए समीकरण निकाय असंगत है

Ans. B

Sol. For infinite solution

$\lambda_1 P_1 + \lambda_2 P_2 = P_3$

$\lambda_1(2x - y + 3z - 5) + (3x + 2y - z - 7)\lambda_2 = (4x + 5y + \alpha z - \beta)$

$2\lambda_1 + 3\lambda_2 = 4] \quad \lambda_1 = -1$

$-\lambda_1 + 2\lambda_2 = 5] \quad \lambda_2 = 2$

$3\lambda_1 - \lambda_2 = \alpha \rightarrow \alpha = -5$

$-5\lambda_1 - 7\lambda_2 = -\beta \rightarrow \beta = 9$

14. If $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix} = \frac{9}{8}(103x+81)$, then $\lambda, \frac{\lambda}{3}$ are the roots of the equation :

[JEE-Main,April,2023]

यदि $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix} = \frac{9}{8}(103x+81)$ है, तो $\lambda, \frac{\lambda}{3}$ किस समीकरण के मूल है : **[JEE-Main,April,2023]**

(A) $4x^2 - 24x + 27 = 0$ (B) $4x^2 - 24x - 27 = 0$

(C) $4x^2 + 24x + 27 = 0$ (D) $4x^2 + 24x - 27 = 0$

Ans. A

Sol. $\begin{vmatrix} x+1 & x & x \\ x & x+\lambda & x \\ x & x & x+\lambda^2 \end{vmatrix} = \frac{9}{8}(103x+81)$

Put $x = 0$

$\begin{vmatrix} 1 & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda^2 \end{vmatrix} = \frac{9}{8} \times 81$

$\lambda^3 = \frac{3^6}{2^3}$

$\lambda = \frac{9}{2}$

$$\frac{\lambda}{3}=\frac{3}{2}$$

Equation : $x^2-\left(\frac{9}{2}+\frac{3}{2}\right)x+\frac{9}{2}\times\frac{3}{2}=0$

$4x^2-24x+27=0$

15. If the system of linear equations

$7x+11y+\alpha z=13$
$5x+4y+7z=\beta$
$175x+194y+57z=361$

has infinitely many solutions, then $\alpha+\beta+2$ is equal to : **[JEE-Main,April,2023]**

यदि रैखिक समीकरण निकाय

$7x+11y+\alpha z=13$
$5x+4y+7z=\beta$
$175x+194y+57z=361$

के अनंत हल हैं तो $\alpha+\beta+2$ बराबर है : **[JEE-Main,April,2023]**

(A) 5 (B) 6 (C) 3 (D) 4

Ans. D

Sol. $7x+11y+\alpha z=13$ ....(i)

$5x+4y+7z=\beta$ ....(ii)

$175x+194y+57z=361$ ....(iii)

$(i)\times10+(ii)\times21-(iii)$

$z(10\alpha+147-57)=130+21\beta-361$

$\therefore 10\alpha+90=0$

$\alpha=-9$

$130-361+21\beta=0$

$\beta=11$

$\alpha+\beta+2=4$

16. For the system of linear equations

$2x+4y+2az=b$
$x+2y+3z=4$
$2x-5y+2z=8$

which of the following is NOT correct? **[JEE-Main,April,2023]**

(A) has unique solution if $a=b=8$

(B) It has infinitely many solutions if $a=3, b=8$

(C) It has infinitely many solutions if $a=3, b=6$

(D) It has unique solution if $a=b=6$

रैखिक समीकरण निकाय

$2x + 4y + 2az = b$

$x + 2y + 3z = 4$

$2x - 5y + 2z = 8$

के लिए निम्न में से कौनसा सही नहीं है? **[JEE-Main,April,2023]**

(A) इसका अद्वितीय हल है यदि $a = b = 8$ हैं

(B) इसके अनंत हल हैं यदि $a = 3, b = 8$ हैं

(C) इसके अनंत हल हैं यदि $a = 3, b = 6$ हैं

(D) इसका अद्वितीय हल है यदि $a = b = 6$ हैं

Ans. C

Sol. $D = \begin{vmatrix} 2 & 4 & 2a \\ 1 & 2 & 3 \\ 2 & -5 & 2 \end{vmatrix}$

$= 2(4 + 15) - 4(2 - 6) + 2a(-5 - 4)$

$= 38 + 16 - 18a = 54 - 18a = 18(3 - a)$

$D_1 = \begin{vmatrix} b & 4 & 2a \\ 4 & 2 & 3 \\ 8 & -5 & 2 \end{vmatrix}$

$= b(4 + 15) - 4(8 - 24) + 2a(-20 - 16)$

$= 19b + 64 - 72a$

$= 19b - 72a + 64$

$D_2 = \begin{vmatrix} 2 & b & 2a \\ 1 & 4 & 3 \\ 2 & 8 & 2 \end{vmatrix}$

$= 2(8 - 24) - b(2 - 6) + 2a(8 - 8)$

$= -32 + 4b = 4(b - 8)$

$D_3 = \begin{vmatrix} 2 & 4 & b \\ 1 & 2 & 4 \\ 2 & -5 & 8 \end{vmatrix}$

$= 2(16 + 20) - 4(8 - 8) + b(-5 - 4)$

$= 72 - 9b = 9(8 - b)$

Now when $a \neq 3, b \in R$ system have unique solution.

when $a = 3, b \neq 8$, system have no solution

when $a = 3, b = 8$, system have infinitely many solutions.

17. If the system of equations

$2x + y - z = 5$

$2x - 5y + \lambda z = \mu$

$x + 2y - 5z = 7$

has infinitely many solutions, then $(\lambda+\mu)^2 + (\lambda-\mu)^2$ is equal to : **[JEE-Main,April,2023]**

यदि समीकरण निकाय

$2x + y - z = 5$
$2x - 5y + \lambda z = \mu$
$x + 2y - 5z = 7$

के अनंत हल हैं, तो $(\lambda+\mu)^2 + (\lambda-\mu)^2$ बराबर है : **[JEE-Main,April,2023]**

(A) 916 (B) 904 (C) 912 (D) 920

Ans. A

Sol. $2x + y - z = 5$

$2x - 5y + \lambda z = \mu$

$x + 2y - 5z = 7$

for infinite solution $D = D_1 = D_2 = D_3 = 0$

$$D = \begin{vmatrix} 2 & 1 & -1 \\ 2 & -5 & \lambda \\ 1 & 2 & -5 \end{vmatrix} = 0$$

$2(25 - 2\lambda) - (-10 - \lambda) - 1(4 + 5) = 0$

$50 - 4\lambda + 10 + \lambda - 9 = 0 \quad \Rightarrow \lambda = 17$

$$D_1 = \begin{vmatrix} 5 & 1 & -1 \\ \mu & -5 & 17 \\ 7 & 2 & -5 \end{vmatrix} = 0$$

$5(25 - 34) - \mu(-5 + 2) + 7(17 - 5) = 0$

$-45 + 3\mu + 84 = 0 \Rightarrow \mu = -13$

Now $(\lambda+\mu)^2 + (\lambda-\mu)^2 = (17-13)^2 + (17+13)^2$

$= 16 + 900 = 916$

18. Let the system of linear equations

$-x + 2y - 9z = 7$
$-x + 3y + 7z = 9$
$-2x + y + 5z = 8$
$-3x + y + 13z = \lambda$

has a unique solution $x = \alpha$, $y = \beta$, $z = \gamma$. Then the distance of the point(α, β, γ) from the plane $2x - 2y + z = \lambda$ is : **[JEE-Main,April,2023]**

यदि रेखिक समीकरण निकाय

$-x + 2y - 9z = 7$
$-x + 3y + 7z = 9$
$-2x + y + 5z = 8$
$-3x + y + 13z = \lambda$

के अद्वितीय हल $x = \alpha, y = \beta, z = \gamma$ हो, तो बिन्दु(α, β, γ) की समतल $2x - 2y + z = \lambda$ से दूरी होगी :

[JEE-Main,April,2023]

(A) 13 (B) 9 (C) 11 (D) 7

Ans. D

Sol. $-x + 2y - 9z = 7$ ...(1)

$-x + 3y + 7z = 9$ ...(2)

$-2x + y + 5z = 8$ ...(3)

(B) – (1)

$y + 16z = 2$ ...(4)

(C) $- 2 \times$ (1)

$-3y + 23z = -6$ ...(5)

$3 \times (4) + (5)$

$71z = 0 \Rightarrow z = 0$

$y = 2$

$x = -3$

$(-3, 2, 0) \rightarrow (\alpha, \beta, \gamma)$

Put in $-3x + y + 13z = \lambda$

$\lambda = 9 + 2 = 11$

$$d = \left|\frac{-6-4-11}{3}\right| = 7$$

19. Let S be the set of values of λ, for which the system of equations

$6\lambda x - 3y + 3z = 4\lambda^2$,

$2x + 6\lambda y + 4z = 1$,

$3x + 2y + 3\lambda z = \lambda$

has no solution. Then $12\sum_{\lambda \in S}|\lambda|$ is equal to _____. **[JEE-Main,April,2023]**

माना λ के मानों, जिनके लिए समीकरण निकाय

$6\lambda x - 3y + 3z = 4\lambda^2$,

$2x + 6\lambda y + 4z = 1$,

$3x + 2y + 3\lambda z = \lambda$

का कोई हल नहीं है, का समुच्चय S है। तो $12\sum_{\lambda \in S}|\lambda|$ बराबर है _____ **[JEE-Main,April,2023]**

Ans. 24

Sol. For no solution

$$\begin{vmatrix} 6\lambda & -3 & 3 \\ 2 & 6\lambda & 4 \\ 3 & 2 & 3\lambda \end{vmatrix} = 0$$

$\Rightarrow 9\lambda^3 - 7\lambda - 2 = 0$

$\Rightarrow (\lambda - 1)(3\lambda + 1)(3\lambda + 2) = 0$

$\Rightarrow \quad 12\sum_{\lambda \in S} |\lambda| = 12 \times \left(1 + \frac{1}{3} + \frac{2}{3}\right) = 24$

20. Let $D_k = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$. If $\sum_{k=1}^{n} D_k = 96$, then n is equal to _____.

[JEE-Main,April,2023]

माना $D_k = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$ है। यदि $\sum_{k=1}^{n} D_k = 96$ तो n बराबर है _____

[JEE-Main,April,2023]

Ans. 6

Sol. $D_k = \begin{vmatrix} 1 & 2k & 2k-1 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix}$

$\sum_{k=1}^{n} D_k = 96 \Rightarrow$

$$\begin{vmatrix} \sum_{k=1}^{n} 1 & \sum 2k & \sum (2k-1) \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix} = 96$$

$$\Rightarrow \begin{vmatrix} n & n^2+n & n^2 \\ n & n^2+n+2 & n^2 \\ n & n^2+n & n^2+n+2 \end{vmatrix} = 96$$

$R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$

$$\begin{vmatrix} n & n^2+n & n^2 \\ 0 & 2 & 0 \\ 0 & 0 & n+2 \end{vmatrix} = 96$$

$\Rightarrow n(2n+4) = 96 \Rightarrow n(n+2) = 48 \Rightarrow n = 6$

21. The values of α, for which $\begin{vmatrix} 1 & \frac{3}{2} & \alpha+\frac{3}{2} \\ 1 & \frac{1}{3} & \alpha+\frac{1}{3} \\ 2\alpha+3 & 3\alpha+1 & 0 \end{vmatrix} = 0$ lie in the interval : **[JEE-Main,Jan.2024]**

α के वह मान, जिनके लिए $\begin{vmatrix} 1 & \frac{3}{2} & \alpha+\frac{3}{2} \\ 1 & \frac{1}{3} & \alpha+\frac{1}{3} \\ 2\alpha+3 & 3\alpha+1 & 0 \end{vmatrix} = 0$ है, किस अंतराल में है ? **[JEE-Main,Jan.2024]**

(A) (0, 3) (B) $\left(-\frac{3}{2},\frac{3}{2}\right)$ (C) (–2, 1) (D) (–3, 0)

Ans. D

Sol. $\begin{vmatrix} 1 & \frac{3}{2} & \alpha+\frac{3}{2} \\ 1 & \frac{1}{3} & \alpha+\frac{1}{3} \\ 2\alpha+3 & 3\alpha+1 & 0 \end{vmatrix} = 0$

$$\Rightarrow (2\alpha+3)\left\{\frac{7\alpha}{6}\right\}-(3\alpha+1)\left\{\frac{-7}{6}\right\}=0$$

$$\Rightarrow (2\alpha+3)\cdot\frac{7\alpha}{6}+(3\alpha+1)\cdot\frac{7}{6}=0$$

$$\Rightarrow 2\alpha^2+3\alpha+3\alpha+1=0$$

$$\Rightarrow 2\alpha^2+6\alpha+1=0$$

$$\Rightarrow \alpha=\frac{-3+\sqrt{7}}{2},\frac{-3-\sqrt{7}}{2}$$

22. Consider the system of linear equations $x+y+z=4\mu, x+2y+2\lambda z=10\mu, x+3y+4\lambda^2 z=\mu^2+15$, where $\lambda, \mu \in R$. Which one of the following statements is NOT correct ? **[JEE-Main,Jan.2024]**

(A) The system is consistent if $\lambda \neq \frac{1}{2}$

(B) The system is inconsistent if $\lambda = \frac{1}{2}$ and $\mu \neq 1$

(C) The system has infinite number of solutions if $\lambda = \frac{1}{2}$ and $\mu=15$

(D) The system has unique solution if $\lambda \neq \frac{1}{2}$ and $\mu \neq 1, 15$

रैखिक समीकरण निकाय $x + y + z = 4\mu, x + 2y + 2\lambda z = 10\mu, x + 3y + 4\lambda^2 z = \mu^2 + 15$, जहाँ $\lambda, \mu \in R$, है, का विचार कीजिए। निम्न कथनों में से कौनसा सही नहीं है ? **[JEE-Main,Jan.2024]**

(A) यदि $\lambda \neq \frac{1}{2}$ है, तो निकाय संगत है।

(B) यदि $\lambda = \frac{1}{2}$ तथा $\mu \neq 1$ है, तो निकाय असंगत है।

(C) यदि $\lambda = \frac{1}{2}$ तथा $\mu=15$ हैं, तो निकाय के अनन्त हल हैं।

(D) यदि $\lambda \neq \frac{1}{2}$ तथा $\mu \neq 1, 15$ हैं, तो निकाय का अद्वितीय हल है।

Ans. B

Sol. $x + y + z = 4\mu, x + 2y + 2\lambda z = 10\mu, x + 3y + 4\lambda^2 z = \mu^2 + 15$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 2\lambda \\ 1 & 3 & 4\lambda^2 \end{vmatrix} = (2\lambda - 1)^2$$

For unique solution $\Delta \neq 0, 2\lambda - 1 \neq 0, \left(\lambda \neq \frac{1}{2}\right)$

Let $\Delta = 0, \lambda = \frac{1}{2}$

$$\Delta_y = 0, \Delta_x = \Delta_z = \begin{vmatrix} 4\mu & 1 & 1 \\ 10\mu & 2 & 1 \\ \mu^2 + 15 & 3 & 1 \end{vmatrix}$$

$= (\mu - 15)(\mu - 1)$

For infinite solution $\lambda = \frac{1}{2}, \mu = 1$ or 15

23. Consider the system of linear equations $x + y + z = 5$, $x + 2y + \lambda^2 z = 9, x + 3y + \lambda z = \mu$, where $\lambda, \mu \in \mathbb{R}$. Then,

which of the following statement is NOT correct? **[JEE-Main,Jan.2024]**

(A) System is consistent if $\lambda \neq 1$ and $\mu = 13$

(B) System has unique solution if $\lambda \neq 1$ and $\mu \neq 13$

(C) System is inconsistent if $\lambda = 1$ and $\mu \neq 13$

(D) System has infinite number of solutions if $\lambda = 1$ and $\mu = 13$

रैखिक समीकरण निकाय $x + y + z = 5,\ x + 2y + \lambda^2 z = 9, x + 3y + \lambda z = \mu$, जहाँ $\lambda, \mu \in \mathbb{R}$ हैं, का विचार कीजिए। तो निम्न में से कौनसा कथन सत्य नहीं है ? **[JEE-Main,Jan.2024]**

(A) निकाय संगत है यदि $\lambda \neq 1$ तथा $\mu = 13$ है

(B) निकाय का अद्वितीय हल है यदि $\lambda \neq 1$ तथा $\mu \neq 13$ हैं

(C) निकाय असंगत है यदि $\lambda = 1$ तथा $\mu \neq 13$ है

(D) निकाय का अनंत हल है यदि $\lambda = 1$ तथा $\mu = 13$ हैं

Ans. B

Sol. $x + y + z = 5$ ...(1)

$x + 2y + \lambda^2 z = 9$ ...(2)

$x + 3y + \lambda z = \mu$ ...(3)

from (1) + (3) – 2(2)

$z\left(\lambda + 1 - 2\lambda^2\right) = \left(\mu - 13\right)$

$\left(\lambda + 1\right)\left(2\lambda + 1\right) Z = \left(13 + \mu\right)$

Infinite solution is $\lambda = 1$ & $\mu = 13$
for unique solution $\lambda \neq 1$
for number of solutions $\lambda = 1$ & $\mu \neq 13$
If $\lambda \neq 1$ & $\mu \neq 13$

In case where $\lambda = \dfrac{-1}{2}$ and $\mu \neq 13$

This will generate number of solutions.

24. If the system of linear equations

$x - 2y + z = -4$

$2x + \alpha y + 3z = 5$

$3x - y + \beta z = 3$

has infinitely many solutions, then $12\alpha + 13\beta$ is equal to : **[JEE-Main,Jan.2024]**

यदि रैखिक समीकरण निकाय

$x - 2y + z = -4$

$2x + \alpha y + 3z = 5$

$3x - y + \beta z = 3$

के अनंत हल हैं, तो $12\alpha + 13\beta$ बराबर है : **[JEE-Main,Jan.2024]**

(A) 54 (B) 64 (C) 60 (D) 58

Ans. D

Sol. $x - 2y + z = -4$

$2x + \alpha y + 3z = 5$

$3x - y + \beta z = 3$

For infinite solution $D = 0, D_1 = 0, D_2 = 0$ & $D_3 = 0$

$$D = 0 \Rightarrow \begin{vmatrix} 1 & -2 & 1 \\ 2 & \alpha & 3 \\ 3 & -1 & \beta \end{vmatrix} = 0$$

$\alpha\beta - 3\alpha + 4\beta = 17$ ...(1)

$$D_1 = \begin{vmatrix} -4 & -2 & 1 \\ 5 & \alpha & 3 \\ 3 & -1 & \beta \end{vmatrix} = 0$$

$$D_2 = \begin{vmatrix} 1 & -4 & 1 \\ 2 & 5 & 3 \\ 3 & 3 & \beta \end{vmatrix} = 0$$

$= (5\beta - 9) + 4(2\beta - 9) + 1(6 - 15) = 0$

$13\beta = 54 \Rightarrow \beta = \dfrac{54}{13}$

Put in (1)

$\dfrac{54}{13}\alpha - 3\alpha + 4 \times \dfrac{54}{13} = 17 \Rightarrow \alpha = \dfrac{1}{3}$

$12\alpha + 13\beta = 12 \times \dfrac{1}{3} + 13 \times \dfrac{54}{13} = 58$

25. If the system of equations

$2x + 3y - z = 5$

$x + \alpha y + 3z = -4$

$3x - y + \beta z = 7$

has infinitely many solutions, then $13\alpha\beta$ is equal to : **[JEE-Main,Jan.2024]**

यदि समीकरण निकाय

$2x + 3y - z = 5$

$x + \alpha y + 3z = -4$

$3x - y + \beta z = 7$

के अनंत हल हैं तो $13\alpha\beta$ बराबर है : **[JEE-Main,Jan.2024]**

(A) 1210 (B) 1220 (C) 1120 (D) 1110

Ans. C

Sol. Given system of equations,

$2x + 3y - z = 5$

$x + \alpha y + 3z = -4$

$3x - y + \beta z = 7$

For infinite solutions,

$\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$

$$\Delta_2 = \begin{vmatrix} 2 & 5 & -1 \\ 1 & -4 & 3 \\ 3 & 7 & \beta \end{vmatrix} = 0 \Rightarrow \beta = \frac{-16}{13}$$

$$\Delta_3 = \begin{vmatrix} 2 & 3 & 5 \\ 1 & \alpha & -4 \\ 3 & -1 & 7 \end{vmatrix} = 0 \Rightarrow \alpha = -70$$

$\Rightarrow 13\alpha\beta = 1120$

26. Let the system of equations $x + 2y + 3z = 5$, $2x + 3y + z = 9$, $4x + 3y + \lambda z = \mu$ have infinite number of solutions. Then $\lambda + 2\mu$ is equal to : **[JEE-Main,Jan.2024]**

माना समीकरण निकाय $x + 2y + 3z = 5$, $2x + 3y + z = 9$, $4x + 3y + \lambda z = \mu$ के अनंत हल हैं। तो $\lambda + 2\mu$ बराबर है :

[JEE-Main,Jan.2024]

(A) 15 (B) 28 (C) 22 (D) 17

Ans. D

Sol. $x + 2y + 3z = 5$

$2x + 3y + z = 9$

$4x + 3y + \lambda z = \mu$

for infinite solution $D = D_1 = D_2 = D_3 = 0$

$$D = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \\ 4 & 3 & \lambda \end{vmatrix} = 0$$

$(3\lambda - 3) - 2(2\lambda - 4) + 3(6 - 12) = 0$

$\lambda = -13$

$$D_3 = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 9 \\ 4 & 3 & \mu \end{vmatrix} = 0$$

$3\mu - 27 - 2(2\mu - 36) + 5(6 - 12) = 0$

$\mu = 15$

$\lambda + 2\mu = -13 + 30 = 17$

27. If the system of equations

$$x + \left(\sqrt{2}\sin\alpha\right)y + \left(\sqrt{2}\cos\alpha\right)z = 0$$

$$x + \left(\cos\alpha\right)y + \left(\sin\alpha\right)z = 0$$

$$x + \left(\sin\alpha\right)y - \left(\cos\alpha\right)z = 0$$

has a non-trivial solution, then $\alpha \in \left(0, \frac{\pi}{2}\right)$ is equal to : **[JEE-Main,April,2024]**

यदि समीकरण निकाय

$$x + \left(\sqrt{2}\sin\alpha\right)y + \left(\sqrt{2}\cos\alpha\right)z = 0$$

$$x + \left(\cos\alpha\right)y + \left(\sin\alpha\right)z = 0$$

$$x + \left(\sin\alpha\right)y - \left(\cos\alpha\right)z = 0$$

का अतुच्छ हल है, तो $\alpha \in \left(0, \frac{\pi}{2}\right)$ बराबर है : **[JEE-Main,April,2024]**

(A) $\frac{3\pi}{4}$ (B) $\frac{7\pi}{24}$ (C) $\frac{11\pi}{24}$ (D) $\frac{5\pi}{24}$

Ans. D

Sol. $$\begin{vmatrix} 1 & \sqrt{2}\sin\alpha & \sqrt{2}\cos\alpha \\ 1 & \sin\alpha & -\cos\alpha \\ 1 & \cos\alpha & \sin\alpha \end{vmatrix} = 0$$

$$\Rightarrow 1-\sqrt{2}\sin\alpha\left(\sin\alpha+\cos\alpha\right)+\sqrt{2}\cos\alpha\left(\cos\alpha-\sin\alpha\right)=0$$

$$\Rightarrow 1+\sqrt{2}\cos 2\alpha-\sqrt{2}\sin 2\alpha=0$$

$$\cos 2\alpha-\sin 2\alpha=-\frac{1}{\sqrt{2}}$$

$$\cos\left(2\alpha+\frac{\pi}{4}\right)=-\frac{1}{2}$$

$$2\alpha+\frac{\pi}{4}=2n\pi\pm\frac{2\pi}{3}$$

$$\alpha+\frac{\pi}{8}=n\pi\pm\frac{\pi}{3}$$

$n = 0,$

$$x=\frac{\pi}{3}-\frac{\pi}{8}=\frac{5\pi}{24}$$

28. If the system of equations

$11x + y + \lambda z = -5$

$2x + 3y + 5z = 3$

$8x - 19y - 39z = \mu$

has infinitely many solutions, then $\lambda^4 - \mu$ is equal to : **[JEE-Main,April,2024]**

यदि समीकरण निकाय

$11x + y + \lambda z = -5$

$2x + 3y + 5z = 3$

$8x - 19y - 39z = \mu$

के अनंत हल हैं, तो $\lambda^4 - \mu$ बराबर है : **[JEE-Main,April,2024]**

(A) 47 (B) 49 (C) 51 (D) 45

Ans. A

Sol. $11x + y + \lambda z = -5$

$2x + 3y + 5z = 3$

$8x - 19y - 39z = \mu$

for infinite sol.

$$D = \begin{vmatrix} 11 & 1 & \lambda \\ 2 & 3 & 5 \\ 8 & -19 & -39 \end{vmatrix} = 0$$

$\Rightarrow 11(-117+95) - 1(-78-40) + \lambda(-38-24)$

$\Rightarrow 11(-22) + 118 - \lambda(62) = 0$

$\Rightarrow 62\lambda = 118 - 242$

$\Rightarrow \lambda = \dfrac{-124}{62} = -2$

$$D_1 = \begin{vmatrix} -5 & 1 & -2 \\ 3 & 3 & 5 \\ \mu & -19 & -39 \end{vmatrix} = 0$$

$\Rightarrow -5(-117+95) - 1(-117-5\mu) - 2(-57-3\mu) = 0$

$\Rightarrow -5(-22) + 117 + 5\mu + 114 + 6\mu = 0$

$\Rightarrow 11\mu = -110 - 231 = -341$

$\Rightarrow \mu = -31$

$\lambda^4 - \mu = (-2)^4 - (-31) = 16 + 31 = 47$

29. The values of m, n, for which the system of equations
$x + y + z = 4$,
$2x + 5y + 5z = 17$,
$x + 2y + mz = n$
has infinitely many solutions, satisfy the equation : **[JEE-Main,April,2024]**

m, n के मान, जिनके लिए समीकरण निकाय
$x + y + z = 4$,
$2x + 5y + 5z = 17$,
$x + 2y + mz = n$
के अनंत हल हैं, इनमें से किस समीकरण को संतुष्ट करते हैं : **[JEE-Main,April,2024]**

(A) $m^2 + n^2 - m - n = 46$ (B) $m^2 + n^2 - mn = 39$

(C) $m^2 + n^2 + m + n = 64$ (D) $m^2 + n^2 - mn = 68$

Ans. B

Sol. Augmented matrix is

$$\begin{bmatrix} 1 & 1 & 1 & : & 4 \\ 2 & 5 & 5 & : & 17 \\ 1 & 2 & m & : & n \end{bmatrix} R_2 \to R_2 - 2R_1, R_3 \to R_3 - R_1$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 1 & : & 4 \\ 0 & 3 & 3 & : & 9 \\ 0 & 1 & m-1 & : & n-4 \end{bmatrix}$$

After dividing R_2 by 3 we obtain

$m - 1 = 1, n - 4 = 3$

$m = 2, n = 7$

$m^2 + n^2 - mn = 4 + 49 - 14 = 49 - 10 = 39$

30. For $\alpha, \beta \in R$ and a natural number n, let $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2}+\alpha \\ 2r & 2 & n^2-\beta \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$. Then $2A_{10} - A_8$ is :

[JEE-Main,April,2024]

$\alpha, \beta \in R$ और प्राकृतिक संख्या n के लिए, माना $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2}+\alpha \\ 2r & 2 & n^2-\beta \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$, तो $2A_{10} - A_8$ है :

[JEE-Main,April,2024]

(A) $4\alpha + 2\beta$ (B) $2n$ (C) 0 (D) $2\alpha + 4\beta$

Ans. A

Sol. $A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2}+\alpha \\ 2r & 2 & n^2-\beta \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$

$R_2 \to R_2 - 2R_1$

$A_r = \begin{vmatrix} r & 1 & \frac{n^2}{2}+\alpha \\ 0 & 0 & -\beta-2\alpha \\ 3r-2 & 3 & \frac{n(3n-1)}{2} \end{vmatrix}$

$Ar = (\beta + 2\alpha)(3r - 3r + 2)$

$Ar = 4\alpha + 2\beta$

Hence $2A_{10} - A_8$

$= A_{10} = 4\alpha + 2\alpha$

31. If $\alpha \neq a, \beta \neq b, \gamma \neq c$ and $\begin{vmatrix} \alpha & b & c \\ a & \beta & c \\ a & b & \gamma \end{vmatrix} = 0$, then $\frac{a}{\alpha - a} + \frac{b}{\beta - b} + \frac{\gamma}{\gamma - c}$ is equal to :

[JEE-Main,April,2024]

यदि $\alpha \neq a, \beta \neq b, \gamma \neq c$ तथा $\begin{vmatrix} \alpha & b & c \\ a & \beta & c \\ a & b & \gamma \end{vmatrix} = 0$ है, तो $\frac{a}{\alpha - a} + \frac{b}{\beta - b} + \frac{\gamma}{\gamma - c}$ बराबर है :

[JEE-Main,April,2024]

(A) 1 (B) 0 (C) 3 (D) 2

Ans. B

Sol. $R_1 \to R_1 - R_2, R_2 \to R_2 - R_3$

$$\begin{vmatrix} \alpha - a & b - \beta & 0 \\ 0 & \beta - b & c - \gamma \\ a & b & \gamma \end{vmatrix} = 0$$

$(\alpha - a)(\gamma(\beta - b) - b(c - \gamma)) - (b - \beta)(-a(c - \gamma)) = 0$

$\gamma(\alpha - a)(\beta - b) - b(\alpha - a)(c - \gamma) + a(b - \beta)(c - \gamma)$

$$\frac{\gamma}{\gamma - c} + \frac{b}{\beta - b} + \frac{a}{\alpha - a} = 0$$

32. If the system of equations $x + 4y - z = \lambda$, $7x + 9y + \mu z = -3$, $5x + y + 2z = -1$ has infinitely many solutions, then $(2\mu + 3\lambda)$ is equal to : **[JEE-Main,April,2024]**

यदि समीकरण निकाय $x + 4y - z = \lambda$, $7x + 9y + \mu z = -3$, $5x + y + 2z = -1$ के अनंत हल हैं, तो

$(2\mu + 3\lambda)$ बराबर है : **[JEE-Main,April,2024]**

(A) 3 (B) 2 (C) –3 (D) –2

Ans. C

Sol. $x + 4y - z = \lambda$

$7x + 9y + \mu z = -3$

$5x + y + 2z = -1$

$$\begin{bmatrix} 1 & 4 & -1 \\ 7 & 9 & \mu \\ 5 & 1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} \lambda \\ -3 \\ -1 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 4 & -1 \\ 7 & 9 & \mu \\ 5 & 1 & 2 \end{bmatrix}, \quad B = \begin{bmatrix} \lambda \\ -3 \\ -1 \end{bmatrix}$$

$AX = B$

$$X = A^{-1}B = \frac{\text{adj}A}{|A|}B$$

If $|A| = 0$ and $(\text{adj}A) \cdot B = 0$, system has infinitely many solutions.

$|A| = 18 - \mu - 4(14 - 5\mu) - 1(7 - 45) = 0$

$\Rightarrow 18 - \mu - 56 + 20\mu + 38 = 0$

$\Rightarrow 19\mu = 0$

$\Rightarrow \mu = 0$

$$\text{Also adj}A = \begin{bmatrix} 18 & -9 & 9 \\ -14 & 7 & -7 \\ -38 & 19 & -19 \end{bmatrix}$$

$(\text{adj}A) \cdot B = 0$

$$\begin{bmatrix} 18 & -9 & 9 \\ -14 & 7 & -7 \\ -38 & 19 & -19 \end{bmatrix}\begin{bmatrix} \lambda \\ -3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$18\lambda + 27 - 9 = 0$

$\Rightarrow 18\lambda = -18$

$\Rightarrow \lambda = -1$

$\Rightarrow 2\mu + 3\lambda = 3(-1) = -3$

33. Let $\lambda, \mu \in \mathbf{R}$. If the system of equations

$3x + 5y + \lambda z = 3$

$7x + 11y - 9z = 2$

$97x + 155y - 189z = \mu$

has infinitely many solutions, then $\mu + 2\lambda$ is equal to : **[JEE-Main,April,2024]**

माना $\lambda, \mu \in \mathbf{R}$ है। यदि समीकरण निकाय

$3x + 5y + \lambda z = 3$

$7x + 11y - 9z = 2$
$97x + 155y - 189z = \mu$
के अनंत हल हैं, तो $\mu + 2\lambda$ बराबर है : **[JEE-Main,April,2024]**

(A) 27 (B) 24 (C) 25 (D) 22

Ans. C

Sol. $3x + 5y + \lambda z = 3$
$7x + 11y - 9z = 2$
$97x + 155y - 189z = \mu$

By carmer's rule, $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$

$$\begin{vmatrix} 3 & 5 & \lambda \\ 7 & 11 & -9 \\ 97 & 155 & -189 \end{vmatrix} = 0$$

$R_3 \to R_3 - 14R_2$

$$\begin{vmatrix} 3 & 5 & \lambda \\ 7 & 11 & -9 \\ -1 & 1 & -63 \end{vmatrix} = 0$$

$C_1 \to C_1 + C_2$

$$2\begin{vmatrix} 4 & 5 & \lambda \\ 9 & 11 & -9 \\ 0 & 1 & -63 \end{vmatrix} = 0$$

$-(-36 - 9\lambda) - 63(44 - 45) = 0$
$36 + 9\lambda + 63 = 0$
$9\lambda = -99$
$\lambda = -11$
$D_3 = 0$

$$\begin{vmatrix} 3 & 5 & 3 \\ 7 & 11 & 2 \\ 97 & 155 & \mu \end{vmatrix} = 0$$

$C_2 \to C_2 + C_1$

$$\begin{vmatrix} 0 & 2 & 3 \\ 5 & 4 & 2 \\ 97-\mu & 58 & \mu \end{vmatrix} = 0$$

$C_2 \to C_2 + C_1$

$$\begin{vmatrix} 0 & 2 & 3 \\ 5 & 4 & 2 \\ 97-\mu & 58 & \mu \end{vmatrix} = 0$$

$-5(2\mu - 174) + (97 - \mu)(4 - 12) = 0$

$10\mu - 870 + 776 - 8\mu = 0$

$2\mu = 94$

$\mu = 47$

$\mu + 2\lambda = 47 - 22$

$= 25$

34. Let $\alpha\beta\gamma = 45$; $\alpha, \beta, \gamma \in R$. If $x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$ for some $x, y, z \in R$, $x\,y\,z \neq 0$, then $6\alpha + 4\beta + \gamma$ is equal to _____. **[JEE-Main,April,2024]**

माना $\alpha\beta\gamma = 45$; $\alpha, \beta, \gamma \in R$ है। यदि कुछ $x, y, z \in R$, $x\,y\,z \neq 0$, के लिए $x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$ है, तो $6\alpha + 4\beta + \gamma$ बराबर है _____

[JEE-Main,April,2024]

Ans. 55

Sol. $\alpha\beta\gamma = 45,\ \alpha\beta\gamma \in R$

$x(\alpha, 1, 2) + y(1, \beta, 2) + z(2, 3, \gamma) = (0, 0, 0)$

$x, y, z \in R, xyz \neq 0$

$\alpha x + y + 2z = 0$

$x + \beta y + 3z = 0$

$2x + 2y + \gamma z = 0$

$xyz \neq 0 \Rightarrow$ non-trivial

$$\begin{vmatrix} \alpha & 1 & 2 \\ 1 & \beta & 3 \\ 2 & 2 & \gamma \end{vmatrix} = 0$$

$\Rightarrow \alpha(\beta\gamma - 6) - 1(\gamma - 6) + 2(2 - 2\beta) = 0$

$\Rightarrow \alpha\beta\gamma - 6\alpha - \gamma + 6 + 4 - 4\beta = 0$

$\Rightarrow 6\alpha + 4\beta + \gamma = 55$

35. If the system of equations

$$2x + 7y + \lambda z = 3$$
$$3x + 2y + 5z = 4$$
$$x + \mu y + 32z = -1$$

has infinitely many solutions, then $(\lambda - \mu)$ is equal to _____. **[JEE-Main,April,2024]**

यदि समीकरणों का निकाय

$$2x + 7y + \lambda z = 3$$
$$3x + 2y + 5z = 4$$
$$x + \mu y + 32z = -1$$

अनन्त हल रखता है, तो $(\lambda - \mu)$ बराबर है _____ **[JEE-Main,April,2024]**

Ans. 38

Sol. $2x + 7y + \lambda z = 3$

$3x + 2y + 5z = 4$

$x + \mu y + 32z = -1$

has infinitely many solution

$D = D_x = D_y = D_z = 0$

$$D_z = \begin{vmatrix} 2 & 7 & \lambda \\ 3 & 2 & 4 \\ 1 & \mu & 1 \end{vmatrix} = 0$$

$-4 + 9\mu + 28 - 6 - 8\mu + 21 = 0$

$$D = \begin{vmatrix} 2 & 7 & \lambda \\ 3 & 2 & 4 \\ 1 & -39 & 32 \end{vmatrix} = 0$$

$\mu = -39$

$\Rightarrow \lambda = -1$

$\therefore \lambda - \mu - 1 + 39 = 38$

Exercise J - A

1. Consider the system of equations

$x - 2y + 3z = -1, -x + y - 2z = k, x - 3y + 4z = 1$

STATEMENT-1 : The system of equations has no solution for $k \neq 3$.

and

STATEMENT-2 : The determinant $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$, for $k \neq 3$. **[JEE-2008]**

(A) Statement-1 is True, Statement-2 is True ; statement-2 is a correct explanation for statement-1

(B) Statement-1 is True, Statement-2 is True ; statement-2 is NOT a correct explanation for statement-1

(C) Statement-1 is True, Statement-2 is False

(D) Statement-1 is False, Statement-2 is True

समीकरणों के निकाय

$x - 2y + 3z = -1$

$-x + y - 2z = k$

$x - 3y + 4z = 1$

कथन-1 : $k \neq 3$ के लिये समीकरणों के निकाय का कोई हल नहीं है।

तथा

कथन-2 : $k \neq 3$ के लिये सारणिक $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} \neq 0$ है।

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण नही है

(C) कथन–1 सत्य है, कथन–2 असत्य है

(D) कथन–1 असत्य है, कथन– 2 सत्य है

Ans. A

Sol. $x - 2y + 3z = -1, -x + y - 2z = k$

and $x - 3y + 4z = 1$

$\Delta = \begin{vmatrix} 1 & -2 & 3 \\ -1 & 1 & -2 \\ 1 & -3 & 4 \end{vmatrix} = 0$ and $\begin{vmatrix} 1 & 3 & -1 \\ -1 & -2 & k \\ 1 & 4 & 1 \end{vmatrix} = 3 - k$

Hence if $k = 3$ then system will have infinite solutions and $k \neq 3$ then system will have no solution. so S(I) and S(II) both are true & (II) is correct explaination for (I).

2. The number of all possinle values of θ, where $0 < \theta < \pi$, for which the system of equations

$(y + z)\cos 3\theta = (xyz)\sin 3\theta$

$$x \sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y \sin 3\theta$

have a solution $(x_0, y_0 z_0)$ with $y_0 z_0 \neq 0$, is. **[JEE-2010]**

θ के उन सभी सम्भावित मानों की संख्या ज्ञात कीजिये, जिनके लिए समीकरण के निकाय

$(y + z)\cos 3\theta = (xyz)\sin 3\theta$

$$x\sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y \sin 3\theta$ का एक हल $(x_0, y_0 z_0)$ है, जहाँ $y_0 z_0 \neq 0$ तथा $0 < \theta < \pi$, है।

[JEE-2010]

Ans. 3

Sol. $(y + z)\cos 3\theta = (xyz)\sin 3\theta$ ...(i)

$$x \sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z} \quad ...(ii)$$

$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y\sin 3\theta$ ...(iii)

where $yz \neq 0$ and $0 < \theta < \pi$

from (i) & (iii)

$(y + z)\cos 3\theta = (y + 2z)\cos 3\theta + y \sin 3\theta$

$\Rightarrow$ $z \cos 3\theta + y \sin 3\theta = 0$ ...(iv)

from eq^n (ii)

$2z \cos 3\theta + 2y \sin 3\theta = xyz \sin 3\theta$ ...(v)

from equation (iv) & (v)

$\Rightarrow$ $xyz \sin 3\theta = 0$

$\Rightarrow$ $x \sin 3\theta = 0$ as $yz \neq 0$

Possible cases are either $x = 0$ or $\sin 3\theta = 0$

Case (1) : if $x = 0$

$\Rightarrow$ $y + z = 0 \Rightarrow y = -z$

from eq^n (iv) $\cos 3\theta = \sin 3\theta$

$$\Rightarrow \quad 3\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}$$

$$\Rightarrow \quad \theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}$$

Case (2) : if $\sin 3\theta = 0$

$\Rightarrow \quad \theta = \frac{\pi}{3}, \frac{2\pi}{3}$

But these values does not satisfy given equations.

Hence, total number of possible values of θ are 3.

3. Which of the following values of α satisy the equation $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\,\alpha$?

[JEE-Adv.-2015]

α के निम्नलिखित मानों में कौन सा मान समीकरण $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\,\alpha$ को संतुष्ट करता है?

[JEE-Adv.-2015]

(A) –4 (B) 9 (C) –9 (D) 4

Ans. BC

Sol. $R_3 \to R_3 - R_2, R_2 \to R_2 - R_1$

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 5+2\alpha & 5+4\alpha & 5+6\alpha \end{vmatrix} = -648\alpha$$

$R_3 \to R_3 - R_2$

$$\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ 3+2\alpha & 3+4\alpha & 3+6\alpha \\ 2 & 2 & 2 \end{vmatrix} = -648\alpha$$

$C_3 \to C_3 - C_2, C_2 \to C_2 - C_1$

$$\begin{vmatrix} (1+\alpha)^2 & \alpha(2+3\alpha) & \alpha(2+5\alpha) \\ 3+2\alpha & 2\alpha & 2\alpha \\ 2 & 0 & 0 \end{vmatrix} = -648\alpha$$

$\Rightarrow \quad 2\alpha^2(2+3\alpha) - 2\alpha^2(2+5\alpha) = -324\alpha$

$\Rightarrow \quad -4\alpha^3 = -324\alpha \Rightarrow \quad \alpha = 0, \pm 9$

4. The total number of distinct $x \in \mathbb{R}$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is. **[JEE-Adv.-2016]**

ऐसे सभी भिन्न $x \in \mathbb{R}$, जिनके लिए $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ है, की कुल संख्या है। **[JEE-Adv.-2016]**

Ans. 2

Sol. $\begin{vmatrix} x & x^2 & 1 \\ 2x & 4x^2 & 1 \\ 3x & 9x^2 & 1 \end{vmatrix} + \begin{vmatrix} x & x^2 & x^3 \\ 2x & 4x^2 & 8x^3 \\ 3x & 9x^2 & 27x^3 \end{vmatrix} = 10$

Using $\begin{vmatrix} a & a^2 & 1 \\ b & b^2 & 1 \\ c & c^2 & 1 \end{vmatrix} = (a-b)(b-c)(c-a)$

Again $\begin{vmatrix} x & x^2 & 1 \\ 2x & 4x^2 & 1 \\ 3x & 9x^2 & 1 \end{vmatrix} + (x)(2x)(3x)\begin{vmatrix} 1 & x & x^2 \\ 1 & 2x & 4x^2 \\ 1 & 3x & 9x^2 \end{vmatrix} = 10$

$\Rightarrow (x-2x)(2x-3x)(3x-x) + 6x^3[(x-2x)(2x-3x)(3x-x)] = 10$

$\Rightarrow 2x^3(1+6x^3) = 10$ (solve by putting $x^3 = t$)

$\Rightarrow x = -1, \left(\frac{5}{6}\right)^{\frac{1}{3}}$

5. Let $a, \lambda, \mu \in \mathbb{R}$. Consider the system of linear equations

$ax + 2y = \lambda$

$3x - 2y = \mu$

Which of the following statement(s) is(are) correct ? **[JEE-Adv.-2016]**

(A) If $a = -3$, then the system has infinitely many solutions for all values of λ and μ

(B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ

(C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$

(D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$

माना कि a, $\lambda, \mu \in \mathbb{R}$ हैं। इन रैखिक समीकरणों के निकाय (system of linear equations) पर विचार कीजिए।

$ax + 2y = \lambda$

$3x - 2y = \mu$

निम्नलिखित में से कौन सा (से) कथन सही है (हैं)? **[JEE-Adv.-2016]**

(A) यदि $a = -3$, तब λ और μ के सभी मानों के लिए निकाय के अनन्त (infinitely many) हल हैं

(B) यदि $a \neq -3$, तब λ और μ के सभी मानों के लिए निकाय का अद्वितीय (unique) हल है

(C) यदि $\lambda + \mu = 0$, तब $a = -3$ के लिए निकाय के अनन्त हल हैं

(D) यदि $\lambda + \mu \neq 0$, तब $a = -3$ के लिए निकाय का कोई हल नहीं है

Ans. BCD

Sol. System has unique solution for $\frac{a}{3} \neq \frac{2}{-2}$

system has infinitely many solution for $\frac{a}{3} = \frac{2}{-2} = \frac{\lambda}{\mu}$

and no solution for $\frac{a}{3} = \frac{2}{-2} \neq \frac{\lambda}{\mu}$

6. Let p, q, r be nonzero real numbers that are, respectively, the 10th, 100th and 1000th terms of a harmonic progression. Consider the system of linear equations **[JEE(Adv.)2022]**

$$x + y + z = 1$$

$$10x + 100y + 1000z = 0$$

$$qr\,x + pr\,y + pq\,z = 0$$

List - I	**List - II**
(I) If $\frac{q}{r} = 10$, then the system of linear equations has	(P) $x = 0, y = \frac{10}{9}, z = -\frac{1}{9}$ as a solution
(II) If $\frac{p}{r} \neq 100$, then the system of linear equations has	(Q) $x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$ as a solution
(III) If $\frac{p}{q} \neq 10$, then the system of linear equations has	(R) infinitely many solutions
(IV) If $\frac{p}{q} = 10$, then the system of linear equations has	(S) no solution
	(T) at least one solution

The correct option is:

ekuk fd $p, q,$ oa r 'kwU;srj okLrfod la[;k;sa gS tks ,d gjkRed Js<+h ds Øe'k% 10ok¡, 100ok¡ ,oa 1000ok¡ in gSA jSf[kd lehdj.kksa ds fudk;

[JEE Adv.-2022]

$$x + y + z = 1$$

$$10x + 100y + 1000z = 0$$

$$qr\,x + pr\,y + pq\,z = 0$$

ij fopkj dhft,A

lwph-I

(I) ;fn $\frac{q}{r} = 10$ gS] rc jSf[kd lehdj.kksa ds fudk; dk

(II) ;fn $\frac{p}{r} \neq 100$ gS] rc jSf[kd lehdj.kksa ds fudk; dk

(III) ;fn $\frac{p}{q} \neq 10$ gS] rc jSf[kd lehdj.kksa ds fudk; dk

(IV) ;fn $\frac{p}{q} = 10$ gS] rc jSf[kd lehdj.kksa ds fudk; dk

lwph-II

(P) gy $x = 0, y = \frac{10}{9}, z = -\frac{1}{9}$ gS

(Q) gy $x = \frac{10}{9}, y = -\frac{1}{9}, z = 0$ gS

(R) vuar gy gS

(S) dksbZ gy ugha gS

(T) de ls de ,d gy gS

lgh fodYi gS :

(A) (I) $\rightarrow$ (T); (II) $\rightarrow$ (R); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (T)

(B) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (S); (III) $\rightarrow$ (S); (IV) $\rightarrow$ (R)

(C) (I) $\rightarrow$ (Q); (II) $\rightarrow$ (R); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (R)

(D) (I) $\rightarrow$ (T); (II) $\rightarrow$ (S); (III) $\rightarrow$ (P); (IV) $\rightarrow$ (T)

Ans. (B)

Sol. $x + y + z = 1$ _____(1)

$10x + 100y + 1000z = 0$ _____(2)

$qrx + pry + pqz = 0$ _____(3)

Equation (3) can be re-written as

$$\frac{x}{p} + \frac{y}{q} + \frac{z}{r} = 0 \qquad (\because p, q, r \neq 0)$$

Let $p = \frac{1}{a+9d}, q = \frac{1}{a+99d}, r = \frac{1}{a+999d}$

Now, equation (3) is

$(a + 9d)x + (a + 99d)y + (a + 999d)z = 0$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 10 & 100 & 1000 \\ a+9d & a+99d & a+999d \end{vmatrix} = 0$$

$$\Delta_x = \begin{vmatrix} 1 & 1 & 1 \\ 0 & 100 & 1000 \\ 0 & a+99d & a+999d \end{vmatrix} = 900(d-a)$$

$$\Delta_y = \begin{vmatrix} 1 & 1 & 1 \\ 10 & 0 & 1000 \\ a+9d & 0 & a+999d \end{vmatrix} = 990(a-d)$$

$$\Delta_z = \begin{vmatrix} 1 & 1 & 1 \\ 10 & 100 & 0 \\ a+9d & a+99d & 0 \end{vmatrix} = 90(d-a)$$

Option I: If $\frac{q}{r} = 10 \Rightarrow a = d$

$\Delta = \Delta_x = \Delta_y = \Delta_z = 0$

And eq. (1) and eq. (2) represents non-parallel planes and eq. (2) and eq. (3) represents same plane
$\Rightarrow$ Infinitely many solutions
I $\rightarrow$ P, Q, R, T

Option II : $\frac{p}{r} \neq 100 \Rightarrow a \neq d$

$\Delta = 0, \Delta_x, \Delta_y, \Delta_z \neq 0$

No solution
II $\rightarrow$ S

Option III : $\frac{p}{q} \neq 10 \Rightarrow a \neq d$

No solution
III $\rightarrow$ S

Option IV : If $\frac{p}{q} = 10 \Rightarrow a = d$

Infinitely many solution
IV $\rightarrow$ P, Q, R, T

7. Let α, β and γ be real numbers. Consider the following system of linear equations

$x + 2y + z = 7$

$x + \alpha z = 11$

$2x - 3y + \beta z = \gamma$

Match each entry in **List-I** to the correct entries in **List-II**.

List-I	**List-II**
(P) If $\beta = \frac{1}{2}(7\alpha - 3)$ and $\gamma = 28$, then the system has	(1) a unique solution
(Q) If $\beta = \frac{1}{2}(7\alpha - 3)$ and $\gamma \neq 28$, then the system has	(2) no solution
(R) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $\gamma \neq 28$, then the system has	(3) infinitely many solutions
(S) If $\beta \neq \frac{1}{2}(7\alpha - 3)$ where $\alpha = 1$ and $\gamma = 28$, then the system has	(4) $x = 11$, $y = -2$ and $z = 0$ as a solution
	(5) $x = -15$, $y = 4$ and $z = 0$ as a solution

The correct option is : **[JEE(Adv.)2023]**

माना कि α, β एवं γ वास्तविक संख्याएं (real numbers) हैं। निम्न रैखिक समीकरण निकाय (system of linear equation) पर विचार कीजिए।

$x + 2y + z = 7$

$x + \alpha z = 11$

$2x - 3y + \beta z = \gamma$

List-I की प्रत्येक प्रविष्ट (entry) का **List-II** की सही प्रविष्टियों (entries) से मिलान कीजिए।

List-I	**List-II**
(P) यदि $\beta = \frac{1}{2}(7\alpha - 3)$ एवं $\gamma = 28$, तब निकाय का(के)	(1) एक अद्वितीय हल (unique solution) है
(Q) यदि $\beta = \frac{1}{2}(7\alpha - 3)$ एवं $\gamma \neq 28$, तब निकाय का(के)	(2) कोई हल नहीं है
(R) यदि $\beta \neq \frac{1}{2}(7\alpha - 3)$ जहाँ $\alpha = 1$ एवं $\gamma \neq 28$, तब निकाय का(के)	(3) अनंत हल हैं
(S) यदि $\beta \neq \frac{1}{2}(7\alpha - 3)$ जहाँ $\alpha = 1$ एवं $\gamma = 28$, तब निकाय का(के)	(4) $x = 11$, $y = -2$ एवं $z = 0$ एक हल है
	(5) $x = -15$, $y = 4$ एवं $z = 0$ एक हल है

सही विकल्प है : **[JEE(Adv.)2023]**

(A) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (1); (S) $\rightarrow$ (4)

(B) (P) $\rightarrow$ (3); (Q) $\rightarrow$ (2); (R) $\rightarrow$ (5); (S) $\rightarrow$ (4)

(C) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (4); (S) $\rightarrow$ (5)

(D) (P) $\rightarrow$ (2); (Q) $\rightarrow$ (1); (R) $\rightarrow$ (1); (S) $\rightarrow$ (3)

Ans. (A)

Sol. $D = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 0 & \alpha \\ 2 & -3 & \beta \end{vmatrix}$

$= 1\,[3\alpha] - 2\,[\beta - 2\alpha] + [-3]$

$= 3\alpha - 2\beta + 4\alpha - 3$

$= 7\alpha - 2\beta - 3$

If $D = 0$

$7\alpha - 2\beta - 3 = 0$

$\beta = \dfrac{7\alpha - 3}{2}$

$D_x = \begin{vmatrix} 7 & 2 & 1 \\ 11 & 0 & \alpha \\ \gamma & -3 & \beta \end{vmatrix}$

if $\gamma = 28$ & $\beta = \dfrac{7\alpha - 3}{2}$

then $D_x = \begin{vmatrix} 7 & 2 & 1 \\ 11 & 0 & \alpha \\ 28 & -3 & \dfrac{7\alpha - 3}{2} \end{vmatrix}$

$7[3\alpha] - 2\left[11\dfrac{(7\alpha - 3)}{2} - 28\alpha\right] + [-33]$

$21\alpha - [11\,(7\alpha - 3) - 56\alpha] - 33$

$21\alpha - (77\alpha - 33 - 56\alpha) - 33$

$= 0$

$D_y = \begin{vmatrix} 1 & 7 & 1 \\ 1 & 11 & \alpha \\ 2 & \gamma & \beta \end{vmatrix}$

Again if $\gamma = 28$ & $\beta = \dfrac{7\alpha - 3}{2}$

$D_y = \begin{vmatrix} 1 & 7 & 1 \\ 1 & 11 & \alpha \\ 2 & 28 & \dfrac{7\alpha - 3}{2} \end{vmatrix}$

$R_1 \to R_1 - R_2$

$$= \begin{vmatrix} 0 & -4 & 1-\alpha \\ 1 & 11 & \alpha \\ 2 & 28 & \dfrac{7\alpha-3}{2} \end{vmatrix}$$

$$4\left[\frac{7\alpha-3}{2}-2\alpha\right]+(1-\alpha)(28-22)$$

$2(7\alpha - 3) - 8\alpha + 6(1 - \alpha)$

$14\alpha - 6 - 8\alpha + 6 - 6\alpha$

$= 0$

$$D_z = \begin{vmatrix} 1 & 2 & 7 \\ 1 & 0 & 11 \\ 2 & -3 & \gamma \end{vmatrix}$$

$1(33) - 2(\gamma - 22) + 7(-3)$

$33 + 44 - 21 - 2\gamma$

$\Rightarrow 56 - 2\gamma$

If $\gamma = 28$

$D_z = 0$

(P) If $\beta = \frac{1}{2}(7\alpha - 3)$ & $\alpha = 28$

then D_x, D_y, D_z are all zero

$\therefore$ Infinite many solution

$P \rightarrow (3)$

(Q) If $\beta = \frac{1}{2}(7\alpha - 3)$ & $\gamma \neq 28$

It can be easily observed that $D_z \neq 0$

$\therefore D = 0$ & $D_z \neq 0$

$\Rightarrow$ No solution

$Q \rightarrow 2$

(R) If $\beta \neq \frac{1}{2}(7\alpha - 3)$, i.e. $D \neq 0$

It will be the case of unique solution but here additional info of $\alpha = 1$ & $\gamma \neq 28$ is provided

(S) If $\alpha = 1$ then

$x + 2y + z = 7$

$x + z = 11$

$2y = 7 - 11$

$y = -2$ will always be the solution.

But $\gamma \neq 28$ then $z \neq 0$

So, $R \rightarrow (1)$

Again

If $\alpha = 1$; $\gamma = 28$

then $y = -2; z = 0$
& $z + 2(-2) = 7$
$x = 7 + 4$
$x = 11$
So we get the solution $S \rightarrow (4)$

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. Let $\Delta_0 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{32} & a_{32} & a_{33} \end{vmatrix}$ and let Δ_1 denote the determinant formed by the cofactors of elements of Δ_0 and Δ_2 denote the determinant formed by the cofactor at Δ_1 similarly Δ_n denotes the determinant formed by the cofactors at Δ_{n-1} then the determinant value of Δ_n is :

माना $\Delta_0 = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{32} & a_{32} & a_{33} \end{vmatrix}$ तथा let Δ_1 सारणिक Δ_0 के सभी सहखण्डों से बनने वाला सारणिक है तथा Δ_2 सारणिक Δ_1 के सभी सह खण्डों से बनने वाला सारणिक है इसी प्रकार Δn, सारणिक Δ_{n-1} के सभी अवयवों के सहखण्डों से बनने वाला सारिणक है तो सारणिक Δ_n का मान होगा :

(A) Δ_0^{2n} (B) $\Delta_0^{2^n}$ (C) $\Delta_0^{n^2}$ (D) Δ_0^2

Ans. B

Sol. $\Delta_1 = \Delta_0^2$

$\Delta_2 = \Delta_1^2 = \Delta_0^4$

$\Delta_3 = \Delta_2^2 = \Delta_0^8$

$\Delta_4 = \Delta_3^2 = \Delta_0^{16}$

and so on $\Delta_n = \Delta_0^{2^n}$

2. Let a determinant is given by $A = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$ and suppose det. $A = 6$. If $B = \begin{vmatrix} p+x & q+y & r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$ then :

माना $A = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$ एक सारणिक दी गयी है तथा det. $A = 6$ है। यदि $B = \begin{vmatrix} p+x & q+y & r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$ तब :

(A) det. $B = 6$ (B) det. $B = -6$ (C) det. $B = 12$ (D) det. $B = -12$

Ans. C

Sol. Consider the det. B, using $R_1 \to R_1 + R_2 + R_3$

$$B = 2\begin{vmatrix} a+p+x & b+q+y & c+r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$$

using $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$

$$= 2\begin{vmatrix} a+p+x & b+q+y & c+r+z \\ -p & -q & -r \\ -x & -y & -z \end{vmatrix}$$

using $R_1 \to R_1 + R_2 + R_3$

$B = 2$ det. $A = 2 \cdot 6 = 12$

3. The system of equations

$2x\cos^2\theta + y\sin2\theta - 2\sin\theta = 0$

$x\sin2\theta + 2y\sin^2\theta = -2\cos\theta$

$x\sin\theta - y\cos\theta = 0$, for all values of θ, can :

(A) have a unique non - trivial solution (B) not have a solution

(C) have infinite solutions (D) have a trivial solution

समीकरणों के निकाय

$2x\cos^2\theta + y\sin2\theta - 2\sin\theta = 0$

$x\sin2\theta + 2y\sin^2\theta = -2\cos\theta$

$x\sin\theta - y\cos\theta = 0$, के लिये θ के प्रत्येक मान लिये :

(A) एक अद्वितीय अशून्य हल है (B) कोई हल नहीं है

(C) अनन्त हल है (D) एक अशून्य हल है

Ans. B

Sol. slope of (1) and (2) is $\cot\theta \Rightarrow$ (1) and (2) are parallel and slope of (3) is $\tan\theta$

$\Rightarrow$ no solution.

Using $R_2 \to R_2 - (2\cos\theta)\,R_3$ and $R_1 \to R_1 + (2\sin\theta)R_3$, the value of determinat is 4.

4. If the system of equations $x - Ky - z = 0$, $Kx - y - z = 0$ and $x + y - z = 0$ has a non zero solution, then the possible values of K are :

यदि समीकरण निकाय $x - Ky - z = 0$, $Kx - y - z = 0$ तथा $x + y - z = 0$ के अशून्य हल है, तो K के सम्भव मान होगें :

(A) –1, 2 (B) 1, 2 (C) 0, 1 (D) –1, 1

Ans. D

Sol. $D = \begin{vmatrix} 1 & -k & -1 \\ k & -1 & -1 \\ 1 & 1 & -1 \end{vmatrix} = 0$

$1(1+1) + k(-k+1) - 1(k+1) = 0$

$2 - k^2 + k - k - 1 = 0$

$1 - k^2 = 0$

$k^2 = 1$

$k = \pm 1$

5. The system of equations

$x - y\cos\theta + z\cos 2\theta = 0$

$-x\cos\theta + y - z\cos\theta = 0$

$x\cos 2\theta - y\cos\theta + z = 0$

has non trivial solution for θ equals :

(A) $n\pi$ only, $n \in I$ (B) $n\pi + \frac{\pi}{4}$ only, $n \in I$

(C) $(2n-1)\frac{\pi}{2}$ only, $n \in I$ (D) all value of θ

यदि समीकरणों के निकाय

$x - y\cos\theta + z\cos 2\theta = 0$

$-x\cos\theta + y - z\cos\theta = 0$

$x\cos 2\theta - y\cos\theta + z = 0$

का अशून्य हल (non trivial) है, तो यह θ के किस मान के लिये सम्भव होगा :

(A) केवल $n\pi$ के लिये, $n \in I$ (B) केवल $n\pi + \frac{\pi}{4}$ के लिये, $n \in I$

(C) केवल $(2n-1)\frac{\pi}{2}$ के लिये , $n \in I$ (D) θ के सभी मान

Ans. D

Sol. for non trivial solution

$$\begin{vmatrix} 1 & -\cos\theta & \cos 2\theta \\ -\cos\theta & 1 & -\cos\theta \\ \cos 2\theta & -\cos\theta & 1 \end{vmatrix} = 0$$

using $C_1 \to C_1 - C_3$

$$\begin{vmatrix} 2\sin^2\theta & -\cos\theta & \cos 2\theta \\ 0 & 1 & -\cos\theta \\ -2\sin^2\theta & -\cos\theta & 1 \end{vmatrix} = 0$$

$$2\sin^2\theta \begin{vmatrix} 1 & -\cos\theta & \cos 2\theta \\ 0 & 1 & -\cos\theta \\ -1 & -\cos\theta & 1 \end{vmatrix} = 0$$

$\sin^2\theta = 0$

or $\quad 1[1 - \cos^2\theta] - 1[\cos^2\theta - \cos 2\theta]$

$\sin^2\theta - [\cos^2\theta - (\cos^2\theta - \sin^2\theta)]$

$\sin^2\theta - \sin^2\theta = 0$

hence $D = 0 \ \forall \ \theta \in R \quad \Rightarrow \quad$ (D)

6. If $D(x) = \begin{vmatrix} x-1 & (x-1)^2 & x^3 \\ x-1 & x^2 & (x+1)^3 \\ x & (x+1)^2 & (x+1)^3 \end{vmatrix}$ then the coefficient of x in D(x) is :

यदि $D(x) = \begin{vmatrix} x-1 & (x-1)^2 & x^3 \\ x-1 & x^2 & (x+1)^3 \\ x & (x+1)^2 & (x+1)^3 \end{vmatrix}$ तो D(x) में x का गुणांक होगा :

(A) 5 (B) – 2 (C) 6 (D) 0

Ans. A

Sol. Coefficient of x in $D(x) = D'(0)$

7. The set of equations

$\lambda x - y + (\cos\theta) z = 0, \quad 3x + y + 2z = 0, (\cos\theta)x + y + 2z = 0$

$0 < \theta < 2\pi$, has non- trivial solution(s) :

(A) for no value of λ and θ (B) for all values of λ and θ

(C) for all values of λ and only two values of θ (D) for only one value of λ and all values of θ

समीकरणों के निकाय

$\lambda x - y + (\cos\theta) z = 0, \quad 3x + y + 2z = 0, (\cos\theta)x + y + 2z = 0$

$(\cos\theta)x + y + 2z = 0$ का अन्तराल $0 < \theta < 2\pi$ में अशून्य हल होगा :

(A) λ तथा θ के किसी भी मान के लिए नहीं (B) λ तथा θ के हर मान के लिए

(C) λ के लिए सभी मान तथा θ के दो मान के लिए (D) λ के केवल एक मान तथा θ को हर मानों के लिए

Ans. A

Sol. $D = \cos\theta - \cos^2\theta + 6 \neq 0$ since $D \neq 0 \Rightarrow$ only trivial solution is possible $\Rightarrow$ (A)

8. The system of equations $(\sin\theta)x + 2z = 0$, $(\cos\theta)x + (\sin\theta)y = 0$, $(\cos\theta)y + 2z = a$ has :
(A) no unique solution
(B) a unique solution which is a function of a and θ
(C) a unique solution which is independent of a and θ
(D) a unique solution which is independent of θ only
समीकरणों के निकाय $(\sin\theta)x + 2z = 0$, $(\cos\theta)x + (\sin\theta)y = 0$, $(\cos\theta)y + 2z = a$:
(A) कोई हल नहीं
(B) एक अद्वितीय हल है जो a तथा θ तथा का फलन है
(C) एक अद्वितीय हल है जो a तथा θ से स्वतन्त्र है
(D) एक अद्वितीय हल है जो θ से स्वतन्त्र है

Ans. B

Sol. $D = \begin{vmatrix} \sin\theta & 0 & 2 \\ \cos\theta & \sin\theta & 0 \\ 0 & \cos\theta & 2 \end{vmatrix}$

$\sin\theta\,(2\sin\theta) + 2\cos^2\theta = 2$; $D_1 = a\sin\theta$

9. The equation $\begin{vmatrix} (1+x)^2 & (1-x)^2 & -(2+x^2) \\ 2x+1 & 3x & 1-5x \\ x+1 & 2x & 2-3x \end{vmatrix} + \begin{vmatrix} (1+x)^2 & 2x+1 & x+1 \\ (1-x)^2 & 3x & 2x \\ 1-2x & 3x-2 & 2x-3 \end{vmatrix} = 0$

(A) has no real solution
(B) has 4 real solutions
(C) has two real and two non-real solutions
(D) has infinite number of solutions , real or non-real

समीकरण $\begin{vmatrix} (1+x)^2 & (1-x)^2 & -(2+x^2) \\ 2x+1 & 3x & 1-5x \\ x+1 & 2x & 2-3x \end{vmatrix} + \begin{vmatrix} (1+x)^2 & 2x+1 & x+1 \\ (1-x)^2 & 3x & 2x \\ 1-2x & 3x-2 & 2x-3 \end{vmatrix} = 0$

(A) कोई वास्तविक हल नही है
(B) 4 वास्तविक हल है
(C) दो वास्तविक तथा अवास्तविक हल हैं
(D) अनन्त हल है, वास्तविक या अवास्तविक

Ans. D

Sol. 1st two columns of 1st determinant are same as 1st two rows of 2nd. Hence transpose the 2nd. Add the two determinants and use $C_1 \to C_1 + C_3 \Rightarrow D = 0$

10. Let $\Delta_1 = \begin{vmatrix} ap^2 & 2ap & 1 \\ aq^2 & 2aq & 1 \\ ar^2 & 2ar & 1 \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} apq & a(p+q) & 1 \\ aqr & a(q+r) & 1 \\ arp & a(r+p) & 1 \end{vmatrix}$ then :

माना $\Delta_1 = \begin{vmatrix} ap^2 & 2ap & 1 \\ aq^2 & 2aq & 1 \\ ar^2 & 2ar & 1 \end{vmatrix}$ और $\Delta_2 = \begin{vmatrix} apq & a(p+q) & 1 \\ aqr & a(q+r) & 1 \\ arp & a(r+p) & 1 \end{vmatrix}$ तो :

(A) $\Delta_1 = \Delta_2$ (B) $\Delta_2 = 2\Delta_1$ (C) $\Delta_1 = 2\Delta_2$ (D) $\Delta_1 + 2\Delta_2 = 0$

Ans. D

Sol. $\Delta_1 = 2a^2 \begin{vmatrix} p^2 & p & 1 \\ q^2 & q & 1 \\ r^2 & r & 1 \end{vmatrix}$

$= 2a^2 (p-q)(q-r)(p-r)$

$= -2a^2 (p-q)(q-r)(r-p)$

$\Delta_2 = a^2 \begin{vmatrix} pq & p+q & 1 \\ qr & q+r & 1 \\ rp & r+p & 1 \end{vmatrix}$

$R_2 \to R_2 - R_1$ & $R_3 \to R_3 - R_2$

$= a^2 \begin{vmatrix} qp & p+q & 1 \\ q(p-r) & (p-r) & 0 \\ r(p-q) & p-q & 0 \end{vmatrix}$

$= a^2 (p-q)(r-p) \begin{vmatrix} qp & p+q & 1 \\ q & 1 & 0 \\ r & 1 & 0 \end{vmatrix}$

$= a^2 (p-q)(r-p)(q-r)$

Now, $\Delta_1 + 2\Delta_2 = 0$

11. The value of λ for which the system of equations $2x - y - z = 12$, $x - 2y + z = -4$, $x + y + \lambda z = 4$ has no solution is :

λ का वह मान जिसके लिये समीकरणों के निकाय $2x - y - z = 12$, $x - 2y + z = -4$, $x + y + \lambda z = 4$ का कोई हल नहीं है, होगा:

(A) 3 (B) –3 (C) 2 (D) –2

Ans. D

Sol. $D = \begin{vmatrix} 2 & -1 & -1 \\ 1 & -2 & 1 \\ 1 & 1 & \lambda \end{vmatrix} = 0$

$2(-2\lambda - 1) + 1(\lambda - 1) - 1(1 + 2) = 0$

$-4\lambda - 2 + \lambda - 1 - 3 = 0$

$-3\lambda - 6 = 0$

$\lambda = -2$

12. The value of $\begin{vmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) & \cos 2\beta \\ \sin\alpha & \cos\alpha & \sin\beta \\ -\cos\alpha & \sin\alpha & \cos\beta \end{vmatrix}$ is independent of :

(A) α (B) β (C) both α and β (D) neither α nor β

$\begin{vmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) & \cos 2\beta \\ \sin\alpha & \cos\alpha & \sin\beta \\ -\cos\alpha & \sin\alpha & \cos\beta \end{vmatrix}$ का मान स्वतंत्र है :

(A) α (B) β (C) α तथा β दोनों (D) न तो α न ही β

Ans. A

Sol. $\begin{vmatrix} \cos(\alpha+\beta) & -\sin(\alpha+\beta) & \cos 2\beta \\ \sin\alpha & \cos\alpha & \sin\beta \\ -\cos\alpha & \sin\alpha & \cos\beta \end{vmatrix}$

$\cos(\alpha + \beta)(\cos(\alpha + \beta) + \sin(\alpha + \beta)\sin(\alpha + \beta) + \cos 2\beta$

$\cos^2(\alpha + \beta) + \sin^2(\alpha + \beta) + \cos 2\beta$

$\cos 2\beta + 1$

independent of α.

13. The values of θ, λ for which the following equations
$\sin\theta x - \cos\theta y + (\lambda+1)z = 0;\ \cos\theta x + \sin\theta y - \lambda z = 0;\ \lambda x + (\lambda+1)y + \cos\theta z = 0$
have non trivial solution, is :

(A) $\theta = n\pi,\ \lambda \in R - \{0\}$ (B) $\theta = 2n\pi$, λ is any rational number

(C) $\theta = (2n+1)\pi,\ \lambda \in R^+,\ n \in I$ (D) $\theta = (2n+1)\frac{\pi}{2},\ \lambda \in R,\ n \in I$

θ तथा λ के मान बताओं जब समीकरणों
$\sin\theta x - \cos\theta y + (\lambda+1)z = 0;\ \cos\theta x + \sin\theta y - \lambda z = 0;\ \lambda x + (\lambda+1)y + \cos\theta z = 0$
का अशून्य हल होगा :

(A) $\theta = n\pi,\ \lambda \in R - \{0\}$ (B) $\theta = 2n\pi$, λ कोई परिमेय संख्या है

(C) $\theta = (2n+1)\pi,\ \lambda \in R^+,\ n \in I$ (D) $\theta = (2n+1)\frac{\pi}{2},\ \lambda \in R,\ n \in I$

Ans. D

Sol. for non trivial solution $\begin{vmatrix} \sin\theta & -\cos\theta & \lambda+1 \\ \cos\theta & \sin\theta & -\lambda \\ \lambda & \lambda+1 & \cos\theta \end{vmatrix} = 0$; this gives $2\cos\theta(\lambda^2+\lambda+1) = 0$

14. Number of triplets of a, b and c for which the system of equations,
$ax - by = 2a - b$ and $(c+1)x + cy = 10 - a + 3b$
has infinitely many solutions and $x = 1,\ y = 3$ is one of the solutions, is :

(A) exactly one (B) exactly two (C) exactly three (D) infinitely many

a, b व c के त्रिक की संख्या बताओ जिसके लिए समीकरणों के निकाय,
$ax - by = 2a - b$ and $(c+1)x + cy = 10 - a + 3b$
के अनन्त हल है तथा $x = 1,\ y = 3$ उनमें से एक हल है :

(A) केवल एक (B) केवल दो (C) केवल तीन (D) अनन्त

Ans. B

Sol. put $x = 1$ & $y = 3$ in 1st equation $\Rightarrow$ $a = -2b$ & from 2nd equation

$c = \frac{9+5b}{4}$; Now use $\frac{a}{c+1} = -\frac{b}{c} = \frac{2a-b}{10-a+3b}$; from first two $b = 0$ or $c = 1$;

if $b = 0 \Rightarrow a = 0$ & $c = 9/4$; if $c = 1$; $b = -1$; $a = 2$

15. For any ΔABC, the value of determinant $\begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin^2 B & \cot B & 1 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$ is :

किसी ΔABC के लिऐ सारणिक $\begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin^2 B & \cot B & 1 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$ का मान है :

(A) 0 (B) 1 (C) $\sin A \sin B \sin C$ (D) $\sin A + \sin B + \sin C$

Ans. A

Sol. $R_1 \to R_1 - R_2$

$R_2 \to R_2 - R_3$

$$\begin{vmatrix} \sin^2 A - \sin^2 B & \cot A - \cot B & 0 \\ \sin^2 B - \sin^2 C & \cot B - \cot C & 0 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$$

$$\begin{vmatrix} \sin(A+B)\sin(A-B) & \dfrac{\sin(B-A)}{\sin A \sin B} & 0 \\ \sin(B-C)\sin(B+C) & \dfrac{\sin(C-B)}{\sin B \sin C} & 0 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$$

$$\Rightarrow \sin(A-B)\sin(B-C)\begin{vmatrix} \sin(A+B) & \dfrac{-1}{\sin A \sin B} & 0 \\ \sin(B+C) & \dfrac{-1}{\sin B \sin C} & 0 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$$

$$\Rightarrow \sin(A-B)\sin(B-C)\left[-\frac{\sin(A+B)}{\sin B \sin C} + \frac{\sin(B+C)}{\sin A \sin B}\right]$$

$$\Rightarrow \sin(A-B)\sin(B-C)\left[\frac{-\sin C}{\sin B \sin C} + \frac{\sin A}{\sin A \sin B}\right]$$

$\sin(A-B)\sin(B-C) \times 0$

0

16. If a, b, c are non-zero real numbers, then $\begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix}$ is equal to :

(A) abc (B) $a^2b^2c^2$ (C) ab + bc + ca (D) None of these

यदि a, b, c अशून्य वास्तविक संख्याएँ हो तो सारणिक $\begin{vmatrix} b^2c^2 & bc & b+c \\ c^2a^2 & ca & c+a \\ a^2b^2 & ab & a+b \end{vmatrix}$ का मान होगा :

(A) abc (B) $a^2b^2c^2$ (C) ab + bc + ca (D) इनमें से कोई नहीं

Ans. D

Sol. $D = \frac{1}{abc}\begin{vmatrix} ab^2c^2 & abc & a(b+c) \\ bc^2a^2 & bca & b(c+a) \\ ca^2b^2 & cab & c(a+b) \end{vmatrix}$

$(R_1 \to aR_1, R_2 \to bR_2, R_3 \to cR_3)$

$= abc\begin{vmatrix} bc & 1 & ab+ac \\ ca & 1 & bc+ab \\ ab & 1 & ca+cb \end{vmatrix}$

$= abc(ab+bc+ca)\begin{vmatrix} bc & 1 & 1 \\ ca & 1 & 1 \\ ab & 1 & 1 \end{vmatrix} \quad (C_3 \to C_3 + C_1)$

$= 0$

17. The determinant $\begin{vmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & \cos 2\phi \\ \sin\theta & \cos\theta & \sin\phi \\ -\cos\theta & \sin\theta & \cos\phi \end{vmatrix}$ is :

(A) 0 (B) independent of θ
(C) independent of ϕ (D) independent of both θ and ϕ

सारणिक $\begin{vmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & \cos 2\phi \\ \sin\theta & \cos\theta & \sin\phi \\ -\cos\theta & \sin\theta & \cos\phi \end{vmatrix}$ का मान होगा :

(A) 0 (B) θ से स्वतंत्र
(C) ϕ से स्वतंत्र (D) θ तथा ϕ दोनों से स्वतंत्र

Ans. B

Sol. $$\begin{vmatrix}\cos(\theta+\phi) & -\sin(\theta+\phi) & \cos 2\phi \\ \sin\theta & \cos\theta & \sin\phi \\ -\cos\theta & \sin\theta & \cos\phi\end{vmatrix} = \frac{1}{\sin\phi\cos\phi}\begin{vmatrix}\cos(\theta+\phi) & -\sin(\theta+\phi) & \cos 2\phi \\ \sin\theta\sin\phi & \sin\phi\cos\theta & \sin^2\phi \\ -\cos\theta\cos\phi & \sin\theta\cos\phi & \cos^2\phi\end{vmatrix}$$

Applying $R_1 \to R_1 + R_2 + R_3$ (संक्रिया $R_1 \to R_1 + R_2 + R_3$ से)

$$= \frac{1}{\sin\phi\cos\phi}\begin{vmatrix}0 & 0 & 2\cos^2\phi \\ \sin\theta\sin\phi & \sin\varphi\cos\theta & \sin^2\phi \\ -\cos\theta\cos\phi & \sin\theta\cos\phi & \cos^2\phi\end{vmatrix}$$

$$= \begin{vmatrix}0 & 0 & 2\cos^2\phi \\ \sin\theta & \cos\theta & \sin\phi \\ -\cos\theta & \sin\theta & \cos\phi\end{vmatrix} = 2\cos^2\phi\,(\sin^2\theta + \cos^2\theta) = 2\cos^2\phi$$

18. The number of distinct real roots of the equation $\begin{vmatrix}\sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x\end{vmatrix} = 0$ in $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ is :

समीकरण $\begin{vmatrix}\sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x\end{vmatrix} = 0$ के अन्तराल $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ में विभिन्न वास्तविक मूल्यों की संख्या है :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol. $$\begin{vmatrix}\sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x\end{vmatrix} = 0$$

$C_1 \to C_1 - C_2 \qquad C_2 \to C_2 - C_3$

$$\begin{vmatrix}\sin x - \cos x & 0 & \cos x \\ -(\sin x - \cos x) & \sin x - \cos x & \cos x = 0 \\ 0 & -(\sin x - \cos x) & \sin x\end{vmatrix}$$

$$(\sin x - \cos x)^2\begin{vmatrix}1 & 0 & \cos x \\ -1 & 1 & \cos x \\ 0 & -1 & \sin x\end{vmatrix} = 0$$

$(\sin x - \cos x)^2 [1(\sin x + \cos x) + \cos x(1)] = 0$

$\sin x = \cos x$ $\quad$ $\sin x + 2\cos x = 0$

$\tan x = 1$ $\quad$ $\tan x = -2$

$x = \pi / 4$

19. If the determinant $\begin{vmatrix} \cos 2x & \sin^2 x & \cos 4x \\ \sin^2 x & \cos 2x & \cos^2 x \\ \cos 4x & \cos^2 x & \cos 2x \end{vmatrix}$ is expanded in powers of sin x, then the constant term is :

यदि सारणिक $\begin{vmatrix} \cos 2x & \sin^2 x & \cos 4x \\ \sin^2 x & \cos 2x & \cos^2 x \\ \cos 4x & \cos^2 x & \cos 2x \end{vmatrix}$ का प्रसार sin x, की घातों के रूप में किया जाता है, तो नियत पद है :

(A) 0 $\quad$ (B) 1 $\quad$ (C) – 1 $\quad$ (D) 2

Ans. C

Sol. $\cos 2x(\cos^2 2x - \cos^4 x) - \sin^2 x(\sin^2 x \cos 2x - \cos 4x \cos^2 x) + \cos 4x(\sin^2 x \cos^2 x - \cos 4x \cos 2x)$

$\Rightarrow \cos^3 2x + 2\cos 4x \cos^2 x \sin^2 x - \cos 2x \cos^4 x - \sin^4 x \cos 2x + \sin^2 x \cos^2 x \cos 4x - \cos^2 4x \cos 2x$

$\Rightarrow (1 - 2\sin^2 x)^3 + 2(2(1 - \sin^2 x) - 1)\sin^2 x(1 - \sin^2 x) - [(1 - 2\sin^2 x)(2(1 - 2\sin^2 x) - 1)^2 + (1 - \sin^2 x)^2 (1 - 2\sin^2 x) + \sin^4 x(1 - 2\sin^2 x)]$

$1 + 0 - (1 + 1 + 0)$

$1 - 2 = -1$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

20. The determinant $\begin{vmatrix} a^2 & a^2 - (b-c)^2 & bc \\ b^2 & b^2 - (c-a)^2 & ca \\ c^2 & c^2 - (a-b)^2 & ab \end{vmatrix}$ is divisible by :

सारणिक $\begin{vmatrix} a^2 & a^2 - (b-c)^2 & bc \\ b^2 & b^2 - (c-a)^2 & ca \\ c^2 & c^2 - (a-b)^2 & ab \end{vmatrix}$ विभाज्य होगा :

(A) $a+b+c$ $\quad$ (B) $(a+b)(b+c)(c+a)$ (C) $a^2+b^2+c^2$ $\quad$ (D) $(a-b)(b-c)(c-a)$

Ans. ACD

Sol. Use $C_2 \to C_2 - C_1 - 2C_3$ then $C_1 \to C_1 + C_2$ take $a^2 + b^2 + c^2$ common from first column

21. Which of the following determinant(s) vanish(es)?
निम्न में से किस सारणिक का मान शून्य हो जाता है/हैं?

(A) $\begin{vmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(c+a) \\ 1 & ab & ab(a+b) \end{vmatrix}$

(B) $\begin{vmatrix} 1 & ab & \frac{1}{a}+\frac{1}{b} \\ 1 & bc & \frac{1}{b}+\frac{1}{c} \\ 1 & ca & \frac{1}{c}+\frac{1}{a} \end{vmatrix}$

(C) $\begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$

(D) $\begin{vmatrix} \log_x xyz & \log_x y & \log_x z \\ \log_y xyz & 1 & \log_y z \\ \log_z xyz & \log_z y & 1 \end{vmatrix}$

Ans. ABCD

Sol. (a) $\begin{vmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(c+a) \\ 1 & ab & ab(a+b) \end{vmatrix}$

$$= a^2b^2c^2 \begin{vmatrix} \frac{1}{bc} & 1 & b+c \\ \frac{1}{ac} & 1 & c+a \\ \frac{1}{ab} & 1 & a+b \end{vmatrix}$$

$$= abc \begin{vmatrix} a & 1 & b+c \\ b & 1 & c+a \\ c & 1 & a+b \end{vmatrix}$$

$C_1 \to C_1 + C_3$

$$= abc(a+b+c) \begin{vmatrix} 1 & 1 & b+c \\ 1 & 1 & c+a \\ 1 & 1 & a+b \end{vmatrix} = 0$$

(b) $\begin{vmatrix} 1 & ab & \dfrac{a+b}{ab} \\ 1 & bc & \dfrac{b+c}{bc} \\ 1 & ca & \dfrac{c+a}{ac} \end{vmatrix}$

$= \dfrac{1}{a^2b^2c^2}\begin{vmatrix} ab & a^2b^2 & a+b \\ bc & b^2c^2 & b+c \\ ca & c^2a^2 & c+a \end{vmatrix}$

$\Rightarrow \begin{vmatrix} 1 & ab & (a+b)c \\ 1 & bc & (b+c)a \\ 1 & ca & (c+a)b \end{vmatrix}$

$C_2 \to C_2 + C_3$

$(ab+bc+ca)\begin{vmatrix} 1 & ab & 1 \\ 1 & bc & 1 \\ 1 & ca & 1 \end{vmatrix} = 0$

Similarly for C and D part

22. If $\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix} = 0$ then $\theta =$

यदि $\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix} = 0$ तो $\theta =$

(A) $\dfrac{7\pi}{24}$ (B) $\dfrac{5\pi}{24}$ (C) $\dfrac{11\pi}{24}$ (D) $\dfrac{\pi}{24}$

Ans. AC

Sol. $R_1 \to 2R_1 - R_2$ and $R_2 \to R_2 - R_3$

Exercise C - II

SUBJECTIVE / INTEGER TYPE QUESTIONS

1. If the equations $a(y + z) = x$, $b(z + x) = y$, $c(x + y) = z$ have nontrivial solutions, then find the value of $\frac{1}{1+a}+\frac{1}{1+b}+\frac{1}{1+c}$.

यदि समीकरणों $a(y + z) = x$, $b(z + x) = y$, $c(x + y) = z$ के अनिरर्थक हल हैं, तो $\frac{1}{1+a}+\frac{1}{1+b}+\frac{1}{1+c}$ का मान ज्ञात कीजिये।

Ans. 2

Sol. $x - ay - az = 0$

$bx - y + bz = 0$

$cx + cy - z = 0$

For non-trivial solution, $D = 0$

$$\begin{vmatrix} 1 & -a & -a \\ b & -1 & b \\ c & c & -1 \end{vmatrix} = 0$$

$$C_1 \to C_1 - C_2$$

$$\begin{vmatrix} 1+a & 0 & -a \\ b+1 & -(1+b) & b \\ 0 & c+1 & -1 \end{vmatrix} = 0$$

$(1 + a)((1 + b) - b(c + 1)) - (1 + b)(a(c + 1)) = 0$

Divide by $(1 + a)(1 + b)(1 + c)$

$$\frac{1}{1+c}-\frac{b}{1+b}-\frac{a}{1+a}=0$$

$$\frac{1}{1+c}+\frac{1}{1+b}+\frac{1}{1+a}=2$$

2. Given $a=\frac{x}{y-z}; b=\frac{y}{z-x}; c=\frac{z}{x-y}$ where x, y, z are not all zero, prove that: $1 + ab + bc + ca = 0$.

दिया गया है $a=\frac{x}{y-z}; b=\frac{y}{z-x}; c=\frac{z}{x-y}$ जहाँ x, y, z सभी शून्य नहीं है, सिद्ध कीजिये कि $1 + ab + bc + ca = 0$.

Ans.

Sol. $ay - az = x, bz - bx = y, cx - cy = z$

$$\begin{vmatrix} 1 & -a & a \\ b & 1 & -b \\ c & -c & -1 \end{vmatrix} = 0$$

$\Rightarrow (-1 - bc) + a(-b + bc) + a(-bc - c) = 0$

$-1 - bc - ab + abc - abc - ac = 0$

$\Rightarrow 1 + ab + bc + ca = 0$

3. If $\sin q \neq \cos q$ and x, y, z satisfy the equations

$x \cos p - y \sin p + z = \cos q + 1$

$x \sin p + y \cos p + z = 1 - \sin q$

$x \cos(p + q) - y \sin (p + q) + z = 2$

then find the value of $x^2 + y^2 + z^2$.

यदि $\sin q \neq \cos q$ तथा x, y, z समीकरणों

$x \cos p - y \sin p + z = \cos q + 1$

$x \sin p + y \cos p + z = 1 - \sin q$

$x \cos(p + q) - y \sin (p + q) + z = 2$

को संतुष्ट करते हैं, तो $x^2 + y^2 + z^2$ का मान ज्ञात कीजियें।

Ans. 2

Sol. $x \cos p - y \sin p + z = \cos q + 1$ _____(1)

$x \sin p + y \cos p + z = 1 - \sin q$ _____(2)

Multiply equation (1) by cos q & equation (2) by sin q & subtracting

$x \cos (p + q) - y \sin (p + q) + z (\cos q - \sin q) = 1 + \cos q - \sin q$ _____(3)

$x \cos (p + q) - y \sin (p + q) + z = 2$ _____(4)

Eq. (3) – (4)

$z(\cos q - \sin q - 1) = -1 + \cos q - \sin q$

$\Rightarrow x \cos p + y \sin p = \cos q$

and $x \sin p + y \cos p = -\sin q$

On solving, $x = \cos(p + q)$

$y = -\sin(p + q)$

$z = 1$

$x^2 + y^2 + z^2 = 2$

4. Investigate for what values of λ, μ the simultaneous equations $x + y + z = 6$; $x + 2y + 3z = 10$ & $x + 2y + \lambda z = \mu$ have :
(a) A unique solution
(b) An infinite number of solutions
(c) No solution

जाँच कीजिये कि λ, μ के किन मानों के लिये युगपत समीकरणों $x + y + z = 6$; $x + 2y + 3z = 10$ तथा $x + 2y + \lambda z = \mu$ के निकाय :
(a) के अद्वितीय हल होंगे
(b) के अन्नत हल होंगे
(c) का कोई हल नहीं होगा

Ans. (a) $\lambda \neq 3$ (b) $\lambda = 3, \mu = 10$ (c) $\lambda = 3, \mu \neq 10$

Sol. $D = \begin{vmatrix} k & 2 & -2 \\ 4 & 2k & -1 \\ 6 & 6 & k \end{vmatrix} = k(2k^2 + 6) - 2(4k + 6) - 2(24 - 12k)$

$\Rightarrow 2k^3 + 6k - 8k - 12 - 48 + 24k$

$\Rightarrow 2k^3 + 22k - 60$

$\Rightarrow 2(k^3 + 11k - 30)$

$\Rightarrow 2(k - 2)(k^2 + 2k + 15)$

For unique solution, $k \neq 2$

$D_1 = \begin{vmatrix} 1 & 2 & -2 \\ 2 & 2k & -1 \\ 3 & 6 & k \end{vmatrix} = (2k^2 + 6) - 2(2k + 3) - 2(12 - 6k)$

$\Rightarrow 2k^2 + 6 - 4k - 6 - 24 + 12k$

$\Rightarrow 2k^2 + 8k - 24$

$\Rightarrow 2(k^2 + 4k - 12) \Rightarrow 2(k + 6)(k - 2)$

$$x = \frac{D_1}{D} = \frac{k+6}{k^2+2k+15}$$

$$D_2 = \begin{vmatrix} k & 1 & -2 \\ 4 & 2 & -1 \\ 6 & 3 & k \end{vmatrix} = k(2k + 3) - (4k + 6) - 2(12 - 12)$$

$\Rightarrow 2k^2 - k - 6$

$\Rightarrow (2k + 3)(k - 2)$

$$y = \frac{2k+3}{2\left(k^2+2k+15\right)}$$

$$D_3 = \begin{vmatrix} k & 2 & 1 \\ 4 & 2k & 2 \\ 6 & 6 & 3 \end{vmatrix} = k(6k - 12) - 2(12 - 12) + (24 - 12k)$$

$\Rightarrow 6k^2 - 12k + 24 - 12k$

$\Rightarrow 6(k^2 - 4k + 4) \Rightarrow 6(k - 2)^2$

$$z = \frac{3(k-2)}{k^2+2k+15}$$

If $k = 2$, infinite solution

Put $x = \lambda$,

we get $y = \frac{1-2\lambda}{2}$, $z = 0$

5. Solve the equations : $Kx+2y-2z=1,\ 4x+2Ky-z=2,\ 6x+6y+Kz=3$

considering specially the case when $K = 2$. If $K = 2$, then $x = \lambda$, $y = \frac{1-2\lambda}{2}$ and $z = 0$, where $\lambda \in R$

समीकरणों : $Kx+2y-2z=1,\ 4x+2Ky-z=2,\ 6x+6y+Kz=3$ को

$K = 2$ के विशिष्ट मान के लिये हल कीजियें। यदि $K= 2$, तो $x = \lambda$, $y = \frac{1-2\lambda}{2}$ तथा $z = 0$ जहाँ $\lambda \in R$

Ans. If $K \neq 2$, $\frac{x}{2(K+6)} = \frac{y}{2K+3} = \frac{z}{6(K-2)} = \frac{1}{2\left(K^2+2K+15\right)}$, If K= 2, then $x = \lambda$, $y = \frac{1-2\lambda}{2}$ and $z = 0$ where $\lambda \in R$

Sol. $D = \begin{vmatrix} k & 2 & -2 \\ 4 & 2k & -1 \\ 6 & 6 & k \end{vmatrix} = k(2k^2 + 6) - 2(4k + 6) - 2(24 - 12k)$

$\Rightarrow 2k^3 + 6k - 8k - 12 - 48 + 24k$

$\Rightarrow 2k^3 + 22k - 60$

$\Rightarrow 2(k^3 + 11k - 30)$

$\Rightarrow 2(k - 2)(k^2 + 2k + 15)$

For unique solution, $k \neq 2$

$D_1 = \begin{vmatrix} 1 & 2 & -2 \\ 2 & 2k & -1 \\ 3 & 6 & k \end{vmatrix} = (2k^2 + 6) - 2(2k + 3) - 2(12 - 6k)$

$\Rightarrow 2k^2 + 6 - 4k - 6 - 24 + 12k$

$\Rightarrow 2k^2 + 8k - 24$

$\Rightarrow 2(k^2 + 4k - 12) \Rightarrow 2(k + 6)(k - 2)$

$x = \frac{D_1}{D} = \frac{k+6}{k^2+2k+15}$

$D_2 = \begin{vmatrix} k & 1 & -2 \\ 4 & 2 & -1 \\ 6 & 3 & k \end{vmatrix} = k(2k + 3) - (4k + 6) - 2(12 - 12)$

$\Rightarrow 2k^2 - k - 6$

$\Rightarrow (2k + 3)(k - 2)$

$y = \frac{2k+3}{2\left(k^2+2k+15\right)}$

$D_3 = \begin{vmatrix} k & 2 & 1 \\ 4 & 2k & 2 \\ 6 & 6 & 3 \end{vmatrix} = k(6k - 12) - 2(12 - 12) + (24 - 12k)$

$\Rightarrow 6k^2 - 12k + 24 - 12k$

$\Rightarrow 6(k^2 - 4k + 4) \Rightarrow 6(k-2)^2$

$$z = \frac{3(k-2)}{k^2 + 2k + 15}$$

If $k = 2$, infinite solution

Put $x = \lambda$,

we get $y = \frac{1 - 2\lambda}{2}$, $z = 0$

6. Let a, b, c, d are distinct numbers to be chosen from the set $\{1, 2, 3, 4, 5\}$. If the least possible positive solution for x to the system of equations $\left.\begin{matrix} ax + by = 1 \\ cx + dy = 2 \end{matrix}\right\}$ can be expressed in the form $\frac{p}{q}$ where p and q are relatively prime, then find the value of $(p + q)$.

समुच्चय $\{1, 2, 3, 4, 5\}$ में से चार भिन्न संख्याओं a, b, c, d का चुनाव किया जाता है। यदि समीकरणों के निकाय $\left.\begin{matrix} ax + by = 1 \\ cx + dy = 2 \end{matrix}\right\}$ के न्यूनतम संभावित धनात्मक हल x को $\frac{p}{q}$ के रूप में प्रदर्शित किया जा सकता है, जहाँ p तथा q सापेक्षतः अभाज्य है, तो $(p + q)$ का मान ज्ञात कीजियें।

Ans. 19

Sol. $$x = \frac{D_1}{D} = \frac{\begin{vmatrix} 1 & b \\ 2 & d \end{vmatrix}}{\begin{vmatrix} a & b \\ c & d \end{vmatrix}} = \frac{d - 2b}{ad - cb}$$

To get least value, $d - 2b$ is minimum & $ad - cb$ will be maximum.

$d - 2b = 1$

Case-I $d = 5, b = 2$

$$x = \frac{1}{5a - 2c}$$

$5a - 2c$ should be maximum.

$a = 4, c = 1$

Case-II $d = 3, b = 1$

$$x = \frac{1}{3a - c}$$

$3a - c$ should be maximum.

$a = 5, c = 2$

$x = \frac{1}{18}$ $x = \frac{1}{13}$

$\frac{1}{18} < \frac{1}{13}$

$x = \frac{1}{18} = \frac{p}{q}$, $p + q = 19$

7. Show that the system of equations

$3x - y + 4z = 3$, $x + 2y - 3z = -2$ and $6x + 5y + \lambda z = -3$

has atleast one solution for any real number λ. Find the set of solutions of $\lambda = -5$.

सिद्ध कीजिये कि समीकरणों के निकाय $3x - y + 4z = 3$, $x + 2y - 3z = -2$ तथा $6x + 5y + \lambda z = -3$ का λ के किसी भी वास्तविक मान के लिये कम से कम एक हल है तथा $\lambda = -5$ के लिए हलों का समुच्चय भी ज्ञात कीजियें।

Ans. If $\lambda \neq -5$ then $x = \frac{4}{7}$; $y = -\frac{9}{7}$ and $z = 0$; If $\lambda = 5$ then $x = \frac{4-5K}{7}$; $y = \frac{13K-9}{7}$ and $z = K$ where $K \in R$

Sol. $D = \begin{vmatrix} 3 & -1 & 4 \\ 1 & 2 & -3 \\ 6 & 5 & \lambda \end{vmatrix} = 3(2\lambda + 15) + (\lambda + 18) + 4(5 - 12)$

$\Rightarrow 7\lambda + 45 + 18 - 28$

$\Rightarrow 7(\lambda + 5)$

for $\lambda = -5$

Put $z = k$

$3x - y = 3 - 4k$ _____(1)

$x + 2y = -2 + 3k$ _____(2)

Multiply eq. (1) by 2 and subtract eq. (2)

$7x = 6 - 8k - 2 + 3k$

$x = \frac{4-5k}{7}$, $y = \frac{13k-9}{7}$

8. Solve the system of equations; $\left.\begin{aligned} z+ay+a^2x+a^3=0 \\ z+by+b^2x+b^3=0 \\ z+cy+c^2x+c^3=0 \end{aligned}\right]$

समीकरणों के निकाय $\left.\begin{aligned} z+ay+a^2x+a^3=0 \\ z+by+b^2x+b^3=0 \\ z+cy+c^2x+c^3=0 \end{aligned}\right]$ को सरल कीजिये।

Ans. $x=-(a+b+c)$, $y=ab+bc+ca$, $z=-abc$

Sol. $D=\begin{vmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{vmatrix}=(a-b)(b-c)(c-a)$

$$D_1=\begin{vmatrix} -a^3 & a & 1 \\ -b^3 & b & 1 \\ -c^3 & c & 1 \end{vmatrix}=-(a-b)(b-c)(c-a)(a+b+c)$$

$$D_2=\begin{vmatrix} a^2 & -a^3 & 1 \\ b^2 & -b^3 & 1 \\ c^2 & -c^3 & 1 \end{vmatrix}=-(a-b)(b-c)(c-a)(ab+bc+ca)$$

$$D_3=\begin{vmatrix} a^2 & a & -a^3 \\ b^2 & b & -b^3 \\ c^2 & c & -c^3 \end{vmatrix}=-abc(a-b)(b-c)(c-a)$$

$$x=\frac{D_1}{D}=-(a+b+c)$$

$$y=\frac{D_2}{D}=-(ab+bc+ca)$$

$$z = \frac{D_3}{D} = -abc$$

9. If $a^2 + b^2 + c^2 = 1$ then show that the value of the determinant

$$\begin{vmatrix} a^2+(b^2+c^2)\cos\theta & ba(1-\cos\theta) & ca(1-\cos\theta) \\ ab(1-\cos\theta) & b^2+(c^2+a^2)\cos\theta & cb(1-\cos\theta) \\ ac(1-\cos\theta) & bc(1-\cos\theta) & c^2+(a^2+b^2)\cos\theta \end{vmatrix}$$ simplifies to $\cos^2\theta$.

यदि $a^2 + b^2 + c^2 = 1$ तो सिद्ध कीजिये सारणिक $\begin{vmatrix} a^2+(b^2+c^2)\cos\theta & ba(1-\cos\theta) & ca(1-\cos\theta) \\ ab(1-\cos\theta) & b^2+(c^2+a^2)\cos\theta & cb(1-\cos\theta) \\ ac(1-\cos\theta) & bc(1-\cos\theta) & c^2+(a^2+b^2)\cos\theta \end{vmatrix}$ का

मान $\cos^2\theta$ में सरल हो जाता है।

Ans.

Sol. $a^2 + b^2 + c^2 = 1$

$$\Delta = \begin{vmatrix} a^2+(b^2+c^2)\cos\theta & ab(1-\cos\theta) & ac(1-\cos\theta) \\ ba(1-\cos\theta) & b^2+(c^2+a^2)\cos\theta & bc(1-\cos\theta) \\ ca(1-\cos\theta) & cb(1-\cos\theta) & c^2+(a^2+b^2)\cos\theta \end{vmatrix}$$

$$\Delta = \frac{1}{abc}\begin{vmatrix} a^3+a(b^2+c^2)\cos\theta & b^2a(1-\cos\theta) & ac^2(1-\cos\theta) \\ ba^2(1-\cos\theta) & b^3+b(c^2+a^2)\cos\theta & bc^2(1-\cos\theta) \\ ca^2(1-\cos\theta) & cb^2(1-\cos\theta) & c^3+c(a^2+b^2)\cos\theta \end{vmatrix}$$

$$\frac{abc}{abc}\begin{vmatrix} a^2+(b^2+c^2)\cos\theta & b^2(1-\cos\theta) & c^2(1-\cos\theta) \\ a^2(1-\cos\theta) & b^2+(c^2+a^2)\cos\theta & c^2(1-\cos\theta) \\ a^2(1-\cos\theta) & b^2(1-\cos\theta) & c^2+(a^2+b^2)\cos\theta \end{vmatrix}$$

$C_1 \to C_1 + C_2 + C_3$

$$= (a^2 + b^2 + c^2)\begin{vmatrix} 1 & b^2(1-\cos\theta) & c^2(1-\cos\theta) \\ 1 & b^2+(c^2+a^2)\cos\theta & c^2(1-\cos\theta) \\ 1 & b^2(1-\cos\theta) & c^2+(a^2+b^2)\cos\theta \end{vmatrix}$$

Applying $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$

संक्रिया $R_2 \to R_2 - R_1$ एवं $R_3 \to R_3 - R_1$ से

$$(a^2 + b^2 + c^2)\begin{vmatrix} 1 & b^2(1-\cos\theta) & c^2(1-\cos\theta) \\ 0 & (a^2+b^2+c^2)\cos\theta & 0 \\ 0 & 0 & (a^2+b^2+c^2)\cos\theta \end{vmatrix} = (a^2 + b^2 + c^2)^3 \cos^2\theta = \cos^2\theta$$

This is independent from a, b, c. यह a, b, c से स्वतन्त्र है।

10. In a Δ ABC, determine condition under which $\begin{vmatrix} \cot\frac{A}{2} & \cot\frac{B}{2} & \cot\frac{C}{2} \\ \tan\frac{B}{2}+\tan\frac{C}{2} & \tan\frac{C}{2}+\tan\frac{A}{2} & \tan\frac{A}{2}+\tan\frac{B}{2} \\ 1 & 1 & 1 \end{vmatrix} = 0.$

एक Δ ABC में, वह प्रतिबन्ध ज्ञात कीजिये जिसके लिये $\begin{vmatrix} \cot\frac{A}{2} & \cot\frac{B}{2} & \cot\frac{C}{2} \\ \tan\frac{B}{2}+\tan\frac{C}{2} & \tan\frac{C}{2}+\tan\frac{A}{2} & \tan\frac{A}{2}+\tan\frac{B}{2} \\ 1 & 1 & 1 \end{vmatrix} = 0$

Ans. Triangle ABC is isosceles

Sol. $C_1 \to C_1 - C_2$, $C_2 \to C_2 - C_3$

$$\begin{vmatrix} \cot\frac{A}{2}-\cot\frac{B}{2} & \cot\frac{B}{2}-\cot\frac{C}{2} & \cot\frac{C}{2} \\ \tan\frac{B}{2}-\tan\frac{A}{2} & \tan\frac{C}{2}-\tan\frac{B}{2} & \tan\frac{A}{2}+\tan\frac{B}{2} \\ 0 & 0 & 1 \end{vmatrix} = 0$$

$$\left(\cot\frac{A}{2}-\cot\frac{B}{2}\right)\left(\cot\frac{B}{2}-\cot\frac{C}{2}\right)\begin{vmatrix} 1 & 1 & \cot\frac{C}{2} \\ \frac{1}{\cot\frac{A}{2}\cot\frac{B}{2}} & \frac{1}{\cot\frac{B}{2}\cot\frac{C}{2}} & \tan\frac{A}{2}+\tan\frac{B}{2} \\ 0 & 0 & 1 \end{vmatrix}=0$$

$A = B$ or $B = C$ or $C = A$

ΔABC is an isosceles triangle.

11. If $ax_1^2 + by_1^2 + cz_1^2 = ax_2^2 + by_2^2 + cz_2^2 = ax_3^2 + by_3^2 + cz_3^2 = d$
and $ax_2x_3 + by_2y_3 + cz_2z_3 = ax_3x_1 + by_3y_1 + cz_3z_1 = ax_1x_2 + by_1y_2 + cz_1z_2 = f$,

then prove that $\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = (d-f)\left[\frac{d+2f}{abc}\right]^{\frac{1}{2}}$ $(a, b, c \neq 0)$

यदि $ax_1^2 + by_1^2 + cz_1^2 = ax_2^2 + by_2^2 + cz_2^2 = ax_3^2 + by_3^2 + cz_3^2 = d$
तथा $ax_2x_3 + by_2y_3 + cz_2z_3 = ax_3x_1 + by_3y_1 + cz_3z_1 = ax_1x_2 + by_1y_2 + cz_1z_2 = f$,

तो सिद्ध कीजिए $\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = (d-f)\left[\frac{d+2f}{abc}\right]^{\frac{1}{2}}$ $(a, b, c \neq 0)$

Ans.

Sol. $\Delta = \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix}$

$$\Delta^2 = \frac{1}{abc}\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} \times \begin{vmatrix} ax_1 & by_1 & cz_1 \\ ax_2 & by_2 & cz_3 \\ ax_3 & by_3 & cz_3 \end{vmatrix}$$

$$= \frac{1}{abc}\begin{vmatrix} d & f & f \\ f & d & f \\ f & f & d \end{vmatrix}$$

$C_1 \to C_1 + C_2 + C_3$

$$= \frac{1}{abc}\begin{vmatrix} d+2f & f & f \\ d+2f & d & f \\ d+2f & f & d \end{vmatrix} = \frac{(d+2f)}{abc}\begin{vmatrix} 1 & f & f \\ 1 & d & f \\ 1 & f & d \end{vmatrix}$$

on solving it we get

$$\Delta^2 = \frac{(d+2f)}{abc}(d-f)^2$$

$$\Delta = (d-f)\left[\frac{d+2f}{abc}\right]^{\frac{1}{2}}$$

12. If $u = ax^2 + 2bxy + cy^2$, $u' = a'x^2 + 2b'xy + c'y^2$. Prove that

यदि $u = ax^2 + 2bxy + cy^2$, $u' = a'x^2 + 2b'xy + c'y^2$ हो, तो सिद्ध कीजिये कि

$$\begin{vmatrix} y^2 & -xy & x^2 \\ a & b & c \\ a' & b' & c' \end{vmatrix} = \begin{vmatrix} ax+by & bx+cy \\ a'x+b'y & b'x+c'y \end{vmatrix} = -\frac{1}{y}\begin{vmatrix} u & u' \\ ax+by & a'x+b'y \end{vmatrix}.$$

Ans.

Sol. $\begin{vmatrix} y^2 & -xy & x^2 \\ a & b & c \\ a' & b' & c' \end{vmatrix}$

$$\Rightarrow y^2(bc'-b'c) + xy(ac'-a'c) + x^2(ab'-a'b)$$

$$\begin{vmatrix} ax+by & bx+cy \\ a'x+b'y & b'x+c'y \end{vmatrix} = (ax+by)(b'x+c'y) - (bx+cy)(a'x+b'y)$$

$$\Rightarrow ab'x^2 + ac'xy + bb'xy + bc'y^2 - a'bx^2 - bb'xy - a'cxy - b'cy^2$$

$$\Rightarrow y^2(bc'-b'c) + xy(ac'-a'c) + x^2(ab'-a'b)$$

$$-\frac{1}{y}\begin{vmatrix} ax^2+bxy+cy^2 & a'x^2+2b'xy+c'y^2 \\ ax+by & a'x+b'y \end{vmatrix}$$

$$\Rightarrow -\frac{1}{y}\left(aa'x^3 + ab'x^2y + 2a'bx^2y + 2bb'xy^2 + a'cxy^2 + b'cy^3 - aa'x^3 - 2ab'x^2y\right.$$

$-ac'xy^2 - a'bx^2y - 2b'bxy^2 - bc'y^3)$

$\Rightarrow -ab'x^2 - 2a'bx^2 - a'cxy - b'cy^2 + 2ab'x^2 + ac'xy + a'bx^2 + bc'y^2$

$\Rightarrow x^2(ab'-a'b) + xy(ac'-a'c) + y^2(bc'-b'c)$

13. Test the consistency and solve them when consistent, the following system of equations for all values of λ

λ के मानों के लिये दिये गये निकाय के संगत तथा असंगत होने की जाँच कीजिए तथा प्रत्येक स्थिति में हल भी ज्ञात किजिए।

$x + y + z = 1$
$x + 3y - 2z = \lambda$
$3x + (\lambda + 2)y - 3z = 2\lambda + 1$

Ans. If $\lambda = 5$, system is consistent with infinite solution given by $z = K$, $y = \frac{1}{2}(3K + 4)$ and $x = -\frac{1}{2}(5K + 2)$ where K $\in$ R. If $\lambda \neq 5$, system is consistent with unique solution given by $z = \frac{1}{3}(1 - \lambda)$; $x = \frac{1}{3}(\lambda + 2)$ and $y = 0$.

Sol. $$D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & -2 \\ 3 & \lambda+2 & -3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 2 & -3 \\ 3 & \lambda-1 & -6 \end{vmatrix}$$

$\Rightarrow -12 + 3\lambda - 3 \Rightarrow 3\lambda - 15 = 3(\lambda - 5)$

If $\lambda \neq 5$, unique solution

$$D_1 = \begin{vmatrix} 1 & 1 & 1 \\ \lambda & 3 & -2 \\ 2\lambda+1 & \lambda+2 & -3 \end{vmatrix} = -9 + 2\lambda + 4 + 3\lambda - 4\lambda - 2 + \lambda^2 + 2\lambda - 6\lambda - 3$$

$\Rightarrow \lambda^2 - 3\lambda - 10 \Rightarrow (\lambda - 5)(\lambda + 2)$

$$D_2 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \lambda & -2 \\ 3 & 2\lambda+1 & -3 \end{vmatrix} = -3\lambda + 4\lambda + 2 + 3 - 6 + 2\lambda + 1 - 3\lambda$$

$\Rightarrow 0$

$$D_3 = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & \lambda \\ 3 & \lambda+2 & 2\lambda+1 \end{vmatrix} \Rightarrow 6\lambda + 3 - \lambda^2 - 2\lambda - 2\lambda - 1 + 3\lambda + \lambda + 2 - 9$$

$\Rightarrow \lambda^2 + 6\lambda - 5$

$\Rightarrow -(\lambda - 1)(\lambda - 5)$

$$x = \frac{1}{3}(\lambda + 2),\ y = 0,\ z = \frac{-1}{3}(\lambda - 1)$$

If $\lambda = 5$, put $z = k$

$x + y = 1 - k$

$x + 3y = 5 + 2k$

$$y = \frac{1}{2}(3k + 4), x = \frac{-1}{2}(5k + 2)$$

14. Prove that the value of the determinant $\begin{vmatrix} -7 & 5+3i & \frac{2}{3}-4i \\ 5-3i & 8 & 4+5i \\ \frac{2}{3}+4i & 4-5i & 9 \end{vmatrix}$ is real.

सिद्ध कीजिये कि सारणिक $\begin{vmatrix} -7 & 5+3i & \frac{2}{3}-4i \\ 5-3i & 8 & 4+5i \\ \frac{2}{3}+4i & 4-5i & 9 \end{vmatrix}$ का मान वास्तविक है।

Ans.

Sol. Let $A = \begin{vmatrix} -7 & 5+3i & \frac{2}{3}-4i \\ 5-3i & 8 & 4+5i \\ \frac{2}{3}+4i & 4-5i & 9 \end{vmatrix}$

$$\overline{A} = \begin{vmatrix} -7 & 5-3i & \frac{2}{3}+4i \\ 5+3i & 8 & 4-5i \\ \frac{2}{3}-4i & 4+5i & 9 \end{vmatrix} = A^T$$

$A = \left|\overline{A}\right| = \left|A^T\right|$

|A| is real.

15. Prove that $\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1 + a^2 + b^2)^3$.

सिद्ध कीजिये $\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1 + a^2 + b^2)^3$.

Ans.

Sol. $\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix}$

$C_1 \to C_1 - bC_3$, $C_2 \to C_2 + aC_3$

$$= \begin{vmatrix} 1+a^2+b^2 & 0 & -2b \\ 0 & 1+a^2+b^2 & 2a \\ b(1+a^2+b^2) & -a(1+b^2+a^2) & 1-a^2-b^2 \end{vmatrix}$$

$$= (1 + a^2 + b^2)^2 \begin{vmatrix} 1 & 0 & -2b \\ 0 & 1 & 2a \\ b & -a & 1-a^2-b^2 \end{vmatrix}$$

$\equiv (1 + a^2 + b^2)^3$

16. Show that the value of the determinant $\begin{vmatrix} \tan(A+P) & \tan(B+P) & \tan(C+P) \\ \tan(A+Q) & \tan(B+Q) & \tan(C+Q) \\ \tan(A+R) & \tan(B+R) & \tan(C+R) \end{vmatrix}$ vanishes for all values of A, B, C, P, Q & R where A + B + C + P + Q + R = 0

सिद्ध कीजिये कि सारणिक $\begin{vmatrix} \tan(A+P) & \tan(B+P) & \tan(C+P) \\ \tan(A+Q) & \tan(B+Q) & \tan(C+Q) \\ \tan(A+R) & \tan(B+R) & \tan(C+R) \end{vmatrix}$ का मान A, B, C, P, Q तथा R जहाँ

`A + B + C + P + Q + R = 0 के प्रत्येक सम्भव संचय के लिए शून्य होगा।

Ans.

Sol. If $X + Y + Z = 0$ then $\tan X + \tan Y + \tan Z = \tan X \tan Y \tan Z$

Expanding by R_1 :

$\tan (A + P) (\tan (B + Q) \tan (C + R) - \tan (B + R) \tan (C + Q))$
$- \tan (B + P (\tan (A + Q) \tan (C + R) - \tan (A + R) \tan (C + Q))$
$+ \tan (C + P) (\tan (A + Q) \tan (B + R) - \tan (A + R) \tan (B + Q)) = 0$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question nos. 17 to 19

Let $x, y, z \in R^+$ & $\Delta = \begin{vmatrix} x & x^3 & x^4 - 1 \\ y & y^3 & y^4 - 1 \\ z & z^3 & z^4 - 1 \end{vmatrix}$

माना $x, y, z \in R^+$ तथा $\Delta = \begin{vmatrix} x & x^3 & x^4 - 1 \\ y & y^3 & y^4 - 1 \\ z & z^3 & z^4 - 1 \end{vmatrix}$

17. If $x \neq y \neq z$ & x, y, z are in GP and $\Delta = 0$ then y is equal to :

(A) 1 (B) 2 (C) 4 (D) None of these

यदि $x \neq y \neq z$ तथा x, y, z गुणोतर श्रेणी में है तथा $\Delta = 0$ तो y होगा :

(A) 1 (B) 2 (C) 4 (D) इनमें से कोई नहीं

Ans. A

Sol. Put $x = \frac{y}{r}, z = yr$

$\Rightarrow y = 1$

18. If x, y, z are roots of $t^3 - 21t^2 + bt - 343 = 0$, $b \in R$ then $\Delta =$

(A) 1 (B) 0

(C) depends on x, y, z (D) data inadequate

यदि x, y, z समीकरण $t^3 - 21t^2 + bt - 343 = 0$, $b \in R$ के मूल है, तो $\Delta =$

(A) 1 (B) 0

(C) x, y, z पर निर्भर करता है (D) अपर्याप्त जानकारी

Ans. B

Sol. $\left(\frac{y}{r}\right)(y)(yr) = 343 \Rightarrow y = 7$

$\frac{7}{r} + 7 + 7r = 21$

$\Rightarrow r = 1 \qquad x = y = z$

$\Delta = 0$

19. If x, y, z are in A.P. and $\Delta = 0$ then $2xy^2z + x^2z^2 =$

(A) 1 (B) 2 (C) 3 (D) None of these

यदि x, y, z A.P. में हो तथा $\Delta = 0$ तो $2xy^2z + x^2z^2 =$

(A) 1 (B) 2 (C) 3 (D) इनमें से कोई नहीं

Ans. C

Sol. $x = y = z = 1 \Rightarrow d = 0$

$2xy^2z + x^2z^2 = 3$

Paragraph for Question nos. 20 to 22

Consider the system of equations $2x + ay + 6z = 8$, $x + 2y + bz = 5$ and $x + y + 3z = 4$ then

माना समीकरण निकाय $2x + ay + 6z = 8$, $x + 2y + bz = 5$ तथा $x + y + 3z = 4$ है, तो

20. The system of equations will have infinite solutions if :

समीकरण निकाय के अनन्त हल होगें, यदि :

(A) $b = 3$ (B) $b = 2$ (C) $a = 3$ (D) $a = 2$

Ans. D

21. The given system of equations will have no solution if :

समीकरण निकाय का कोई हल नहीं होगा, यदि :

(A) $a = 2, b \neq 3$ (B) $a \neq 2, b \neq 3$ (C) $a \neq 2, b = 3$ (D) $a = 2, b = 3$

Ans. C

22. The given system of equations will have a unique solution if :

समीकरण निकाय का अद्वितीय हल होगा, यदि :

(A) $a \neq 2, b \neq 3$ (B) $a \neq 2$ (C) $a = 2, b = 3$ (D) $a = 2, b \neq 3$

Ans. A

Sol. $$D = \begin{vmatrix} 2 & a & 6 \\ 1 & 2 & b \\ 1 & 1 & 3 \end{vmatrix} = 2(6-b) - a(3-b) + 6(1-2)$$

$\Rightarrow 12 - 2b - 3a + ab - 6$

$\Rightarrow a(b-3) - 2(b-3) \Rightarrow (a-2)(b-3)$

$$D_1 = \begin{vmatrix} 8 & a & 6 \\ 5 & 2 & b \\ 4 & 1 & 3 \end{vmatrix} = 8(6-b) - a(15-4b) + 6(5-8)$$

$\Rightarrow 48 - 8b - 15a + 4ab - 18 \Rightarrow 4b(a-2) - 15(a-2)$

$\Rightarrow (4b-15)(a-2)$

$$D_2 = \begin{vmatrix} 2 & 8 & 6 \\ 1 & 5 & b \\ 1 & 4 & 3 \end{vmatrix} = 2(15-4b) - (24-24) + (8b-30)$$

$\Rightarrow 0$

$$D_3 = \begin{vmatrix} 2 & a & 8 \\ 1 & 2 & 5 \\ 1 & 1 & 4 \end{vmatrix} = 2(8-5) - a(4-5) + 8(1-2)$$

$\Rightarrow 6 + a - 8$

$\Rightarrow a - 2$

For unique solution; $D \neq 0$; $a \neq 2, b \neq 3$

For infinitely many solutions; $D = D_1 = D_2 = D_3 = 0$

$\Rightarrow a = 2$

For no solution; $D = 0$ & either one of $D_1, D_2, D_3 \neq 0$

$a \neq 2, b = 3$

Paragraph for Question nos. 23 to 24

Given $f(x)=\begin{vmatrix} 5+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 5+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 5+4\sin 2x \end{vmatrix}$, then :

दिया गया है कि $f(x)=\begin{vmatrix} 5+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 5+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 5+4\sin 2x \end{vmatrix}$, तो

23. The domain of the function f(x) :

फलन f(x) का प्रांत होगा :

(A) $(-\infty, 0)$ (B) $(-\infty,\infty)$ (C) $(0, \infty)$ (D) $(0, 1)$

Ans. B

24. The range of the function f(x) :

(A) [50, 250] (B) [25, 250] (C) [50, 50] (D) None of these

फलन f(x) का परिसर होगा :

(A) [50, 250] (B) [25, 250] (C) [50, 50] (D) इनमें से कोई नहीं

Ans. A

Sol. $R_2 \to R_2 - R_1, R_3 \to R_3 - R_1$

$$f(x)=\begin{vmatrix} 5+\sin^2 x & \cos^2 x & 4\sin 2x \\ -5 & 5 & 0 \\ -5 & 0 & 5 \end{vmatrix}$$

$$\Rightarrow 5\left(5\cos^2 x\right)+5\left(25+5\sin^2 x+20\sin 2x\right)$$

$$\Rightarrow 25+125+100\sin 2x$$

$$\Rightarrow 150+100\sin 2x$$

Domain : R

Range : [50, 250]